ABSTRACT

Title of Dissertation:	AGE-STRUCTURED MODELS OF TUMOR GROWTH AND EVOLUTION OF DRUG RESISTANCE
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Spatial heterogeneity plays a critical role in anti-tumor treatment response, influencing how therapies are delivered and interact with tumors. While there have been numerous studies using age-structured models to capture tumor cell behavior in drug response and proliferation that depends on cell cycles, most of these models remain spatially homogeneous. This thesis presents age-structured spatio-temporal tumor cell density models. These models incorporate spatial structure that allows model behaviors such as pressure-driven cell migration and inhibition of proliferation, and delivery of nutrients and drugs.

We develop an age-structured spatio-temporal model of tumor growth, in which cell movement and proliferation is largely driven by pressure. The age of cells is incorporated in the model as a continuous variable, where birth and death mechanisms of tumor cells are allowed to vary by cell age. We demonstrate that the asymptotic behavior of the solutions of our model depends on

the presence of cell death. The fronts of model solutions are shown to propagate in finite speed. Additionally, age distributions and the spatial mitotic patterns are investigated. In model simulations, we observe the emergence of a generic structure in early solid tumor, consisting of the non-proliferating tumor core, actively proliferating rim slightly inside the periphery, and 'surfing' cells at the edge of tumor.

We further expand the model by introducing chemotherapy and drug resistance. The proliferation mechanism in this model mainly depends on the nutrient concentration and uptake rate, while proliferation is limited at the homeostatic pressure. Tumor growth and resistance evolution dynamics under varying doses of chemotherapy are observed from several aspects, including the age distribution across tumors. The results demonstrate drug-induced evolution of resistance, where the selection force of drug is greater under higher doses.

Age-Structured Models of Tumor Growth and Evolution of Drug Resistance

by

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List of Abbreviations

ABC	ATP-binding cassette
ABM	Agent-Based Model
AML	Acute myeloid leukemia
ATP	Adenosine Triphosphate
CAF	Cancer-Associated Fibroblast
CDK	Cyclin-Dependent Kinase
CSC	Cancer Stem Cells
DNA	Deoxyribonucleic Acid
ECM	Extracellular Matrix
EMT	Epithelial Mesenchymal Transition
ERCC	Excision Repair Cross-Complementing
ERK	Extracellular-Regulated-Kinase
Fisher-KPP	Fisher-Kolmogorov-Petrovskii-Piskounov
FU	Fluorouracil
HIF	Hypoxia-Inducible Factor-1
MDR	Multi-Drug Resistance
MMP	Matrix Metalloproteinases
MOMP	Mitochondrial Outer Membrane Permeabilization
MTX	Methotrexate
ODE	Ordinary Differential Equation
OIS	Oncogene-Induced Senescence
PCNA	Proliferating Cell Nuclear Antigen
PDE	Partial Differential Equation
RFC	Reduced Folate Carrier
Rho	Ras homolog

ROCK	Rho-associated protein kinase
ROS	Reactive Oxygen Species
SASP	Senescence-associated secretory phenotype
Src	Sarcoma proto-oncogene tyrosine kinase
TGF	Transforming Growth Factor
TIS	Therapy-Induced Senescence
VEGF	Vascular Endothelial Growth Factor

Chapter 1: Introduction

1.1 Cancer — An Overview

1.1.1 Hallmarks of Cancer

Cancer, especially with its varying types and locations where they develop, is a complex disease to identify. It is often characterized by the uncontrolled proliferation of abnormal cells that can invade surrounding tissues, and metastasize to distant organs in advanced stages. This characterization provides an accessible description, yet it is an incomplete characterization. Many tumors exhibit periods of dormancy, slow growth, or even regression, and their growth dynamics are often much more complicated than simple uncontrolled growth. It is widely known that tumor growth is shaped by complex interactions with the microenvironment, immune responses, and other mechanical or metabolic constraints. In some tumors, such as indolent lymphomas, benign neoplasms, or dormant micrometastases, the proliferating fraction can remain low or balanced by cell death. Thus, the description ”uncontrolled proliferation” captures the general difference between tumor cells and normal tissue, although it oversimplifies the diverse behaviors observed across tumor types.

The tumor initiation and progression happens through the progressive accumulation of genetic and epigenetic alterations that disrupt the regulatory balance between cell proliferation, differentiation, and death. These alterations affect key signaling pathways that control growth factors, DNA repair, apoptosis, and cell to cycle checkpoints [1]. The interplay of these changes results not merely in unchecked cell division, but in dysregulated adaptation, where cells gain selective advantages that allow survival and proliferation under conditions where normal cells would die or halt their progression through cell cycle control mechanisms.

The influential work of Hanahan and Weinberg (2000), The Hallmarks of Cancer [1], posed and answered fundamental questions about the characterization of traits in the development of tumors, while encompassing this diversity in various types of malignancies. These hallmarks include self-sufficiency in growth signals, insensitivity to growth inhibition, evasion of apoptosis, limitless replicative potential, sustained angiogenesis, and the ability to invade and metastasize. The deregulated cellular energetics, evasion of immune destruction, and genome instability were added as additional hallmarks of cancer in a follow up paper [2]. As noted by Hanahan and Weinberg, not every tumor displays all eight hallmarks, but these properties represent a set of organizing principles that underlie malignant behavior in different combinations and to different degrees.

1.1.2 Tumor Initiation and Progression

It was proposed in [3] that most tumors do not appear fully malignant initially, but rather are results of a single cell acquiring genetic variability that allows for sequential aggressive evolution. Tumor development is widely understood as a multistage process in which normal cells gradually acquire malignant traits through biological and genetic alterations. In [4], the stages of cancer development observed at a cellular level, are classified into initiation, promotion, and progression, which is a framework also supported by decades of experimental and clinical observations [3, 5]. The three stages identified in [4] are explained below.

The formation of tumor begins with the initiation of a normal cell undergoing a mutation toward malignancy. It involves a rapid, irreversible alteration in the DNA of a normal cell, where this cell gains a potential growth or survival advantage. These alterations can result from exposure to chemical carcinogens, radiation, replication errors, or inherited mutations that affect key regulatory genes, including proto-oncogenes that regulate cell growth and proliferation, and tumor suppressors [5]. Although such cells of origin carries a permanent alteration in the DNA that is inherited to its daughter cells, they typically remain normal at this stage. It is also noted that the initiated cell might not proliferate without additional external stimuli.

In the promotion stage that follows, clones selectively expand from the initially mutated cell under the influence of external factors that are called promoters. Promoters, such as chronic inflammation, hormonal imbalance, and other chemical signals, encourage cell division, alter cell to cell signaling, and suppress apoptosis of cells, to favor the expansion of the initiated cells. At this stage, a visible pre-malignant lesion or benign tumor is produced. This phase typically does not involve new stable changes genetically, and is a reversible phase by removing the stimulus that promotes growth, which then leads to a halt in expansion, or even regression.

The progression stage is where benign or pre-malignant lesion acquire additional genetic or phenotypic changes so that it develops into invasive and malignant tumor [6]. It is an irreversible and genetically unstable stage, in which tumor cells modify their gene expressions, and continue to accumulate sequential mutations along with large scale changes in chromosome arrangements. As a result, a tumor becomes largely heterogeneous in this phase, with the emerging clones having diverse traits, such as proliferation capacities, metabolic behaviors, and drug sensitivities [7]. During progression, the tumor gains the ability to invade surrounding tissues and spread to distant organs by producing cells with invasive traits, and unlike in the promotion stage, cells at this stage can proliferate and invade independently of the external promoting stimulus.

Restricting our attention to solid tumors, how tumor progresses at the tissue level is commonly divided into the avascular, vascular, and metastatic stages. Along with changes in the physical structure of tumors, the main biological constraints on the growth of tumors also evolves over these stages.

Avascular stage is the early growth stage of tumor, which begins with a small cluster of abnormal cells. This cluster expands by consuming oxygen and nutrients from nearby capillaries. Avascular tumor growth is frequently modeled by tumor spheroids, and is often seen as the simplest stage to model mathematically. Extensive studies on avascular growth of tumor spheroids have been available experimentally [8,9]. One main discovery established by these experimental studies is the structure an avascular tumor develops as it increases in volume. Since there are only limited amount of oxygen and nutrients available nearby, the initial exponential growth where all tumor cells proliferate has a limited size of diameter 50-$200\,\mu\text{m}$ [10–12]. Then the deprivation of oxygen and nutrients causes starvation and quiescence of cells in the center of tumor spheroid, which is referred to as the necrotic core. The proliferating cells are located in the peripheral parts of the tumor, and the diameter of tumor spheroid at this stage is known to exhibit linear increase with time [9]. The tumor then expands in volume to eventually have a maximum diameter of 1-4 mm available for an avascular tumor depending on the types of tumor and the underlying environments, where the volume growth can be put on a hold even when nutrients are replenished. These experimental observations inspired the mathematical models that incorporate growth inhibition in solid avascular tumor [13].

Tumors can remain dormant for months or years after saturation in the avascular stage before starting a new phase of exponential growth [8]. The vascular stage of tumor begins with tumor cells releasing vascular endothelial growth factor (VEGF) that stimulates the growth of blood vessels [14, 15]. Then the newly built capillaries near the tumor function as new delivery routes for oxygen and nutrients, allowing tumor to grow beyond the diffusion-limited size. It was established through animal experiments that tumors in vascular stage rely on angiogenesis to supply nutrients and oxygen, and thus sufficient vascularization is crucial in progressive tumor growth [14, 16]. Angiogenesis is widely used as an indicator of malignant tumor, and numerous mathematical models of vascularization have been studied, both continuous and discrete [17].

After the vascular stage, some tumor cells gain the ability to migrate to distant locations in the body, which is referred to as metastasis of tumor. The understanding about what sets apart the tumors that enter this phase mainly began with the 'seed and soil' hypothesis of Stephen Paget, in 1889 [18]. The observation made from his autopsy records was that the occurrence of metastasis depends on the compatibility of tumor cell (seed) and the organ where this tumor cell is located (soil). This was later supported by multiple studies that date to modern days [7, 19–21] which shows that tumor cells that migrate to different locations may or may not grow as a tumor colony depending on the expression of endothelial cell surface receptors in vasculature that vary by several factors, including the organs they migrated to, the growth factors affecting metastatic phenotype, and the tumor cell traits that allow this cell to survive through the steps toward metastasis [7]. The discussions about steps of cancer metastasis in [7, 21] reveal that due to inefficiencies that arise in multiple steps of metastasis, most tumor cells fail to successfully turn into a metastatic tumor colony, which emphasizes the role of tumor cell traits. Mathematical models of invasive tumors incorporate environmental factors such as the degradability of an extracellular matrix (ECM) and vasculature environment [22], and host tissue stiffness [23].

The progression of tumor growth stages, both in the cellular and tissue level, may not always appear in a sequential manner, strictly separated in time. For instance, in the work of Qian et al. (2016) [24], they focus on the fact that full vascularization is not necessary for invasive tumor growth. This study mentions how most angiogenesis contributes ineffectively to tumor expansion, suggesting that more attention should be drawn to alternative angiogenic mechanisms in designing targeted therapies. However, classifying tumor growth into the aforementioned stages helps from in a modeling viewpoint, as tumors at different stages must be represented with different shapes, and require significantly distinct environmental factors in describing their growth.

1.1.3 Interactions with Microenvironment

Tumor microenvironment is a broad concept that refers to the surroundings of tumor that contains stromal cells such as immune cells and fibroblasts, ECM, growth factors, and cytokines. As was emphasized in the empirical findings about metastasis, progression of tumor depends not only on the tumor cell traits, but also heavily on the surrounding tissue. The nineteenth century discovery of Paget [18] connects to a broader realization that the tumor microenvironment plays an important role in the development of cancer.

Oxygen and nutrient supply are influential microenvironmental factors in a growing tumor at any stage. In the avascular tumor stage, it is now widely accepted that there is a growth size cap for such tumors, since oxygen and nutrients fail to diffuse far enough to nourish cells beyond a certain radius. Thomlinson and Gray in 1955 [25] showed through photomicrographs that all tumor cords of size greater than 180μ in radius under their observations exhibited necrotic cores. They further demonstrated a case that a tumor cord can continue to grow through wide areas of necrosis, when nourished by a capillary. A later work by Folkman in 1971 also argues that new capillary formation is required for tumor growth beyond a micro size, indicating that there is a saturation point of avascular tumor size due to the diffusion-limited supply of nutrient and oxygen [26]. These discoveries led to the experimental findings of Mueller-Klieser in 1987 [8], showing that avascular tumor spheroids form a layered structure of proliferating outer rims and necrotic centers.

Upon facing this limitation in growth, typically the rapid proliferation of tumor cells that outpaces supply of nutrients and oxygen lead to hypoxia, which has deep connections with tumor angiogenesis. Forsythe *et al.* tested whether Hypoxia-Inducible Factor-1 (HIF-1) is involved in increased expression of VEGF in hypoxic regions [27]. The study showed that cotransfected cells of a dominant negative form of HIF-1α inhibits the VEGF activation in hypoxic cells, and further showed that mutant cells not expressing the HIF-1β did not induce VEGF response in hypoxia. Both of these findings revealed that oxygen deprived environment can promote angiogenesis by activating HIF-1.

Bissell and Hines (2011) [28] mention surprising results from routine examinations and autopsies that *in situ* cancers are highly prevalent in most organs, and even considered normal to be present in some locations such as thyroid gland. Then a natural question arises: why do more people not go on to develop malignant tumors? A 1940s a study showed that skin microenvironment can act as a tumor/mutation suppressor [28, 29]. Indeed, numerous studies demonstrate that one of the barriers that *in situ* cancers face in developing to be a clinically significant cancer is the microenvironment, which can actively suppress tumor progression by restraining inappropriate proliferation and normalizing aberrant cells. [30, 31].

However, there also is a completely different side to the microenvironment. The same work in 2011 of Bissell and Hines [28] discusses how microenvironment can also promote and induce cancer. They argue that although healthy microenvironment can act as a tumor-suppressor, ”unhealthiness” in the microenvironment such as the gradual change and damage accumulated to stroma (the tissue surrounding tumor cells) by aging can turn it to be permissive of tumor growth. These discoveries suggest that the microenvironment is not a static barrier but is rather dynamically evolving during the stages of tumorigenesis [30]. We now examine several important microenvironment interactions with tumor growth, in order to understand the dual roles of the microenvironment in the development of cancer.

The ECM is one of the key components of the microenvironment that interacts with tumor. It is mostly composed of collagen, but also contains other structural and regulatory molecules such as fibronectin and hyaluronic acid. The ECM directs cell growth, death, migration, and immune cells. In its normal state, the ECM limits uncontrolled growth of tumor by giving mitogenic signals and inducing cell apoptosis [32]. It is a highly dynamical matrix that goes through constant remodeling to maintain homeostatic state, and this remodeling can involve changes in the composition and absolute amount of ECM materials [33]. The stiffness in this matrix may arise through these compositional or quantity changes, and the interactions between ECM stiffness and malignant tumor cells are widely recognized as an important step in tumor progression.

Tumor cells are known to be stiffer, which is attributable to excessive accumulation of ECM components, most often various collagen types and hyaluronic acid [34, 35]. Tumor cells themselves can synthesize ECM components to increase rigidity in ECM. [36]. Then tumor cells and fibroblasts sense this tissue stiffness through mechanotransduction and secrete chemokines that recruit immune cells through inflammatory signals, and growth factors such as TFG-β to activate fibroblasts, and VEGF [37]. The recruited immune cells release growth factors and also activates fibroblasts which exacerbate the situation by strengthening this loop and worsening inflammation [38]. Then the tumor cells recruit and transform these fibroblasts into cancer-associated fibroblasts (CAFs) which is a dominant force in excessive ECM deposition, and these processes altogether reinforce this ECM stiffness feedback loop [35, 38].

ECM stiffness actively drives malignancy in tumor by directing altered proliferation and survival signals, promoting tumor invasion and migration, and enhancing resistance to apoptosis in tumor cells. The consistent release of growth factors involving VEGF in the stiffness feedback loop itself promotes angiogenesis and proliferation of cells. It was also shown in [39] that increased collagen matrix density promotes cell proliferation. In addition, cells sense matrix stiffness through integrins and focal adhesion kinase (FAK) signaling, which triggers downstream activation of FAK, Src, Rho/ROCK, and related pathways. This signaling leads to increased VEGF expression and enahnced endothelial migration where both promote angiogenesis [40]. This signaling also results in raised expression of Cyclin D1, which accelerates cell cycle progression, and PCNA, which is associated with faster proliferation [40]. In terms of cell motility, type I collagen that accumulates in abnormal ECM, is known to promote cancer cell migration through the intracellular signals triggered by the binding of $\alpha 2 \beta 1$ integrin and type 1 collagen [41]. Then this increased motility of tumor cells allow them to migrate and invade, which is essential in metastasis.

ECM stiffness also impacts tumor by slowing down drug delivery and promoting drug resistance in tumor cells. Using synthetic substrates, both substrate stiffness [42] and collagen-rich environment [43] were shown to confer drug resistance to tumor cells, lowering chemotherapeutic responses. How ECM stiffness impairs efficient delivery of drugs has been studied in various aspects. The reduced porosity and permeability in tumor stroma due to the stiff, dense ECM sets a physical barrier in drug transportation [40, 44]. The discovery that platinum based chemotherapy such as cisplatin binds to collagen fibers excessively [45], also indicates that drug binding in tumor cells with excessive collagen will disrupt drug delivery. In another aspect explained in [40], growing volume of tumor increases pressure in the host tissue, and in return stiff ECM resists this pressure to push back the vasculature in tumor cells. In addition, poorly formed lymphatic vasculature contributes to an increase in interstitial fluid that puts more pressure on tumor blood vessels. The overly pressured tumor blood vessels result in poor perfusion in tumor, which hinders drug delivery.

1.2 Mathematical models of Tumor Growth

Tumor growth has been studied extensively, from the early characterization of avascular tumors to models that focus on vascularization and the metastatic progression of malignant tumors. In this work, we restrict our attention to solid avascular tumors, which represent an important stage of early tumor development before vascularization. In simple mechanistic models of tumor growth, the tumor volume is described by exponential [46,47], logistic [48], or Gompertz [49,50] growth functions. These exponential, logistic, and Gompertzian models are ordinary differential equations (ODEs) of the form

$$\frac{dN(t)}{dt} = k_g N - \mu N \tag{1.1}$$

$$\frac{dN(t)}{dt} = k_g N \left(1 - \frac{N}{K}\right) \tag{1.2}$$

$$\frac{dN(t)}{dt} = k_g N \ln\left(\frac{K}{N}\right) \tag{1.3}$$

where $N(t)$ denotes the tumor volume, K is the carrying capacity, μ is the death rate, and k_g represents the intrinsic growth rate. The exponential growth law assumes that tumor grows unbounded in size, without a decline in growth rate, which can only describe the early tumor growth. Logistic growth, on the other hand, assumes linear slow down in the growth rate as tumor volume increases, eventually approaching a carrying capacity. Gompertzian growth assumes logistically decreasing growth rate depending on the tumor size, where a tumor volume approaches a carrying capacity.

While these foundational models (1.1) - (1.3) provided simple descriptions of overall growth, they have limitations in capturing the heterogeneity of tumors. In 1955, Thomlinson and Gray developed equations describing the spatial variation in oxygen availability based on the biological observation of a necrotic core [25]. They argued that oxygen concentration towards the center of the tumor is significantly lower than at the periphery, and this creates spatial variance in the efficacy of radiotherapy. Although their study did not describe tumor growth, such observations of tumor heterogeneity in space motivated the models that incorporate diffusion.

Introducing spatial variables allowed models to incorporate the interactions of tumor with its microenvironment, such as nutrient diffusion and competition with surrounding healthy tissue. Such change in the modeling framework brought the spatial mitotic patterns in a solid avascular tumor to attention. To describe the spatial structure, early spatio-temporal models, such as Greenspan (1972) assumed a spherical tumor which maintains spherical symmetry throughout its growth [13]. The clear advantage of these models compared to the ODE models was that they can distinguish necrotic core and proliferating rim. Greenspan, in his foundational work in 1972, divided avascular tumor into three layers: the proliferating rim, quiescent intermediate region, and necrotic core inside [13]. Each of these layers was represented as a discrete region in his model with a moving boundary. Greenspan hypothesized that the formation of such layers come from inhibitory chemical signals produced either by living cells or from a necrotic region, and assumed enough concentration of the inhibitory chemical induces quiescence and necrosis.

While later researches attributed the emergence of proliferating rim at the periphery and necrotic core to nutrient gradients [51], the growth inhibitory signal hypothesis developed to the idea that the spatial mitotic pattern in avascular tumor arises from a cellular mechanism of producing a chemical called *chalones*, which gives negative feedback to cell proliferation. Glass constructs a model of tumor size regulation in 1973 based on this chemical signal hypothesis [52]. Incorporating the diffusion of chalones, Glass showed that instability in tumor size can arise [52]. Although the tumor growth in this work does not involve a spatial element, the diffusion of chalones was described by a reaction-diffusion PDE.

These attempts of including spatial structure in the microenvironment that tumor interacts with, became a foundation of reaction-diffusion models [53–55] that describe tumor as an incompressible fluid interacting with microenvironment. In 1976, Greenspan modeled tumor growth with proliferation regulated by nutrient availability and included diffusion [55]. Greenspan used the concept of expansion force from the tumor interior due to proliferation and death induces cell motion, and this idea remains widely used. A counteracting force, the surface tension caused by cell to cell adhesion, balances the internal pressure to restrict tumor expansion.

Later in 1996, Byrne and Chaplain constructed a mathematical model of tumor growth extending the work of Greenspan (1976), where they considered a radially symmetric tumor interacting with diffusing nutrient to study how cell to cell adhesion affects tumor growth [54]. The main differences between their model and Greenspan's are that no explicit necrosis is modeled except for the setting that proliferation depends on the nutrient availability, and nutrient concentration on the tumor boundary depends on the curvature, satisfying the Gibbs-Thomson relation. Byrne and Chaplain observed that even when all cells are in the proliferative state, nutrient availability and cell to cell adhesion forces can inhibit tumor growth.

In the same year, Gatenby and Gawlinski developed a reaction-diffusion tumor growth model to study how pH alteration in the microenvironment induced by tumors affect the emergence of tumor malignancy [53]. The important biological difference between tumor cells and normal cells in their energy-generation mechanism, known as the *Warburg effect*, is what causes this pH alteration. While normal cells rely on glycolysis only in oxygen-scarce environments, cancer cells use glycolysis even when oxygen is sufficiently available. As a result of this glycolytic metabolism, cancer cells generate excess lactate and acidic (H^+) byproducts, altering pH levels in the microenvironment. Gatenby and Gawlinski (1996) incorporated this biological phenomenon by modeling tumor cells as a continuous source of acid [53]. Their model used a Lotka-Volterra type competition between normal and tumor cells, where the proliferation of both cells depend on local PH of the microenvironment, with a survival advantage given to tumor cells. The spatial heterogeneity in their model enabled discussions of the role of microenvironmental constraints in invasive tumor front propagation. From this study, Gatenby and Gawlinski proposed the so called *acid mediated tumor invasion hypothesis*. The idea is that acid generated by tumor cells diffuses in the microenvironment to preferentially kill normal cells, facilitating tumor invasion. This is observed by a traveling wave solution of their model where the front of acid precedes the wave of tumor cells, creating retreat of the wave for normal cells.

Research about moving tumor fronts was also developed by Sherratt and Chaplain in 2001, that integrated the distinction of proliferating and quiescent cells into their model [56]. This work was based on an earlier work of Sherratt in 2000 that investigated the contact inhibition of migration by competing cell subpopulations where one has growth advantage over the other [57]. However, Sherratt and Chaplain moved away from the discrete layer structures of tumor and instead introduced a continuum model in which proliferating, quiescent, and necrotic compartments are modeled as spatially varying continuous cell densities. The transition rates from proliferating cells to quiescent, and quiescent to necrotic cells were taken as a function of the local nutrient concentration.

The basic reaction-diffusion partial differential equation (PDE) used in these tumor growth models is given by

$$\frac{\partial n}{\partial t} = D\Delta n + f(n, c) - d(n, c), \tag{1.4}$$

where $n(x, t)$ is the cell population density at the location x, D is the cell motility coefficient, $f(n, c)$ is the proliferation term as a function of cell density n and nutrient concentration c, and $d(n, c)$ is the death term. The notable difference from the previous ODE models is the cell movement term $D\Delta n$ usually modeled by linear diffusion.

In addition to spatial structure, continuum models can incorporate tumor heterogeneity through internal cellular variables that describe individual cell states. One such variable is cell age, defined as the time elapsed since the last division of a cell. In tumor cells, cellular age is closely related to proliferative capacity. As tumor cells age, DNA damage accumulates through repeated cell-cycle progression, including processes such as telomere shortening, which can lead to telomere dysfunction that inhibits clonal expansion [58]. This motivates the use of age-structured mathematical models of tumor growth [59], which will be discussed in detail in section 2.1.

Mechanistic models focus on the description of tumor dynamics at a population level, but observations revealed that stochasticity and heterogeneity are some of the intrinsic features of tumor. Agent Based Models (ABMs) address this by representing each or a group of cells as an individual entity. Since integrating multiple biological factors such as nutrients, drugs, and ECM, which occur in varying time scales is easier with ABMs, hybrid models are popular among ABMs. ABMs have been used to generate predictions and motivate experiments as computational modeling approaches became accepted in cancer biology.

Depending on how cells are represented and which biological interactions are included, a range of approaches have been developed. In a cellular automaton model, [60] cells grouped by their states are represented by points on a lattice. The states of each cell are updated each time step according to the biological assumptions in the model. For instance, Patel *et al.* developed a cellular automaton model based on the work of Gatenby and Gawlinski (1996). The acid-mediated tumor invasion hypothesis suggested in Gatenby and Gawlinski (1996) was revisited with an individual based type of model, allowing cells to die, remain quiescent, or proliferate to move to adjacent space [60]. Instead of the competition between tumor and normal cells that was modeled by the Lotka-Volterra type equation in Gatenby and Gawlinski (1996), Patel *et al.* used the nature of cellular automaton model, where each lattice can be occupied by one cell at a time, creating competition for space between cells. The advantage of using cellular automaton model instead of the previous RDE descriptions was that early stage dynamics become more accurate since averaging over small number of cancer cells can create relatively large errors in model prediction.

However, the limitation of traditional lattice-based cellular automaton model is clear. Since cells move on a discrete lattice, realistic representation of cell velocities and trajectories is challenging, pressure induced cell movement is limited, crowding effect cannot be reflected, and growth patterns are not well represented. In 2000, Kansal *et al.* developed a new approach that addresses these limitations of traditional cellular automaton model, by using Voronoi tessellation [61]. This approach did not assume a fixed grid and allowed the grid size to vary. In Voronoi tessellation, a set of points is placed in the domain to serve as centers, and the domain is divided into regions around these central points so that each region contains all locations closer to each central point. In terms of tumor modeling, the space occupied by a cell, or a group of cells is represented by such region. As cells grow or move, the corresponding central point and region updates, varying in shape and free from lattice, providing a good geometric representation. Kansal *et al.* (2000) addressed the limitation of traditional models, which described transitions between cell states purely stochastically rather than based on biological mechanisms, and also noted that these models may unrealistically restrict proliferation by the assumption that only one cell may occupy a grid point at a time. With the benefits of this novel approach, they were able to represent cell density at each lattice site, where higher density was observed near the center of the domain. Transition between proliferative, quiescent, and necrotic cell phases was modeled to be dependent on the distance from vasculature so that proliferation rate depends on nutrient supply.

Particle-based models, while still representing a cluster of cells as a point, distribute agents in a continuous space. These models [62] are therefore off-lattice, making them well-suited to simulate irregular tissue structures and cell-motility. In 2003, Drasdo and Höhme constructed a particle-based model where each cell is represented as a slightly deformable sphere, and cell movement is modeled by the repulsive and adhesive forces between cells [62]. With mechanical stress being the major mechanism that prevents exponential tumor growth after the initial phase, the model of Drasdo and Höhme exhibits generic tumor growth dynamics of the initial exponential growth, followed by linear growth in the diameter. Byrne and Drasdo further developed this idea in 2009 by replacing the regulatory force on the tumor expansion with random cell migration [63]. Byrne and Drasdo showed that for monolayers or spheroids, the particle-based model produce results aligning with that of the continuum counterpart, the Fisher-Kolmogorov-Petrovskii-Piskounov (Fisher-KPP) reaction diffusion equation. Both of these studies emphasize that the cell level behaviors in single-cell mechanics are captured in best details by choosing particle-based model forms.

Cellular Potts models [64,65] treat cells as collections of pixels or voxels on a lattice, which allows each representation of cell to have varying shapes. In 1993, Glazier and Graner used a cellular Potts model with mainly two mechanisms: the adhesion strength between cell to cell and cell to medium, and the area constraint that keeps cells from shrinking or expanding boundlessly [65]. Glazier and Graner modeled cells to be isotropic, and the cell behaviors are governed by the adhesion forces. Without further biological properties included in the model, they found that this cellular Potts model can reproduce the cell sorting phenomenon. Using a cellular Potts model in their study is beneficial since it best describes adhesion force that mainly governs the mechanisms in the model, by allowing cells to freely change shapes and contact surfaces and thus interacting in a dynamical manner.

Immersed boundary models (IBCell) [66] let cell representations have deformable elastic boundaries in a fluid domain. These models are off-lattice, continuous in nature unlike the other aforementioned discrete approaches, and thus are useful for studying how tumor cells deform and migrate. Rejniak used IBCell model to describe early tumor growth from cellular level mechanisms in 2007 [66]. In this work, cells were modeled as elastic membranes immersed in a viscous, incompressible fluid. Processes such as cell growth and division, necrosis, cell to cell adhesion, and microenvironment interactions were all incorporated for each cell, allowing for a realistic representation of a tumor at the cellular level [66]. Their model contributes to the comprehension of how individual cell mechanisms affect tumor expansion. The use of IBCell model was particularly beneficial in this study due to several reasons. Cells are fully deformable and the deformation process is continuous contrary to the cellular Potts models, allowing growth and proliferation to be modeled as continuous processes. In addition, under IBCell model, cells can sense their microenvironment (such as nutrient concentration and cell to cell interaction forces) at the cell boundaries, which gives more accurate local descriptions than averaging over a region.

1.3 Evolution of Drug Resistance in Cancer

1.3.1 Resistance is Evolutionary

Drug resistance is a common and major problem that anti-tumor therapies undergo, and has therefore been extensively studied. In 1943, Luria-Delbrück designed a fluctuation experiment where they grew parallel bacteria cultures to later expose them to bacteriophage. High variation in virus resistance was observed across bacteria cultures, providing the groundbreaking evidence that resistance arises prior to exposure through mutations rather than is induced by exposure [6]. Later in 1976, a foundational study of Nowell [5] proposed that tumors evolve through Darwinian selection over time among genetically unstable cells. This idea explains the heterogeneity in tumor in different hosts, organs, and tumor regions, and moreover provided the idea that drug resistance is evolutionary. These early landmark studies established the understanding that therapeutic resistance in tumor is an outcome of clonal mutations and natural selections during tumor growth.

As a result of mutations, tumor is largely heterogeneous. The dynamics of drug resistance in tumor evolves according to the fitness of cells, which again is shaped by numerous factors, such as cell to cell interactions, microenvironment, and interventions if there are any. Particularly, the treatment driven evolution of drug resistance has been suggested in the 1980s [67, 68], arguing that gene amplification occurs in resistant tumor cells under therapies. Later, this extended to the understanding that treatments select resistant cells [69]. Studies also suggest that this therapy induced evolution of resistance happens through the selection of pre-existing resistant cells, rather than the emergence of new mutations induced by treatments. Almendro et al. (2014) reported that genetic diversity in tumor cell populations did not change before and after neoadjuvant chemotherapy on breast cancer patients, while relative genetic distributions in cells with specific phenotypes significantly changed after treatment [70]. This result implies that chemotherapy did not induce new mutations in the population, but can select genetic properties depending on the cell phenotypes.

However, without any treatment intervention, the preference in natural selection can be very different. It was found in many studies that resistant cells tend to pay fitness cost, showing decreased proliferative potentials in drug-free conditions [71–73]. Using such property, a 2017 study proposed the strategy of maintaining a certain amount of sensitive cells during Cisplatin treatment to provide a target for future continued treatments [74]. They observed both in vitro and in vivo that sensitive cells out-compete the resistant cells in drug-free conditions, and suggested a treatment plan of delaying the dominance of resistant cells by utilizing the intratumoral competition between resistant and sensitive cells.

1.3.2 Cell Cycle Phases and Cytotoxic Drugs

Cell phases are particularly important in cytotoxic treatments, since unlike the targeted anti-cancer therapies, cytotoxic drugs do not target cancer cells, but rather, their selectivity comes from the preference of killing the rapidly proliferating cells [75]. Drugs that focus on killing actively dividing cells, in their nature, spare dormant or slowly cycling cells to allow their relapse in the long term [76,77]. Thus cells in slow cycle or dormancy is conferred resistance over most cytotoxic drugs [78,79].

The cell cycle consists of the following phases. In G_1 phase, cell grows to a certain size and then enters the S phase where DNA synthesis happens, then G_2 where cells prepare for mitosis, and then M phase follows where cell division is completed. Then there are cells which either temporarily or permanently exit the cycle and enter the G_0 phase [80].

The temporary rest from cell cycle progression among G_0 is referred to as quiescence, and cells in this state may re-enter the cell cycle and progress toward mitosis. For normal cells, this re-entry requires external mitogenic signals. In contrast, many tumor cells can re-enter the cell cycle from G_0 without such regulation, by producing growth factors on their own and ignoring growth inhibitory signals [2]. Other than quiescence and terminally differentiated cells (who cannot re-enter cell cycle), G_0 phase also includes senescence. In 1960s, Hayflick observed that human diploid somatic cells ceased to proliferate after a certain amount of time, to discover that there is a finite replicative capacity for these cells regardless of culture conditions [81,82]. Then these cells that reached its intrinsic replicative capacity goes into a proliferation inhibited state, which is called the senescence.

Much has been discovered about senescent cells since the first discovery, including the fact that senescence can be also induced in cancer cells by several factors, such as DNA damage, telomere shortening, oncogene activation, metabolic stress, and anti-cancer therapies [83,84]. It has been shown that low dose of chemotherapy can induce senescence in tumor, and there has been attempts to utilize this in therapy plans, which comes with the additional benefit of less toxicity [85]. The point in inducing senescence is not merely in its low proliferative potential. Multiple studies have reported that senescent tumor cells, especially the oncogene-induced senescence (OIS) has shown to disrupt tumor formation by halting mitosis of cells that are likely to go through transformation [86], and arrest neighboring cells [87–89]. Moreover, senescent cells release senescence-associated secretory phenotype (SASP), which contains cytokines, chemokines, proteases, matrix-remodeling enzymes (MMPs), and some growth factors. The immune-modulating factors among SASP recruit immune cells which can clear detectable tumor cells [86]. This immune response triggering mechanism has shown its effectiveness in reducing the size of tumor in mice [90].

However, tumor cells have proven their potential to escape therapy-induced senescence (TIS) and proliferate again [91]. An even more alarming observation made in their same work and in Triana-Martínez *et al.* (2020) [92] is that during TIS, a transcriptional reprogramming is also triggered. They compared virtually identical populations exposed to chemotherapy that underwent TIS with those that never experienced senescence, and found that the once senescent cells, upon their return to the cycle, had returned with strongly enhanced tumor initiating potentials. This result raised a concern in utilizing TIS in therapies, and added an important interpretation of senescence as a potential reservoir of a more resistant and aggressive tumor.

Contrary to normal senescent cells, senescent tumor cells may persist, and their persisting secretion of SASP becomes problematic. The pro-inflammatory cytokines and chemokines secreted in senescent tumor cells induce senescence in fibroblasts, leading them to produce SASP as well, and thereby amplifies inflammation [93]. This exacerbates the already existing inflammatory side effect of cytotoxic drugs. Studies also suggest that SASP does not have a single fixed profile but varies greatly depending on the cell types and how senescence was induced. The SASP factors secreted by tumor cells in particular, are known to have pro-tumor aspects. It has been shown in several studies that SASP materials promote tumor cell proliferation [94, 95]. For instance, SASP factors can activate stromal fibroblasts to stimulate hyperproliferation, and induce senescence in fibroblasts, which accelerate the progression of pre-cancerous cells to strongly facilitate tumorigenesis [94, 95]. The pro-tumor aspects of SASP factors also include immunosupression, [96] angiogenesis and metastasis [97], acquisition of drug resistance, and induction of stem cell-like cancer cells.

Cancer Stem Cells (CSCs), or stem cell-like cancer cells, are slowly cycling tumor cells with the ability to generate heterogeneous cell types to form a tumor. Early foundational studies such as Bonnet and Dick (1997) [98] established that not all tumor cells can start a new tumor. In their experiment of transplanting human acute myeloid leukemia (AML) into mice, Bonnet and Dick discovered that only a rare subpopulation of cells, characterized by certain cell surface phenotype of stem cells, could succeed in reforming new tumors in mice. CSC has received much attention since the discovery of their existence. Research over the past two decades has revealed several mechanisms of how CSC arises. They were shown to originate from a normal stem cell in some studies [99, 100], while some other cases suggested that CSC can originate by a tumor cell acquiring stem cell-like characteristics [101]. Importantly, epithelial-mesenchymal transition (EMT) has been shown to have a key contribution to the emergence of CSCs by inducing the acquisition of CSC phenotypes [102]. Researchers have found evidence of EMT-inducing transcription factors (TFs) promoting stemness [103]. It has been also found through xenotransplantation that cells with tumor initiating capacity are much more common than the traditional expectation. The measured proportion of cells with CSC properties among cancer cells was around 25% for melanoma in [104] and over 10% in [105].

As expected from their slow cell cycle, CSCs are difficult to treat with cytotoxic drugs. In a mouse glioma experiment, Chen and colleagues observed that the slow cycling cells could survive chemotherapy and cause relapse of tumor after treatment [76]. However, given that CSCs are resistant to drugs and are the only cells with initiating capacity, CSCs are the cells largely responsible for the propagation of drug resistance in tumor [106]. This highlights the importance of CSCs in any anti-tumor therapies and models of drug resistance. Gupta, Chaffer and Weinberg in 2009 [102] emphasized the importance of EMT in CSC modeling, and posed a question about the possibility of CSC cells losing its stem cell-like properties. Some later research on this loss of stem cell functionality has shown that differentiation therapies can reduce stemness in CSCs [107, 108], suggesting new dynamics to incorporate in CSC models.

1.3.3 Multi-drug Resistance

Tumor cells can acquire resistance across multiple, often structurally unrelated drugs, which is referred to as multi-drug resistance (MDR). In 1976, Julaino and Ling discovered that surface P-glycoprotein (P-gp) in drug resistant cells grants cross resistance to multiple drugs in these cells, defining the concept of MDR [109]. They observed that drug resistant cancer cells overexpressed P-gp. P-gp, also called MDR1, was later identified as an ATP-binding cassette (ABC) efflux pump, and in their experiment, less accumulation of drug in resistant cancer cells was observed, suggesting that the overexpressed P-gp in these cells were attributable. In order for any chemotherapeutic drug to be effective, it must accumulate to a certain concentration inside the tumor cells to cause damage. Tumor cells avoiding this accumulation by expelling drugs can protect themselves against a wide range of chemotherapies and therefore acquires MDR. The complete realization of what this correlation between P-gp and MDR in the study of Juliano and Ling meant, was made through numerous other studies that followed. P-gp was recognized as MDR1 gene by [110] and the role of MDR1 in MDR of pumping drug out of the cells, was understood through a series of studies in 1980s including the study of Horio, Gottesman and Pastan (1988) [111–113].

Reducing absorption of drugs is not only viable through increased release of drug from the cells. Some mutations of tumor cells can diminish the binding affinity of drugs. For instance, it was shown that when an anti-tumor drug Methotrexate (MTX) was given at an insufficient dose, some of the tumor cells who survived mutated to decrease MTX binding with the reduced folate carrier (RFC) protein which transports MTX into cells [114]. Drug accumulation in cancer cells can also be reduced due to decreased expression of drug import carriers as a result of cancer cell mutations. Defective transport of MTX drug was observed in tumor cells having acquired MTX resistance through suppressed expression of RFC [115]. Further, poor drug treatment responses are also associated with the decreased expression of drug import carriers in multi-drug resistant cancer cells. For instance, it was clinically observed in [116] that 65% of patients suffering from poor chemotherapeutic responses in this study showed downregulated RFC expression.

Some other major tumor cell mechanisms of avoiding therapeutic effects include epigenetic altering, cell death evasion, DNA repair, and deactivating or detoxifying drugs [117]. Epigenetic alterations have shown to confer drug resistance by DNA repair [118], upregulation of MDR1 [119], and histone modification that enables the cells to enter a drug-tolerant persister (DTP) state [120]. A correlation between epigenetic silencing of miR-200c expression and the promotion of both metastasis and chemo-resistance has also been observed [**?**]. These epigenetic alterations can shift cellular phenotypes to become less responsive to therapies and more adaptable under drug pressure.

The DNA repair mechanism of tumor cells help them survive the DNA damage from treatments such as chemotherapy. Tumor cells can enhance this ability by increasing the efficiency of their DNA repair systems such as nucleotide excision repair system (NER) and homologous recomination repair system (RRM) [117]. For instance, overexpression of the excision repair cross-complementing 1 (ERCC1), a part of the signaling pathway of NER, has shown to decrease the efficiency of cisplatin [**?**].

In a broader view, cytotoxic drugs induce apoptosis through a shared mechanism involving mitochondrial outer membrane permeabilization (MOMP). Tumor cells can avoid cell death by interfering with this pathway. For instance, the anti-apoptotic BCL-2 family proteins can block MOMP, suppressing chemotherapy induced apoptosis [**?**]. On the other hand, downregulation of pro-apoptotic proteins such as Bax and Bak are also known to occur in tumor cells to decrease the apoptotic pressure [112].

Finally, tumor cells may acquire MDR by altering how drug affects the cells. They are known to change the target of drugs, which can occur through mutations in the target of a therapy that reduces or abolishes drug binding [?]. Some other important effects on drugs occur through enzymes. They play crucial roles in the metabolism of chemotherapies. Tumor cells can deactivate chemotherapies by upregulating or downregulating such enzymes. For instance, platinum drugs are known to deactivate by thiol glutathione [?, ?], and the activation of chemotherapies such as MTX and 5-fluorouracil (5-FU) requires certain cellular enzyme activities [?, ?].

Then again we revisit CSCs for the discussion of their MDR to more general drugs. CSCs are important factors in drug resistance, not only because of their slow cell cycles, but also due to their ability to evade apoptosis, repair DNA damage, and their strong association with EMT. For instance, ROCK activates Rho signaling that leads to upregulated survivin expression [?]. This grants CSCs drug resistance since survivin protects CSCs from therapy-induced apoptosis. CSCs also can resist cell death from DNA damage, via enhanced reactive oxygen species (ROS) scavenging and DNA repair. High expression of the CSC markers CD44 and CD133 is shown to result in greater ROS defense and enhanced DNA repair [?]. Other than inducing and maintaining stemness in cells, and promoting tumor invasion, EMT is also known to enhance drug resistance in CSCs. The EMT inducing TFs such as ZEB-1, Twist, FOXC2, and FOXM1 contributes to enhancing drug resistance in cells [?], although the exact pathways of their contributions are still elusive. ZEB-1 has shown to reduce apoptotic cell death [?]. Silencing ZEB-1 sensitized mesenchymal-like pancreatic cancer cells to several chemotherapies such as gemcitabine, 5-FU, and cisplatin [?], which could be highly related with ZEB-1 being a suppressor of miR-200 and E-Cadharin, as miR-200 has shown to re-sensitize cells [?] and silencing E-Cadherin results in increased drug resistance [?].

1.4 Outline of Thesis

In this thesis, we develop and analyze mathematical models of tumor growth and models for the evolution of drug resistance. We focus on integrating age-structure and spatial heterogeneity in the models. The rest of this thesis is organized as following.

In Chapter 2, we introduce an age-structured spatio-temporal model of a solid tumor growing under no therapeutic interventions. The work presented here is based on the preprint with Levy, Mellet, and Wildes [?]. The chapter begins with biological motivation and a review of the related previous modeling approaches for the main model of Chapter 2 in section 2.1. In section 2.2, the mathematical formulation and analysis of the model are given, including the model equations in section 2.2.1. Mathematical analysis about the existence of a weak solution and finite propagation of the support is given in section 2.2.2. Some possible extensions of the model are given in section 2.2.3. The numerical simulations of the model are shown in section 2.3. The asymptotic behavior of the solutions to the age-structured model are shown in section 2.3.4, starting with the simplest motivational space-homogeneous case without cell death, leading to the space-heterogeneous case with death process. Concluding comments for this chapter are given in section 2.4.

Chapter 3 extends the model of the previous chapter to study chemotherapy response and the evolution of drug resistance during chemotherapy. The model incorporates spatially heterogeneous distribution of nutrients and drugs, and includes the resistance phenotype-structure. The model equations are described in section 3.2 along with the functional specifications given in 3.2.1. Numerical results are shown in the section 3.3. The numerical results explore how distributions of cell density, mitosis events, mean resistance level, average age, and total cell population changes. Three different cases distinguished by the initial resistance distributions and the boundary conditions of drug and nutrients. In section 3.3.1, the initial distribution and parameters are specified. In section 3.3.2, tumor growth without drug intervention is studied numerically, and the dynamics under continuous infusion of cytotoxic drug with varying intensities are presented in section 3.3.3. The discussion of this work follows in section 3.4.

Chapter 4 concludes the dissertation, with the summary of the main findings and suggested future directions.

Chapter 2: Age-structured Model of Tumor Growth

2.1 Background

Cell-cycle dynamics are widely recognized as an important factor for understanding tumor growth, since the durations of different phases influence proliferative potential and response to therapies. Therefore, explicit representations of cell cycle transitions are often used to capture phase specific rates of division, death, and quiescence. [**?**, **?**, **?**]

One consistent difficulty that the theoretical models of tumor growth face, however, is the shortage of clinical data, which is essential in validating the models and determining the model parameters. Cell cycle transitions are known to have huge variation in lengths, demanding for the use of some other factors that can explain this variance. Furthermore, direct measurement of the transition rates between proliferative and quiescent states in human tumor is a challenging data to obtain. For instance, the boundary between quiescence and very slow cycle is already very obscure.

Given the difficulty of measuring the transition rates between cell-cycles robustly in vivo, an age-structured formulation, which uses "time since birth", can be an alternative that captures cycle-dependent effects while avoiding to specify explicit phase switching. In this regard, age-structured models can yield useful and simple representations of cellular populations. A simple such model is the growth-fragmentation equation [?, ?]:

$$\begin{cases} \partial_t n + \partial_\theta n = -\beta(\theta)\, n - \mu(\theta) n & \theta > 0,\ t > 0 \\ n(0,t) = 2\displaystyle\int_0^\infty \beta(\theta)\, n(\theta,t)\, d\theta & t > 0, \end{cases} \tag{2.1}$$

Here $n(t, \theta)$ denotes the cell density at time t and age θ. The function $\beta(\theta)$ is the proliferation rate, and $\mu(\theta)$ is the death rate. Age-structure was used in Johnston et al. (2007) [59] to model the cell population of stem cells, semi-differentiated cells, and fully differentiated cells in colorectal cancer, to account for the cell cycles (death, differentiation, and proliferation). Gabriel et al. (2012) [?] also discuss how age-structured models can capture cell age-dependent responses to targeted therapeutics. Since this study used clinical measurements of cell lifespans and intermitotic times to validate the model, an age-structure was a natural framework for their model.

Age-structures have been used to capture cell division timing and cycle dependent therapy responses, by letting division and death rates depend on the time since last mitosis. Early works such as [59] analyzed the growth of cell populations, mentioning divisions happen asynchronously across cells, and allowed the birth and death events depend on the age of the cells. In [?], the proliferative and quiescent cell cycle was studied using cell age as the main influential factor in general cell population. The connection between the age and cycle transition was made by assigning age intervals to cell phases, where suspension in the progress through phases happens for quiescent cells. The study discovered that age distribution stabilized under their model and this limiting distribution was independent of the initial distribution in cell ages and proliferating-quiescent states.

Several extensions of age-structured growth model were made to include other tumor biological details. In [**?**], the tumor immune model is analyzed:

$$\begin{cases} \partial_t n + \partial_\theta n = -\beta(\theta)\, n - \mu(\theta) n - \sigma(\theta) E(t) n & \theta > 0,\ t > 0 \\ n(0,t) = 2\displaystyle\int_0^\infty \beta(\theta)\, n(\theta,t)\, d\theta & t > 0, \end{cases} \tag{2.2}$$

where $E(t)$ denotes the number of effector cells (immune cells responsible for identifying and clearing tumor cells), whose evolution is modeled by

$$\frac{dE(t)}{dt} = s - \mu_E E(t) + \tau N(t) E(t), \qquad N(t) = \int_0^\infty n(\theta,t)\, d\theta.$$

A similar model is developed in [**?**], while proliferative and quiescent subpopulations are considered instead of the immune reactions. Most of the papers including the two mentioned above, however, do not take into account how pressure drives cell motion and affect division rates. Pressure can drive the evolution of the tumor, where increasing pressure inhibits proliferation in the tumor core and only the cells near the outer rim of the tumor contribute significantly to the growth of the tumor. This leads to linear, instead of exponential growth.

In [**?**, **?**], Gurtin and MacCamy study a nonlinear version of the growth fragmentation equation (**??**) without immune reaction term, which considers the number of cells to account for overcrowding effect. In the context of tumor growth, Liu et al. [**?**] consider a nonlinear age-structured model in which the proliferation rate is a decreasing function of the number of tumor cells (which is a particular case of our model (**??**)). In these work, however, the migration of cells toward less crowded regions is not taken into account. Other studies have focused on this migration and on the role of mechanical stress in the growth of the tumor (but without the age-structure) [63]. The resulting models have been studied by the mathematical community for a decade or so (see for example [**?**,**?**,**?**,**?**,**?**,**?**,**?**]). A classical model, introduced in [63] (see also [**?**,**?**,**?**]) and studied for example in [**?**, **?**, **?**, **?**, **?**] models the evolution of the cell population distribution function $\rho(x,t)$ by the nonlinear diffusion equation

$$\partial_t \rho - \mathrm{div}\,(\rho \nabla_x p) = \rho F(p), \qquad p = \frac{m}{m-1}\left(\frac{\rho}{\rho_M}\right)^{m-1} \tag{2.3}$$

where p is the pressure function attaining $p = p_M$ at $\rho = \rho_M$, and $F(p)$ is the growth rate which is a decreasing function of pressure, vanishing at $p = p_M$.

Taking into account the cells' migration in the physical growth of the tumor is crucial, especially in order to account for the interactions with the surrounding tissues (competition for space with other healthy and cancerous cells, availability of nutrients etc.). We point out that when $1 < m < \infty$, migration is modeled in (**??**) by degenerate diffusion (porous media type equation) which leads to finite speed of propagation of the support of ρ (and thus the support of n). The limit $m \to \infty$ has been at the topic of numerous papers since the original work of Perthame, Quiros and Vazquez [**?**] (see for example [**?**, **?**, **?**, **?**, **?**] and references therein). The resulting mathematical model is a Hele-Shaw free boundary problem which describes the motion of the interface (the edge of the tumor). We also refer to [**?**] for a review of other free boundary problems appearing in the context of tumor growth modeling.

In [63] and [?] both agent-based models and continuum models (similar to (??) and its singular limit $m \to \infty$) have been studied numerically. Our model is an intermediate model between these, as it does not track the fate of every individual cell, but nevertheless allows us to take into account the fact that the age distribution in the cell is far from homogeneous. For example, experimental findings have suggested that most mitosis events occur near the edge of the tumor (this is also the case with the individual based models considered in [63]) and point to the development of regions of quiescent cells and of necrosis in the core of the tumor. Finally, we point out that our model (??) also shares some similarities with models with phenotypic heterogeneity studied for instance in [?], but in that case, each cell is associated to a phenotypic variable $y \in [0, 1]$ which affects their proliferation rate but does not change with time.

There can thus be significant advantages in keeping track of both the age and the position of the cells. The boundary value problem (??) carries more information than (??), since the distribution n encodes the age structure of the cell population and can thus give critical insight into the model's prediction. Such a model can then be used to model the effects of various therapies since the response of the cell to the proposed treatment depends on its life cycle (most therapies target cells that are at some specific stage of their life cycle).

The rest of the chapter is organized as follows: In section 2.2, our main equations are introduced. The existence of a weak solution of the model is given in 2.2.2. A few extensions, mainly about the volume density that accounts for the volume growth of a cell, are also discussed. In

2.3, the numerical results about tumor growth and mitosis events, age distributions, and asymptotic behavior are given. The algorithms and discretization of the model are detailed in 2.3.0.2. Finally, discussion about our model follows in 2.4.

2.2 Model Properties and Settings

2.2.1 Model Equations

In this chapter, the maximum packing density ρ_M is assumed to be 1 for simplicity. Then the pressure in (**??**) reduces to

$$p = \frac{m}{m-1}\rho^{m-1} \tag{2.4}$$

and $p_M = \frac{m}{m-1}$.

The life cycle of tumor cells mainly consist of two phases in our model: The interphase where a cell grows and copies its genetic material, often followed by the mitotic phase where a cell divides into two daughter cells. The newborn daughter cells start from the interphase cycle again until they proliferate. These phases can be refined by cell age, denoted by $\theta \geq 0$. A cell ages until its proliferation. At the birth of the daughter cells (who enter interphase), the age of each daughter cell is reset to 0 and increases until this cell reproduces again. The cell distribution function $n(x, \theta, t)$ is the cell density of cells at age θ, position $x \in \mathbb{R}^d$ and time t.

$$
\begin{cases}
\partial_t n + \partial_\theta n - \operatorname{div}(n\nabla p) = -\nu(\theta,p)\,n - \mu(\theta)\,n & x \in \mathbb{R}^d,\ \theta > 0,\ t > 0 \\
n(x,0,t) = 2\displaystyle\int_0^\infty \nu(\theta,p)\,n(x,\theta,t)\,d\theta & x \in \mathbb{R}^d,\ \ t > 0 \\
n(x,\theta,0) = n_{in}(x,\theta) & x \in \mathbb{R}^d,\ \theta > 0.
\end{cases} \tag{2.5}
$$

The term $\partial_\theta n$ accounts for the aging of the cells, and $\nu(\theta,p)$ is the proliferation rate of the cells of age θ under the pressure p, which is a decreasing function of p that vanishes at $p = p_M$. The death rate μ of cells is also taken as a function of the age of the cells. The right hand side of the first equation in (**??**) accounts for the outflow of the cells of age θ at each location x and time t, either from proliferation ($\nu(\theta,p)n$) or death. The cells of age 0 are given by the number of newborn daughter cells, which is double the amount of cells who proliferate.

In general, ν depends on both the age of the cell θ and the local pressure $p = p(x,t)$ (we note that in experiments, there appears to be a lot of variability in the typical time between two mitosis events [**?**]). Pressure-limited proliferation corresponds to the assumptions

$$
\partial_p \nu(\theta,p) < 0, \qquad \nu(\theta,p_M) = 0. \tag{2.6}
$$

The coefficient $\mu(\theta)$ denotes the death rate, which can be taken to be constant or more generally a function of the age. A maximum age θ_m can be imposed by requiring that $\int_0^{\theta_m} \mu(\theta)\,d\theta = +\infty$.

Then the population density function ρ is given by

$$\rho(x,t) = \int_0^\infty n(x,\theta,t)\ d\theta.$$

Thus the following holds for ρ:

$$\begin{cases} \partial_t \rho - \operatorname{div}(\rho \nabla p) = \int_0^\infty v(\theta,p) n(x,\theta,t)\ d\theta - \int_0^\infty \mu(\theta) n(x,\theta,t)\ d\theta \\ \rho(x,0) = \int_0^\infty n_{in}(x,\theta)\ d\theta \end{cases} \tag{2.7}$$

2.2.2 Existence of a Weak Solution

For simplicity, we take $\mu = 0$ in 2.2.2. Then the system (**??**) reduces to

$$\begin{cases} \partial_t n + \partial_\theta n - \operatorname{div}(n \nabla p) = -\nu(\theta,p)\, n & x \in \mathbb{R}^d,\ \theta > 0,\ t > 0 \\ n(x,0,t) = 2\displaystyle\int_0^\infty \nu(\theta,p)\, n(x,\theta,t)\, d\theta & x \in \mathbb{R}^d,\ \ t > 0 \\ n(x,\theta,0) = n_{in}(x,\theta) & x \in \mathbb{R}^d,\ \theta > 0. \end{cases} \tag{2.8}$$

The following will be assumed about the reproduction rate in this section. There exists a positive constant C such that

$$0 \leq \nu(\theta,p) \leq C \qquad \forall \theta,\ p \geq 0. \tag{2.9}$$

Theorem 2.2.1. *For all $m > 2$ and for any initial condition $n_{in}(x,\theta)$ such that*

$$n_{in} \in L^1 \cap L\log L(\mathbb{R}^d \times (0,\infty)), \quad (|x|^2+\theta)n_{in} \in L^1(\mathbb{R}^d \times (0,\infty)), \quad \rho_{in} \in L^\infty(\mathbb{R}^d) \tag{2.10}$$

there exists a weak solution $n(x,\theta,t)$ of (**??**)*. The function $n(x,\theta,t)$ is such that*

$$n \in L^\infty(0,T; L^1 \cap L\log L(\mathbb{R}^d \times (0,\infty))), \quad (|x|^2+\theta)n \in L^\infty(0,T; L^1(\mathbb{R}^d \times (0,\infty)))$$

while the density $\rho(x,t)$ and pressure $p(x,t) = \frac{m}{m-1}\rho(x,t)^{m-1}$ are such that

$$\rho \in L^\infty(0,T; L^1 \cap L^\infty(\mathbb{R}^d)), \qquad \nabla p \in L^2(0,T; L^2(\mathbb{R}^d))$$

and the equation (**??**) *is satisfied in the following distributional sense:*

$$\begin{aligned}\int_0^T\!\!\int_{\mathbb{R}^d}\!\int_0^\infty n(x,\theta,t)\big[-\partial_t\psi(x,\theta,t) - \partial_\theta\psi(x,\theta,t) + \nabla_x p(x,t)\cdot\nabla_x\psi(x,\theta,t)\\ + \nu(\theta,p(x,t))\psi(x,\theta,t)\big]\,d\theta\,dx\,dt\\ = \int_{\mathbb{R}^d}\!\int_0^\infty n_{in}(x,\theta)\psi(x,\theta,0)\,d\theta\,dx + \int_0^T\!\!\int_{\mathbb{R}^d}\psi(x,0,t)2\int_0^\infty \nu(\theta,p(x,t))n(x,\theta,t)\,d\theta\,dx\,dt\end{aligned} \tag{2.11}$$

for all $\psi \in \mathcal{D}(\mathbb{R}^d \times [0,\infty) \times [0,T))$. Furthermore, the density $\rho(x,t)$ solves (**??**)*.*

This proves the existence of a weak solution to the system (**??**). The theorem above is proven in Chapter 4 of [**?**].

Note that from the assumptions in (**??**), the population density function ρ satisfies

$$\partial_t \rho - \operatorname{div}(\rho \nabla p) \leq C\rho.$$

Thus multiplying this equation by $m\rho^{m-2}$, the following equation for the pressure p can be deduced:

$$\partial_t p - (m-1)p\Delta p - |\nabla p|^2 \leq (m-1)Cp.$$

Using the above, the following proposition can be proved.

Proposition 2.2.1 (Finite speed of propagation of the support)**.** *Under the assumptions of Theorem* **??***, assume further that there exists R_0 such that $n_{in}(x,\theta) = 0$ for all $|x| \geq R_0$ and $\theta \geq 0$. Then*

$$n(x,\theta,t) = 0 \quad \textit{for all } |x| \geq \overline{R} + e^{(m-1)Ct} \textit{ and for all } \theta \geq 0$$

for some $\overline{R}$ depending on R_0 and $\|\rho_{in}\|_{L^\infty}$.

This proposition is proven in Chapter 2 of [**?**].

2.2.3 Extensions of the Model

As in (**??**), the redistribution of the cells away from crowded regions is modeled by Darcy's law, but the pressure p now depends not on the local number of cells $\int_0^\infty n(x,\theta,t)\, d\theta$ but on the volume occupied by the cells. Indeed, we wish to take into account the first phase of the cells life cycle by assuming that a cell of age $\theta \geq 0$ occupies a volume $V(\theta)$, where $\theta \mapsto V(\theta)$ is a non-decreasing function. The volume density is then defined as

$$\rho(x,t) = \int_0^\infty V(\theta) n(x,\theta,t)\, d\theta.$$

2.2.3.1 Pressure-Limited Growth

In (**??**), we assumed (see (**??**)) that high pressure (or high volume density) prevents the cells from dividing, thus limiting the growth of the tumor. Alternatively, we can assume that high pressure will slow down the whole life cycle of the cell (potentially preventing a newborn cell from growing and reaching its "adult" size). To write such a model, we can change slightly the meaning of the variable θ, to represent a parameter indicating how far along its life cycle a cell has been able to go (i.e. the cell physiological age). In this model, a cell no longer "ages" linearly in time, since its growth is limited by a parameter depending on the pressure p, leading to the system:

$$\begin{cases} \partial_t n + r(p)\partial_\theta n - \operatorname{div}_x(n\nabla p) = -r(p)\nu(\theta,p)n - \mu(\theta)n & x \in \mathbb{R}^d,\ \theta > 0,\ t > 0 \\ n(x,0,t) = 2\displaystyle\int_0^\infty \nu(\theta,p) n(x,\theta,t)\, d\theta & x \in \mathbb{R}^d,\ \ t > 0 \\ n(x,\theta,0) = n_{in}(x,\theta) & x \in \mathbb{R}^d,\ \theta > 0, \end{cases} \tag{2.12}$$

where $r(p)$ is a decreasing function of p such that $r(0) = 1$ and $r(p) = 0$ for $p \geq p_M$.

Equations (**??**) and (**??**) are related to the classical model (**??**) as follows: When $\nu = \nu(p)$, $\mu = \mu_0$ and $V = V_0$ are all independent of the age θ, we can integrate the first equation in (**??**)

with respect to θ to get the following equation for $\rho(x,t) = V_0 \int_0^\infty n(x,\theta,t)\, d\theta$:

$$\begin{cases} \partial_t \rho - \operatorname{div}(\rho \nabla p) = \rho\,(\nu(p) r(p) - \mu_0), \qquad p = \frac{m}{m-1}\rho^{m-1} & x \in \mathbb{R}^d,\ t > 0 \\ \rho(x,0) = V_0 \int_0^\infty n_{in}(x,\theta)\, d\theta & x \in \mathbb{R}^d, \end{cases} \tag{2.13}$$

which is (**??**) with growth rate $F(p) = \nu(p)r(p) - \mu_0$.

In this simple case, it is thus possible to solve for $\rho(x,t)$ without determining $n(x,\theta,t)$. In general (when ν, μ and V do depend on θ), that is not possible: After multiplying the first equation in (**??**) by $V(\theta)$ and integrating with respect to θ we find the following equation for the volume density $\rho(x,t)$:

$$\begin{aligned} \partial_t \rho - \operatorname{div}(\rho \nabla p) = r(p) \int_0^\infty V'(\theta) n(x,\theta,t)\, d\theta + r(p) \int_0^\infty \nu(\theta,p)(2V(0) - V(\theta)) n(x,\theta,t)\, d\theta \\ - \int_0^\infty \mu(\theta) V(\theta) n(x,\theta,t)\, d\theta. \end{aligned} \tag{2.14}$$

The first term in the right hand side describes the expansion of the tumor due to the volume change of individual cells, while the second term accounts for the change of volume during mitosis. Such events are typically volume preserving (i.e. the total volume of the two daughter cells is equal to the volume of the dividing cell), which can be enforced by assuming that $\nu(\theta,\rho)(2V(0) - V(\theta)) = 0$ for all $\theta > 0$. A simple form for the coefficients ν and V describing this situation is as follows:

$$V(\theta) = \begin{cases} V_0 + \alpha\theta & \text{if } \theta \in [0, V_0/\alpha] \\ \\ 2V_0 & \text{if } \theta \geq V_0/\alpha, \end{cases} \tag{2.15}$$

$$\nu(\theta, p) = 0 \quad \text{for } \theta < V_0/\alpha, \tag{2.16}$$

in which case (**??**) reduces to

$$\partial_t \rho - \operatorname{div}(\rho \nabla p) = \alpha r(\rho) \int_0^{V_0/\alpha} n(x,\theta,t)\, d\theta - \int_0^\infty \mu(\theta) V(\theta) n(x,\theta,t)\, d\theta.$$

2.2.3.2 Quiescent Subpopulations

An important feature of many tumor growth models is the fact that not all cells behave similarly and that one should make a distinction between the proliferating cells, with distribution function still denoted by $n(x, \theta, t)$ and the quiescent cells, with distribution function $q(x, \theta, t)$ [**?**, **?**, **?**].

After mitosis, a fraction λ of the new cells will be quiescent (and the rest proliferating). In addition, cells can switch back and forth from one state to the other with probabilities σ_1 and σ_2.

This leads to the following system of equations:

$$\begin{cases} \partial_t n + r_1(p)\partial_\theta n - \operatorname{div}_x(n\nabla_x p) = -\mu(\theta)n - r_1(p)\nu(\theta,p)n + \sigma_1 q - \sigma_2 n \\ \partial_t q + r_2(p)\partial_\theta q - \operatorname{div}_x(q\nabla_x p) = -\mu(\theta)q - \sigma_1 q + \sigma_2 n \\ n(x,0,t) = 2(1-\lambda)\displaystyle\int_0^\infty \nu(\theta,p)n(x,\theta,t)\,d\theta \\ q(x,0,t) = 2\lambda\displaystyle\int_0^\infty \nu(\theta,p)n(x,\theta,t)\,d\theta, \end{cases} \tag{2.17}$$

with r_1 and r_2 the different "aging speeds" for the two types of cells and

$$p(x,t) = \frac{m}{m-1}\rho(x,t)^{m-1}, \qquad \rho(x,t) = \int_0^\infty V_n(\theta)n(x,\theta,t) + V_q(\theta)q(x,\theta,t)\,d\theta.$$

Here, V_n and V_q denote the volumes of the proliferating and quiescent cells. Note that the quiescent cells do not duplicate and might have a slower evolution if $r_1 \neq r_2$. Taking $r_2(p) = 0$ amounts to assuming that they have a frozen life cycle where growth and duplication do not happen unless and until they transition to the proliferating state. The rates of transition between the proliferative and quiescent compartments, σ_1 and σ_2 can depend on the age variable θ as well as on the pressure [**?**].

2.3 Numerical Simulation

2.3.1 Model Parameters

We fix $m = 4$ in the equation (**??**) for pressure so that $p(x,t) = \frac{4}{3}\rho(x,t)^3$ in this section. For the proliferation rate, we take

$$v(\theta, p) = \beta(\theta)\left(\frac{4}{3} - p\right) \tag{2.18}$$

where β satisfies $\beta(\theta) = 0$ for some small θ. This reflects the biological fact that daughter cells, upon birth, undergo a growth phase prior to their subsequent division. Following [**?**], we take:

$$\beta(\theta) = \begin{cases} 0 & \text{if } \theta \leq \theta_0 \\ \frac{(\theta-\theta_0)^2}{\sigma(2\sigma^2+2\sigma(\theta-\theta_0)+(\theta-\theta_0)^2)} & \text{if } \theta \geq \theta_0 \end{cases} \tag{2.19}$$

where $\theta_0 = 1$ and $\sigma = 20$. We restrict ourselves to the case $V(\theta) \equiv V_0$. Note that for computation purposes, we need to take the age variable θ in a bounded interval $[0, \theta_{max}]$. When we can take θ_{max} larger than the maximum computation time, this maximum age does not play a role in determining the solution. But since cells do not live forever, it can be relevant (and numerically less costly) to take $\theta_{max} \leq T$.

In this setting, one must determine how cells are treated once they reach the age $\theta = \theta_{max}$. For the numerical computations that follow, we assume that these 'old enough' cells remain within the system and contribute to the pressure, but they cease to proliferate. This is represented by imposing $\beta(\theta) = 0$ for $\theta > \theta_{max}$.

Note that it would be easy to assume instead that these cells are proliferating with a constant rate ($\beta(\theta) = \beta(\theta_{max})$ for $\theta > \theta_{max}$). In practice, we actually see very little difference between these two settings in the simulations: Indeed, older cells are primarily found in the center of the tumor, where the proliferation is already negligible because of the high pressure $p \sim \frac{4}{3}$. This is illustrated in Figure **??** which shows the density of the cells of age $\theta = \theta_{max}$ inside the tumor. We also point to Figure **??**, which shows that when the death of cells is taken into account, the number of cells of age θ converges to zero exponentially fast and taking θ_{max} large enough will again lead qualitatively to the same results as when $\theta_{max} = +\infty$.

In the following simulations, we take $\theta_{max} = 20$. Setting the death rate $\mu(\theta)$ as in (**??**) had little impact on the simulation results, except in the long-term behaviors. Thus the death process will be investigated in the section 2.3.3.

2.3.2 Algorithms and Discretization

Focusing on the discretization of the variables θ and t, we write $n_{k,i}(x) = n(x, k\Delta t, i\Delta\theta)$. Using first-order approximations, Equation (**??**) leads to

$$\frac{n_{k+1,i} - n_{k,i}}{\Delta t} + \frac{n_{k,i} - n_{k,i-1}}{\Delta\theta} - \operatorname{div}_x(n_{k,i}\nabla_x p_k) = -\nu_{k,i}n_{k,i} - \mu_i n_{k,i}, \tag{2.20}$$

for $i = 1, \ldots, i_M - 1$. Setting $\lambda \coloneqq \Delta t/\Delta\theta$, and rearranging (**??**), we obtain

$$n_{k+1,i} = n_{k,i} + \lambda(n_{k,i-1} - n_{k,i}) + \operatorname{div}_x(n_{k,i}\nabla_x p_k)\Delta t - \nu_{k,i} n_{k,i}\Delta t - \mu_i n_{k,i}\Delta t. \tag{2.21}$$

Since cells that reach the age $\theta_{max} = i_M\Delta\theta$ do not age or duplicate, the equation for n_{k+1,i_M} is:

$$n_{k+1,i_M} = n_{k,i_M} + \lambda n_{k,i_M-1} + (\Delta t)[\operatorname{div}_x(n_{k,i_M}\nabla_x p_k) - \nu_{k,i_M} n_{k,i_M} - \mu_{i_M} n_{k,i_M}] \tag{2.22}$$

The density at time $k\Delta t$ is then given by

$$\rho_k = \sum_{i=0}^{i_M} n_{k,i}\Delta\theta$$

and using (**??**)-(**??**), we get

$$\rho_{k+1} - n_{k+1,0}\Delta\theta = \rho_k - n_{k,0}\Delta\theta + n_{k,0}\Delta t + (\Delta t\Delta\theta)\sum_{i=1}^{i_M}[\operatorname{div}_x(n_{k,i}\nabla_x p_k) - \nu_{k,i} n_{k,i} - \mu_i n_{k,i}]. \tag{2.23}$$

With a slight abuse of notation, we write $\nu_k\rho_k = \sum_{i=0}^{i_M} \nu_{k,i} n_{k,i}\Delta\theta$ and $\mu\rho_k = \sum_{i=0}^{i_M} \mu_i n_{k,i}\Delta\theta$, so that equation (**??**) can be rewritten as

$$\begin{aligned}\rho_{k+1} =& \rho_k + (n_{k+1,0} - n_{k,0})\Delta\theta + n_{k,0}\Delta t + \operatorname{div}_x(\rho_k\nabla_x p_k)\Delta t - \operatorname{div}_x(n_{k,0}\nabla_x p_k)\Delta t\Delta\theta \\ & - \nu_k\rho_k\Delta t + \nu_{k,0} n_{k,0}\Delta t\Delta\theta - \mu\rho_k\Delta t + \mu_0 n_{k,0}\Delta t\Delta\theta. \end{aligned} \tag{2.24}$$

With the same notation, the equation for the density (**??**) can be discretized as $\rho_{k+1} =$

$\rho_k + \text{div}_x(\rho_k \nabla_x p_k)\Delta t + \nu_k \rho_k \Delta t - \mu \rho_k \Delta t$. Combining these two equations leads to

$$2\nu_k \rho_k \Delta t = (n_{k+1,0} - n_{k,0})\Delta\theta + n_{k,0}\Delta t - \text{div}_x(n_{k,0}\nabla_x p_k)\Delta t\Delta\theta + \nu_{k,0}n_{k,0}\Delta t\Delta\theta + \mu_0 n_{k,0}\Delta t\Delta\theta,$$

which gives the appropriate discretization of the condition at $\theta = 0$ ($i = 0$):

$$\begin{aligned} n_{k+1,0} &= 2\sum_{i=0}^{i_M} \nu_{k,i} n_{k,i}\Delta t + n_{k,0}\,(1-\lambda) + \text{div}_x(n_{k,0}\nabla_x p_k)\Delta t - \nu_{k,0}n_{k,0}\Delta t - \mu_0 n_{k,0}\Delta t \\ &= 2\sum_{i=1}^{i_M} \nu_{k,i} n_{k,i}\Delta t + n_{k,0}\,(1-\lambda+\nu_{k,0}\Delta t - \mu_0\Delta t) + \text{div}_x(n_{k,0}\nabla_x p_k)\Delta t. \end{aligned} \tag{2.25}$$

Equations (**??**)-(**??**)-(**??**) provide the numerical scheme for the aging/duplication/death of the cells. Importantly, we note that in order to reduce the computational cost, it might be necessary to take $\Delta\theta$ to be much larger than Δt ($\lambda \ll 1$). The discretization with respect to the spatial variable x is done with a uniform grid $\Delta x_1 = \Delta x_2 = \Delta x$ and the gradients are calculated by the centered difference

$$(\nabla_{x_i} p)_{l,m} = \frac{p_{l+1,m} - p_{l-1,m}}{2\Delta x}, \qquad i = 1, 2.$$

The algorithm **??** provides the pseudocode for the algorithm used for the simulation.

2.3.3 Tumor Growth and Mitosis Events

2.3.3.1 Growth

Figure **??** shows the evolution of the density $\rho(x,t) = \int_0^\infty n(x,\theta,t)\,d\theta$ with $\theta_{\max} = 20$ for some non-symmetric initial data. It is numerically observed that when starting with small enough

Algorithm 1 Pseudocode for the simulation

$T \leftarrow$ total time

$i_M \leftarrow$ (Final age)$/\Delta\theta$

Calculate the gradient operators for x_1 and x_2 each.

for $k = 1$ **to** $T/\Delta t$ **do**

Calculate $\nabla_{x_1} p$ and $\nabla_{x_2} p$.

$D_{i_M} \leftarrow \nabla_{x_1}(n_{k,i_M} \nabla_{x_1} p) + \nabla_{x_2}(n_{k,i_M} \nabla_{x_2} p)$ ▷ Diffusion term for the oldest

$n_{k+1,i_M} \leftarrow n_{k,i_M}[1 - (\Delta t)\mu(i_M)] + \lambda n_{k,i_M - 1} + (\Delta t) D_{i_M}$

for $i = i_M - 1$ **to** 1 **do**

$D_i \leftarrow \nabla_{x_1}(n_{k,i} \nabla_{x_1} p) + \nabla_{x_2}(n_{k,i} \nabla_{x_2} p)$

$\text{Pro}_i \leftarrow \nu_{k,i} n_{k,i}$ ▷ Proliferation at each age

$n_{k+1,i} \leftarrow n_{k,i}[1 - (\Delta t)\mu(i)] + \lambda(n_{k,i-1} - n_{k,i}) + (\Delta t)(\text{Pro}_i + D_i)$

end for

$\text{Newborn}_k \leftarrow 2 \sum_{i=1}^{i_M} \text{Pro}_i(\Delta t)$ ▷ Newborn cells

$n_{k+1,0} \leftarrow \text{Newborn}_k + n_{k,0}[1 - \lambda - (\Delta t)\mu(0)]$

$\rho_k \leftarrow \sum_{i=0}^{i_M} n_{k,i}(\Delta\theta)$

$p_k \leftarrow \frac{m}{m-1} \rho_k^{m-1}$

end for

asymmetry, the cells grew symmetrically over time.

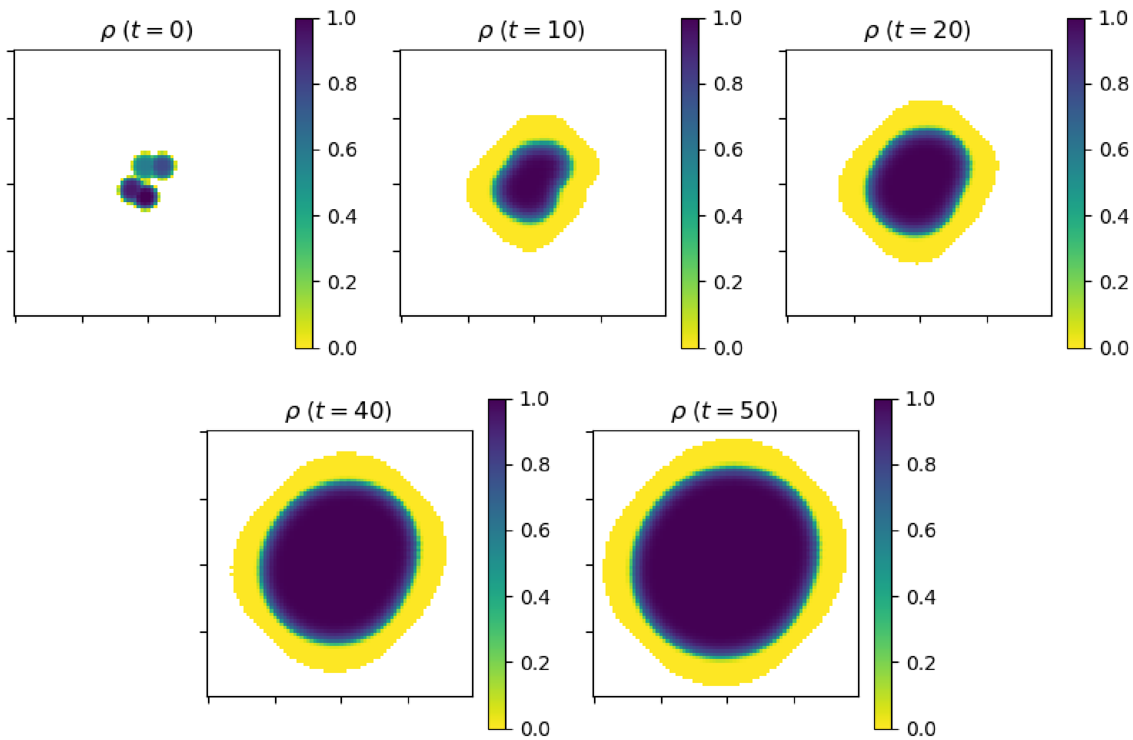

Figure 2.1: The density $x \mapsto \rho(x,t)$ for a 2-dimensional tumor at time $t = 0$, 10, 20, 40 and 50

The square rooted growth of the overall population under our model is shown in figure **??**. This implies that the total mass of the tumor grows quadratically after the initial phase of faster growth.

2.3.3.2 Mitosis Events

One of the most interesting features of our model, compared to the macroscopic models that only describes the evolution of the density $\rho(x,t)$ is that it keeps track of where proliferation is taking place. Indeed, in vitro experiments with cancerous cells suggest that the growth of the tumor is mostly due to the proliferation of the cells in a small region near, but not at, the

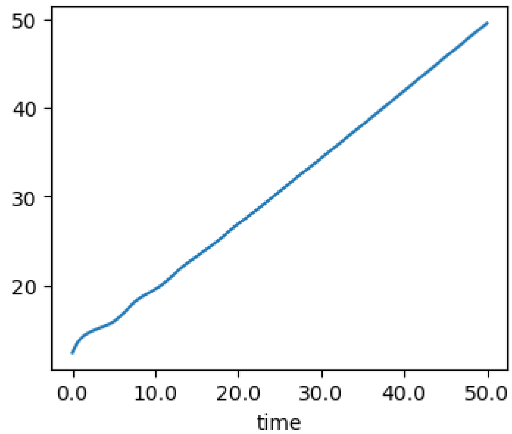

Figure 2.2: $\left(\int_\Omega \rho(x,t)\ dx\right)^{\frac{1}{2}}$ over time for a two dimensional tumor.

outer edge of the tumor. Figure **??** shows the distribution of cells that just experienced mitosis $n(x, \theta = 0, t)$ and clearly shows that the model is consistent with this observation. The regions where mitosis mainly takes place in Figure **??** is in an annulus close to the edge of the support of n.

2.3.4 Age Distributions

As mentioned in 2.3.0.1, an upper bound θ_{max} was enforced on the age variable θ for the numerical simulations. Figure **??** tracks the distribution of the cells of age θ_{max} over time. The cells of this age no longer proliferate in our model, and are shown to be concentrated at the center of the tumor in the figure. This can be interpreted as a necrotic core.

The average age of the cells, as a function of the cell location x and time t can be defined as

$$\Theta(x,t) := \frac{\int_0^\infty \theta n(x,\theta,t)\ d\theta}{\int_0^\infty n(x,\theta,t)\ d\theta}.$$

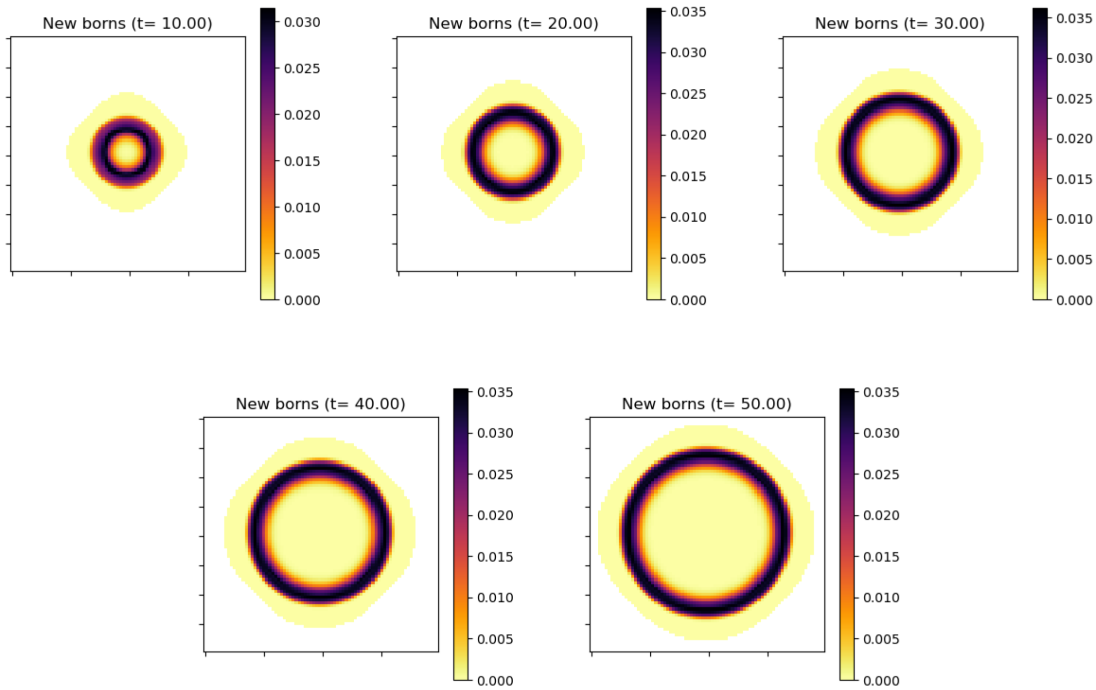

Figure 2.3: Mitosis events inside the tumor

Figure **??** illustrates the average age of the cells. The lowest average age is concentrated slightly inside the tumor, but not at the edge. This coincides with the results in 2.3.1.2, as this region is where most mitosis events take place. Overall, the age distributions observed in our model indicate an important feature often observed in experiments, the notion of surfing cells. The leading edge of the tumor are being pushed by the growing bulk density so that the minimum average age of the cells is not found at the edge.

2.3.5 Asymptotic Behavior of Solutions

We now turn our attention to the behavior of the solutions for large time. Two important features of tumor growth are the speed of spreading of the tumor and the asymptotic age distribu-

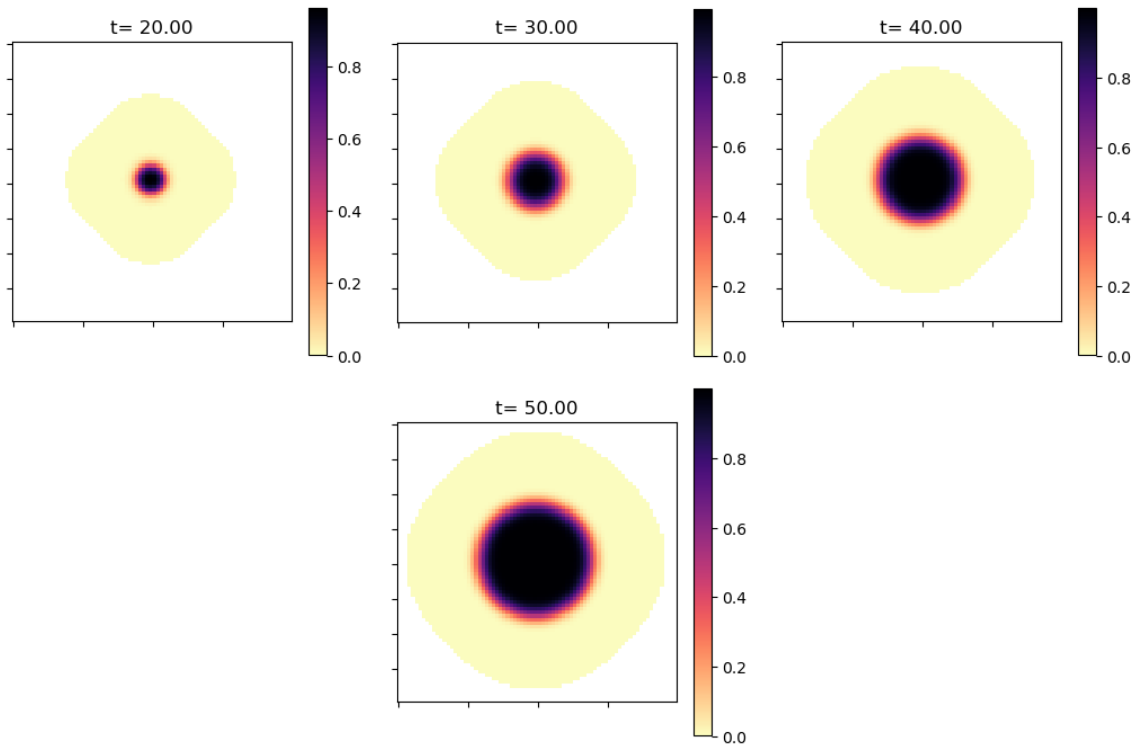

Figure 2.4: Spatial distribution of the oldest cells

tion of the cells across the tumor. Indeed, a motivation for the study of age-structured models is the development and improvement of therapies. Since cells in different phases of their life cycle are affected differently by therapies, a precise description of the age distribution of the tumor cells can help develop effective strategies to limit their growth.

2.3.5.1 Motivation : Space-Homogeneous Model

There is an extensive literature (see for example [**?**]) devoted to space-homogeneous age-structured models such as (**??**). The long time asymptotic for (**??**) is a classical problem: This system has particular solutions of the form

$$n_\infty(\theta, t) = ce^{\lambda t}\varphi(\theta) \tag{2.26}$$

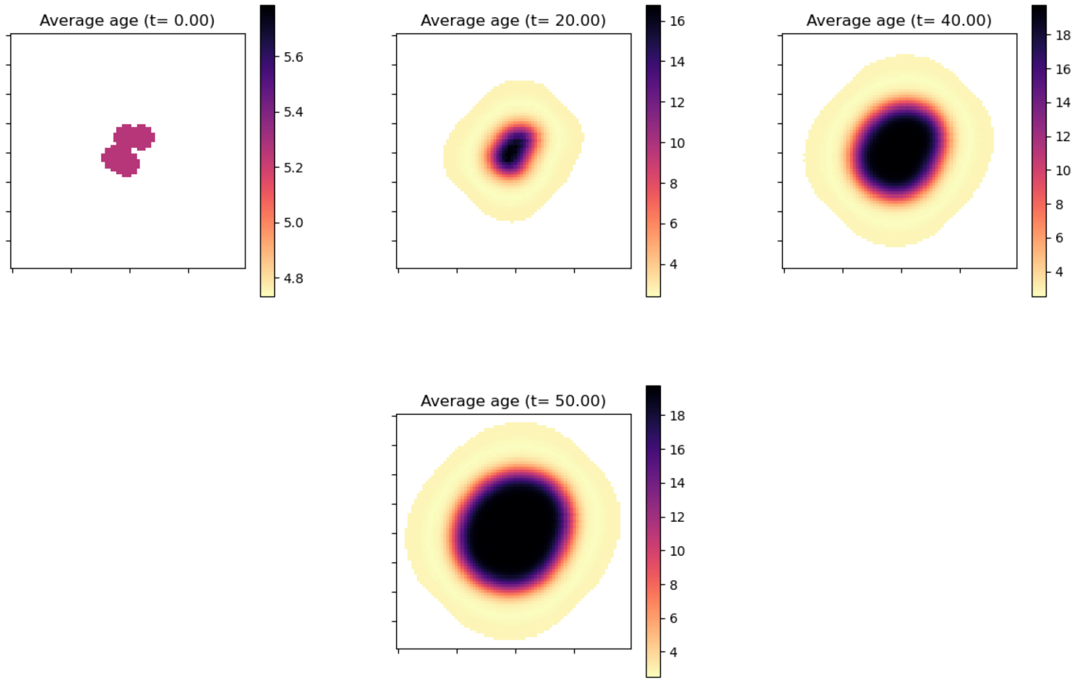

Figure 2.5: Average age of the cells across the tumor

where φ solves

$$\lambda\varphi + \varphi' = -\beta(\theta)\,\varphi - \mu(\theta)\varphi, \qquad \varphi(0) = 1$$

and λ is determined by the condition

$$2\int_0^\infty \beta(\theta)\,\varphi(\theta)\,d\theta = 1.$$

The formula (**??**) shows that the parameter λ characterizes the *exponential growth* of the tumor, while the function φ describes the asymptotic age distribution of the cells. In particular, it can be proved (see [**?**]) that for any nonzero initial data, the solution of (**??**) satisfies:

$$\frac{n(\theta,t)}{\int_0^\infty n(\theta',t)d\theta'} \to \varphi(\theta) \qquad \text{as } t \to +\infty.$$

2.3.5.2 Space-Dependent Model : Traveling Wave Solutions

When pressure and space variable are taken into account, the asymptotic behavior of the solutions is very different. An important property of the reaction-diffusion equation (**??**) is the existence of traveling wave solutions and the finite speed of spreading (see for example [**?**, **?**] and references therein for traveling wave solutions in the context of tumor growth). In particular, we recall that when $F(p) = p_M - p$, (**??**) admits solutions of the form $\bar{\rho}(x - ct)$ for all $c \geq c^*$ where the smallest speed c^* is also the spreading speed of compactly supported solutions.

The existence of traveling wave solutions for the simple age-structured model

$$\begin{cases} \partial_t n + \partial_\theta n - \operatorname{div}_x(n \nabla_x p) = -\nu(\theta, p)\, n \\ n(x, 0, t) = 2 \displaystyle\int_0^\infty \nu(\theta, p)\, n(x, \theta, t)\, d\theta \end{cases} \tag{2.27}$$

with $\nu(\theta, p) = \beta(\theta)(p_M - p)$ can be observed numerically in dimension 1 as shown in Figure **??**: After some initial transition time, the solution is asymptotically close to a traveling wave, which is a solution of (**??**) of the form $n(x, \theta, t) = \bar{n}(x - ct, \theta)$ where the corresponding pressure variable $\bar{p}(x)$ satisfies the boundary condition

$$\lim_{x \to -\infty} \bar{p}(x) = p_M, \qquad \lim_{x \to +\infty} \bar{p}(x) = 0. \tag{2.28}$$

A rigorous mathematical justification of this fact is the object of a forthcoming work [**?**].

This implies that the growth of the tumor is no longer exponential: The speed of the traveling wave characterizes the growth of the diameter of the tumor. In dimension 2, for example, this implies that the total mass of the tumor grows quadratically, a phenomenon that can be confirmed

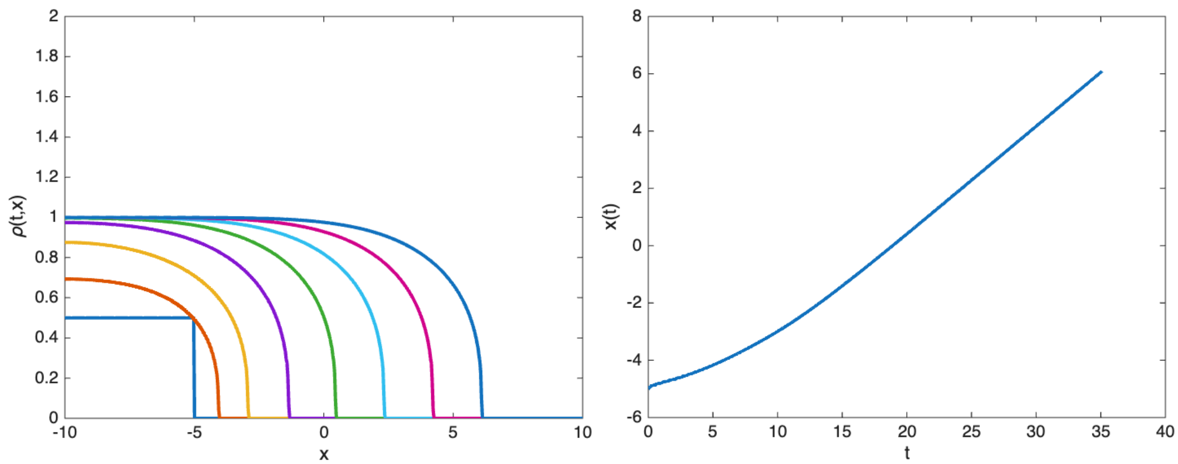

Figure 2.6: Development of a front propagating with constant speed: The left figure shows the evolution of the one-dimensional density $\rho(x,t)$ starting from a characteristic function $\frac{1}{2}\chi_{x\leq -5}$ at time $t = 0, 5, 10, 15, 20, 25, 30$ and 35. The right figure shows the position of the front $x(t)$ as a function of time (defined as $\rho(x(t),t) = 1/2$). As expected from traveling waves, the front motion is asymptotically linear in time.

numerically (see Figure **??**).

Focusing on the behavior of $n(x,\theta,t)$ for large t, we see numerically that the traveling wave solution satisfies

$$n(x,\theta,t) = \bar{n}(x-ct,\theta) \sim \phi(x+c(\theta-t)) \text{ as } t \to \infty \tag{2.29}$$

for fixed x and θ and for some profile ϕ. This observation is consistent with the equation since (**??**) implies $\partial_t n + \partial_\theta n = -c\partial_x \bar{n} + \partial_\theta \bar{n} \to 0$ as $x \to -\infty$. This can be explained as follows: As the tumor is spreading, the pressure at a given x converge to p_M when $t \to \infty$ so that both diffusion and proliferation become negligible. The asymptotic behavior (**??**) simply captures the aging of the cells.

Death Rate : Expansion vs. Regression of the Tumor

The long time behavior of the solutions is quite different when we include the death rate $\mu(\theta)$ in the model:

$$\begin{cases} \partial_t n + \partial_\theta n - \operatorname{div}_x(n\nabla_x p) = -\nu(\theta, p)\, n - \mu(\theta) n \\ n(x, 0, t) = 2\displaystyle\int_0^\infty \nu(\theta, p)\, n(x, \theta, t)\, d\theta. \end{cases} \tag{2.30}$$

First, we observe numerically that for large enough μ, the tumor might shrink and eventually disappear ($\lim_{t\to\infty} n(t) = 0$), while for small μ, a moving front with positive speed develops as in the case $\mu = 0$. We also note that traveling wave solutions cannot satisfy (**??**) when $\mu > 0$ since the right-hand side of (**??**) no longer vanishes for $p = p_M$. Finally, when the tumor spreads, numerical simulations show that there exists a unique profile $\phi_\infty(\theta)$ such that

$$n(x, \theta, t) \to \phi_\infty(\theta) \quad \text{ as } t \to \infty$$

for all $x \in \mathbb{R}^d$ (which is very different from the asymptotic behavior (**??**)). Examples of this profile $\phi_\infty(\theta)$ are shown in Figure **??** when μ is given by

$$\mu(\theta) = \begin{cases} 0 & \text{if } \theta < 10 \\ 0.1 & \text{if } \theta \geq 10 \end{cases} \tag{2.31}$$

and for two different choices of $\beta(\theta)$.

This behavior can be investigated analytically: Going back to (**??**) we see that the asymp-

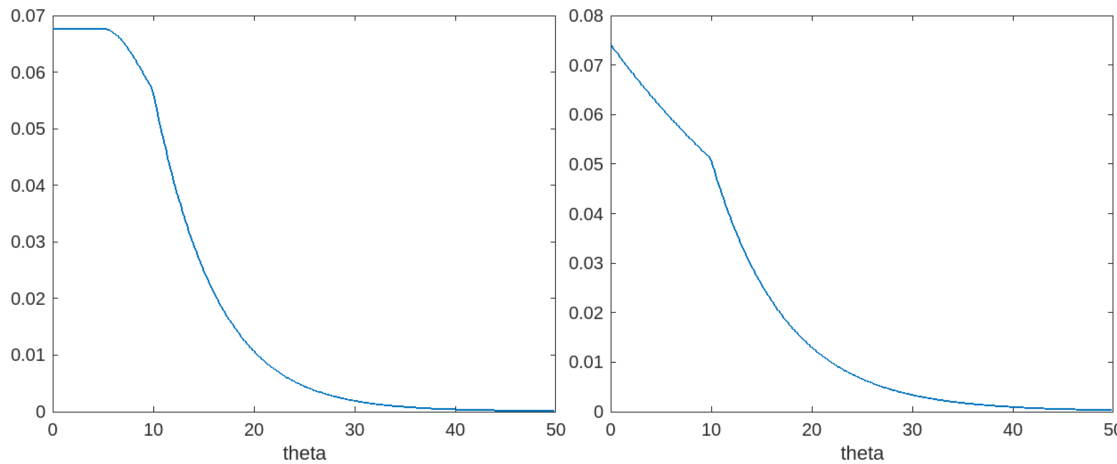

Figure 2.7: The function $\phi_\infty(\theta)$ when $\mu(\theta)$ is given by (**??**) and $\beta(\theta)$ is given by (**??**) (left) or $\beta(\theta) \equiv 1$ (right)

totic profile $\phi_\infty(\theta)$ (if it exists) must solve

$$\begin{cases} \phi'(\theta) = -\nu(\theta, p_0)\phi(\theta) - \mu(\theta)\phi(\theta), \qquad \theta > 0 \\ \phi(0) = 2\displaystyle\int_0^\infty \nu(\theta, p_0)\phi(\theta)\, d\theta \\ p_0 = \dfrac{m}{m-1}\left(\displaystyle\int_0^\infty V(\theta)\phi(\theta)\, d\theta\right)^{m-1} \end{cases} \tag{2.32}$$

and the following result is shown in the work of [**?**].

Proposition 2.3.1. *Assume that $m > 1$, that $\mu(\theta)$ satisfies $\int_0^\infty \mu(\theta)\, d\theta = \infty$ (every cell dies eventually) and that $\nu(\theta, p)$ satisfies* (**??**)*. Then* (**??**) *has a unique positive solution $\phi_\infty(\theta)$ if and only if*

$$\int_0^\infty \mu(\theta) e^{-\int_0^\theta \nu(s,0)+\mu(s)\, ds}\, d\theta < \frac{1}{2}. \tag{2.33}$$

In view of this proposition, we conjecture that (**??**) identifies the threshold that characterizes the long time behavior (expansion vs. regression) of the tumor mass and that when (**??**) holds, the long time dynamic of the tumor will be described by traveling wave solutions of (**??**)

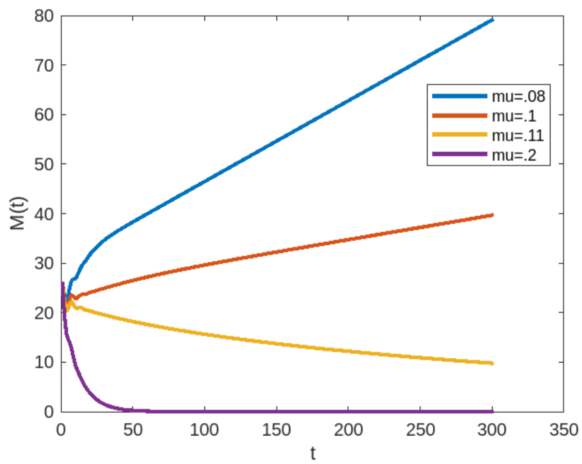

Figure 2.8: Evolution of the mass of the tumor when $\beta(\theta)$ is given by (**??**) and for different values of the death rate μ. The first two values of μ are below the threshold $\mu = 0.1076$ determined by condition (**??**), while the last two values are above.

which satisfy the boundary conditions

$$\lim_{x\to-\infty} \bar{p}(x) = p_0, \quad \lim_{x\to-\infty} \bar{n}(x,\theta) = \phi_\infty(\theta), \qquad \lim_{x\to+\infty} \bar{p}(x) = \lim_{x\to+\infty} \bar{n}(x,\theta) = 0.$$

When μ and $\nu(p) = \beta(p_M - p)$ are independent of θ, condition (**??**) is equivalent to $\mu < \beta\, p_M$ which is indeed a necessary and sufficient condition for the growth of the tumor. This is easy to show since in that case the equation for the density reduces to

$$\partial_t \rho - \Delta\rho^m = \rho(\beta p_M - \mu - p).$$

When $\nu(p,\theta) = \beta(\theta)(p_M - p)$ with $\beta(\theta)$ given by (**??**), Figure **??** shows the evolution of the mass of the tumor for different values of μ (independent of θ). The behavior is consistent with condition (**??**) which identifies $\mu = 0.1076$ as the threshold value when the death rate is constant.

2.4 Discussion

In this chapter, we developed an age-structured spatial model of tumor growth that incorporates the time daughter cells spend growing before their next mitosis. Extensions of this model in a few directions were investigated, such as one that allows the length of the cell cycle to depend on the local pressure, and a model that distinguishes the proliferating and quiescent subpopulations. The existence of traveling wave solutions to our model was examined, which is crucial in investigating the asymptotic behavior of the solution. The numerical simulations in the long-term supported the existence of an asymptotic age distribution. The simulations provided further insights into the growth pattern, speed of the growth of the population size, and the spatial distribution of mitosis. In particular, the two dimensional population growth was shown to scale quadratically with time, while it was linear in one dimension. This matches with the previous findings [13, 54] that the growth of radius in a radially symmetric tumor is linear. Mitosis events were concentrated in an annulus surrounding the saturated tumor core, rather than at the very outer edge. Finally, the effect of cell death was investigated, and this showed that sufficiently high death rates can eliminate the tumor, which is consistent with our understanding.

One observation about the growth pattern of tumor under this model was that the tumor core reached saturation, causing proliferation to occur outside of this region where pressure is lower. This resembles the experimental findings from multicellular spheroids, which often display a densely packed necrotic core surrounded by a proliferating rim [**?**, 8]. However, numerical simulations show that the density grows to a uniform saturation point and rapidly drops around the saturated core. This is a natural result following from our assumption on the pressure dependent proliferation rate, which is proportional to $p_M - p$ when age is fixed. This assumption was important for establishing the asymptotic behavior and for carrying out our analysis of the model, but by incorporating factors such as nutrient availability, richer and more interesting dynamics are expected.

In summary, we studied a mathematical model of tumor growth, combining age structures and spatial heterogeneity in this chapter. While we focused on untreated tumor growth in this chapter, the model can be used in several directions for future work. One natural extension is to add anti-tumor therapies. For example, by adding resistance structure on tumor, discussions about tumor responses to drugs can be made, not limited to the change in the tumor size, but also how age structure influences the emergence of drug resistance or the spatial distribution of the resistant cells. Thus, using our findings as a foundation, we extend this model to include chemotherapy and drug resistance in the following chapter.

Chapter 3: Evolution of Drug Resistance in Age-structured Tumors under Chemotherapy

3.1 Background

Drug resistance is a major barrier to successful chemotherapy. This resistance to drugs arises from numerous biological mechanisms such as genetic mutations, epigenetic alterations, tumor dormancy, and plasticity in phenotypes. Then these mechanisms limit treatment efficacy against the resistant cells. Along with the enhanced understanding of tumor heterogeneity both within tumors and across patients, the former models of tumor dynamics under treatments that assumed homogeneous sensitivity across tumor, developed into models that incorporate the biological structures which significantly affect drug resistance. These studies aim to capture treatment response and the evolution of drug resistance more accurately with the heterogeneity in drug resistance among tumor cell populations, and built the foundation of the age-structured model we introduce in the main sections.

3.1.1 Assumptions on Drug Efficacy

The classical modeling of the chemotherapy response in tumor originates from the Skipper-Schabel studies of leukemia in 1960s and 1970s. They proposed the log-kill hypothesis, which states that cytotoxic drugs eliminate a constant fraction of tumor cells in each course of treatment [?, ?]. Early mathematical models that considered drug interventions were built on this foundational hypothesis [?, ?]. In 1977, Norton and Simon refined this idea and suggested from experimental observations that the efficacy of chemotherapy is not constant. They proposed what is now known as the Norton-Simon hypothesis, which states that the rate of therapy induced mortality is proportional to the intrinsic growth rate of the tumor at its size [?].

One of the main difference between these two hypothesis is how the efficacy of chemotherapy relates to the tumor size and growth rate. Under the log-kill hypothesis, the sensitivity of tumor is independent of the tumor volume and its growth rate, while the idea of Norton and Simon asserts that rapidly growing tumors are more sensitive to treatments, which leads to the claim that drug efficacy decreases in a sufficiently large tumor under most growth laws that assume the saturation of tumor size. Considering that most cytotoxic chemotherapies tend to target proliferating cells and spare slow-cycling or dormant cells, the Norton-Simon hypothesis incorporates the biological observation that rapidly growing tumors are more vulnerable while the slowly cycling or quiescent tumors are more resistant. This shift in the theoretical assumption influenced the later mathematical models that include drugs and drug resistance in optimal control formulations [?], growth law based dose-response models [?], and many others, and remain central to modern approaches in the drug resistance models, and treatment optimization and scheduling.

3.1.2 Early models of drug resistance in tumor

Early mathematical models of drug resistance in cancer were developed from the seminal Luria Delbrück experiment. The discovery that mutation pre-exists in the cell population before exposure to drugs inspired mathematical models for estimating mutation rates in a cell population [**?**]. The work of Goldie and Coldman established the foundational drug resistance model in 1970s to 1980s, with a stochastic model that describes how resistant cell clones arise due to mutation during tumor growth. They assumed a constant mutation rate per cell division, and used a Poisson process to model the occurrence of mutation events. These assumptions led to their inferences that the probability of a resistant cell existing in the cell population saturates to 1 in a short time interval as the cell population grows, while the fraction of resistant cells in the tumor is not proportional to the tumor volume but varies, depending on the timing of the earliest mutation event [**?**]. Their models assumed exponential growth in the cell population, in both resistant and sensitive subpopulations. The probability of a resistant cell being present in the cell population was modeled by

$$P(N_R > 0) = 1 - e^{-\alpha N} \tag{3.1}$$

where N is the cell population, α is the mutation rate per cell division, and N_R is the resistant cell population.

This model formulation developed into the follow up work of Goldie and Coldman in 1983, where they incorporated therapies explicitly and explored how varying combinations and scheduling of multiple drugs affect treatment success [**?**]. An implication of their early models is that delayed therapy increases the likelihood of pre-existing resistance, and that alternating or combining treatments which tumors have no cross resistance to, increases the chance of eradicating the resistant subpopulations to individual treatment, leading to a successful eradication of tumor. These insights influenced clinical chemotherapy protocols, supporting early treatment initiation, alternating schedule therapy, and combination therapy.

The early deterministic approaches of drug resistance models, which stems from the earlier and simpler works of Swan (1987), Birkhead *et al.* (1987) and Birkhead and Gregory (1984) assumed two compartments, resistant and sensitive, and allowed mutations from the sensitive to resistant compartment, similar to the works of Goldie and Coldman [**?**, **?**, **?**]. The classical form of these models were given by the following equations:

$$\frac{dS}{dt} = (\beta_S - \kappa_S(D))S - \alpha S \tag{3.2}$$

$$\frac{dR}{dt} = (\beta_R - \kappa_R(D))R + \alpha S \tag{3.3}$$

where each of S and R represent the sensitive and resistant cell subpopulation respectively, β_S and β_R represent the growth rate of each cell subpopulation, κ_S and κ_R each models the killing rate of the drug typically dependent on the dose D, and α is the mutation rate modeled continuously. This model continues to be used widely in the studies of chemotherapy optimization [**?**,**?**].

In particular, the work of Swan in 1987 assumed Verhulst-Pearl equation type of tumor growth. Thus the tumor growth dynamics in this work follows the logistic growth law unlike

Coldman and Goldie's, allowing saturation of the population and slower growth as tumor grows. The growth rates assumed in his model were of the form

$$\beta_S = \beta_{S,0}\left(1 - \frac{S+R}{K}\right), \qquad \beta_R = \beta_{R,0}\left(1 - \frac{S+R}{K}\right) \tag{3.4}$$

where K is the carrying capacity of the total population, and $\beta_{i,0}$ is the intrinsic growth rate of the resistant and sensitive subpopulation for each $i = R, S$. This model has its advantage in distinguishing the resistant cells from the sensitive cells, in their growth rate and drug response in the simplest form. Moreover, the growth rate assumption made in this work shifted the attention of mathematical models of drug resistance in tumor to the competition between resistant and sensitive cells. This inspired the later models focused on competitive interactions between different cell subpopulations, such as Panetta (1996) which describes the competition between normal cells, resistant tumor cells, and sensitive tumor cells under drug administrations [**?**].

3.1.3 Spatially Heterogeneous Deterministic Models

Early ODE models of drug resistance treat tumors as a homogeneously mixed populations. However, clinical experiments demonstrated the crucial factors that influence resistance in tumor, which are often spatially heterogeneous. The tumor microenvironment is known to have complex interactions with tumor cells that are important in both successful treatments and design of anti-tumor therapies. Limited nutrition was shown to shape avascular tumor growth in different layers, which impacts the proliferative and quiescent states of tumor cells. Hypoxia has significant connection with treatment efficacy, due to its role in inducing drug resistance, including its impact on cell cycles, and the drug delivery issue caused by poor vascularization in the oxygen deprived environment. Spatially heterogeneous mathematical models of drug resistance were introduced to capture the impact of spatial variability in the dynamics of drug resistance in tumor.

Deterministic models of drug resistance that are spatially explicit commonly use the reaction-diffusion PDEs framework of (1.4). For instance, Jackson and Byrne in 2000 developed a model of rapidly dividing sensitive and resistant subpopulations that include drug diffusion, clearance, and uptake [**?**]. The basic form of these spatial models with sensitive and resistant subpopulation and drug concentration is given below:

$$\frac{\partial S}{\partial t} = D_S \Delta S + \beta_S(S,R) - \kappa_S(D)S - \delta_S S, \tag{3.5}$$

$$\frac{\partial R}{\partial t} = D_R \Delta R + \beta_R(S,R) - \kappa_R(D)R - \delta_R R, \tag{3.6}$$

$$\frac{\partial D}{\partial t} = D_D \Delta D - \delta_D D - \eta(S,R)D + u(t,x), \tag{3.7}$$

where S and R represents the sensitive and resistant cell population respectively, D is the drug concentration, D_i is the diffusion coefficients of S, R, and D for each $i = S, R$, and D, each of κ_i and β_i is the drug induced death and proliferation term for the resistant and sensitive population for $i = S$ and R respectively. δ_i is the natural death rate for each R and S, δ_D is the drug decay rate, η accounts for the drug lost by uptake, and $u(t,x)$ represents the drug influx. Cell mortality rate induced by drugs are often taken as a product of the drug uptake rate and the killing rate of therapy per unit drug.

Jackson and Byrne (2000) assumed a spherically symmetric tumor, with a moving boundary driven by cell proliferation and death. Additionally, the drug is assumed to enter through blood vessels, and thus the cells have different uptake rates depending on the location of the tumor. This is modeled using a blood-tissue transfer coefficient in the equation for drug, which is a function of the radial position in the tumor. The model of Jackson and Byrne reflects the delivery issue of the drug at the core of a tumor by incorporating drug penetration and vascular supply. They observed that the presence of resistant cells delay cure and accelerates tumor relapse, and changed the preferable strategy of treatment. In addition, their model allowed analysis of the how drug administration strategies, the degree of vascularization, and the tumor size influence treatment response.

3.1.4 PDE Models with Continuous Resistance Phenotype

While resistance can be considered binary, experimental observations suggest that tumor cells have a continuum of drug sensitivities, rather than being purely sensitive or resistant. Each tumor cell has continuously varying levels of drug efflux transporter expression, local vascularization, availability of nutrients and oxygen, and intracellular mechanisms such as DNA repair and altered metabolism, where all of these contribute to treatment tolerance. One challenge that the traditional discrete resistance compartmental models face, is that this gradual variations in tumor may be oversimplified. Better resolution is required in order to capture the resistance dynamics and the interaction between the environmental forces and tumor resistance in a more accurate way. This limitation can be addressed by including a continuous resistance phenotype variable in the model instead of using discrete compartments to represent resistance levels.

A class of spatially homogeneous PDE models that incorporate resistance as a continuous variable generally describe the cell density $n(t,\theta)$ by the following equation form:

$$\frac{\partial}{\partial_t} n(t,\theta) = [R(\theta,\rho(t)) - C(t,\theta) - d(\theta)]n(t,\theta) + \int R(\theta',\rho(t))M(\theta',\theta)n(t,\theta')\, d\theta', \quad (3.8)$$

where θ represents the resistance phenotype of a cell, $\rho(t) = \int n(t,\theta)\, d\theta$ is the total cell density, R is the intrinsic growth rate depending on the resistance level of the cell, C is the mortality rate induced by drugs, d is the natural death rate, and $M(\theta',\theta)$ is the probability of mutation from resistance level θ' to θ. Depending on the biological assumptions, studies use modified functional forms of the growth, drug-response, or mutation kernel to account for the mechanisms in the model. The mutation term is also often chosen as $D_\theta \Delta_\theta n$ that represents the variation in resistance due to mutation, plasticity and epigenetic alteration. This takes account of the biological assumption that small epimutations are much more frequent than large epimutations. Such choice has an advantage in mathematical analysis, and allows to avoid choosing the often unknown mutation kernel M.

Lorz *et al.* investigated the evolutionary dynamics of drug resistance and therapy effects in 2013 with a PDE model having cell adaptations and mutations [**?**]. They model resistance as a continuous phenotype variable, and thus the epigenetic adaptations and mutations are included as continuous events rather than jumps in discrete steps. The model in [**?**] is shown to reflect the biological observation that therapeutic pressure shapes the resistance dynamics in tumors.

The model (3.8) can be extended to spatially heterogeneous models that include a continuous resistance variable. The general form of the equation for cell density $n(t, x, \theta)$ in such models is given by:

$$\frac{\partial n}{\partial t} = D_x \Delta_x n + [R(t,x,\theta) - C(t,x,\theta) - d(t,x,\theta)]n + \int R(t,x,\theta')M(\theta',\theta)n(t,x,\theta')\, d\theta' \tag{3.9}$$

where the mutation term represented as the integral are often replaced by $D_\theta \Delta_\theta$, and D_x is the diffusion coefficient in space.

Cho and Levy investigate the drug resistance evolution and various drug administration strategies with a radially symmetric tumor assumption. Their model incorporates mutations, and spatial diffusion of drugs and nutrients [**?**]. They observe that higher drug dose induces stronger selection of more resistant traits, and on-off therapy schedules also select more resistant traits than continuous schedules. The work [**?**] was extended to a resistance evolution work in 2018 [**?**]. This is the model on which Chapter 3 largely builds on. Although mutations are not incorporated in this model, the model is freed from the radially symmetric assumption and fixed boundary by choosing 2-dimensional space, and the resistance selection force shown in this model is more clearly attributable to the selection force coming from chemotherapies.

3.1.5 Cell Cycle and Age Structured Models

As clinical outcomes accumulated with various chemotherapies, it became increasingly clear that many cytotoxic drugs eliminate cells that are actively dividing, but fail to eradicate slow-cycling or quiescent tumor cells. These survivors who persist through treatments have proven in experimental observations that they may later resume proliferation, driving tumor recurrence after treatment. By the time of relapse, the tumor has enhanced drug resistance, which contributes to therapeutic failure. These biological observations motivated the development of mathematical models that incorporate cell cycles or cell ages to better describe the tumor response to drugs and resistance dynamics.

The idea of growth fraction in mathematical model began in 1960s, in the work of Skipper, Schable, and Wilcox which distinguished the proliferative state from non-cycling cells [**?**]. This was later developed in Panetta (1997) where proliferating cells are taken as more sensitive cells than the resistant quiescent cells [**?**]. The model aims to study therapeutic scheduling strategies, and also find that the treatment outcomes were heavily sensitive to the proliferating fraction in the tumor. Page and Uhr (2005) distinguish the quiescent cell subpopulation from the proliferating ones. Their work includes antibody in their model to study how the interaction between tumor cells and antibody leads to tumor dormancy [**?**]. Numerous mathematical models have been using this framework of two compartmental model of quiescent and proliferating subpopulations, which is essentially almost similar to the compartmental model of resistant and sensitive subpopulations.

However, most cytotoxic agents have varying efficacy across much finer subdivisions of the cell cycle than the simple quiescent and proliferating states. As most mathematical models that distinguished growth fraction, used this framework to account for the resistance gained in the non-proliferating cells, incorporating explicit cell cycle structure in the model was only a natural choice. Kimmel and Axelrod establishes cell cycle branching process with compartments to investigate various drug scheduling strategies and suggest designing drugs based on the cell cycle distribution of tumor, while addressing the main barrier of utilizing cell cycles in modeling that experimental observation of such distribution under chemotherapies is difficult to obtain [**?**]. In a later work, based on the biological observation that cancer cells undergo hypoxia induced quiescence, Alarcon, Byrne, and Maini constructed a cell cycle model with local ODE defined on each lattice points, where cell cycle is modeled as a continuous variable and the progression of cell cycle interacts with cyclin-dependent kinase (CDK) and p27 [**?**].

In 1987, Webb focuses on the resonance phenomenon expected to be produced by matching the timing of chemotherapy with when the target phase aligns with tumor cells but not normal cells, using that tumor cells tend to have longer cycle lengths compared to the normal ones. This motivates the age-size structured cell population models, where cells are allowed for heterogeneous size and their progresses through cell cycles are modeled by transition functions distinct for normal and cancer cells, which depend on the cell age [**?**, **?**]. While the presumed synchronization of cell cycles among normal or cancer cells where cytotoxic drug affects seems challenging to adapt clinically, these works contributed significantly to the development of age-structured models. This later develops into models that pairs age and cell cycle together to better explain cell cycle transition, such as the later work of Dyson and Villella-Bressan and Webb [**?**], and the models that focus on the heterogeneous impact of chemotherapy on cell cycle phases [**?**, **?**].

3.2 Model Equations

In this section, we present our model equations for the tumor cell density $n(t,x,\theta,\tau)$ at age θ and resistance level τ. The cell age ranges from 0 to ∞. The resistance level to cytotoxic drug ranges from 0 to 1, where $\tau = 0$ represents no resistance (fully sensitive) and $\tau = 1$ represents maximum resistance. The cell population density is defined as

$$\rho(t,x) = \int_0^\infty \int_0^1 n(t,x,\theta,\tau)\, d\theta\, d\tau. \tag{3.10}$$

The pressure is given by

$$p(t,x) = \frac{m}{m-1}\rho^{m-1}(t,x) \tag{3.11}$$

for $m > 1$ as in Chapter 2.

The cell density $n(t,x,\theta,\tau)$ at age θ and resistance level τ solves the following system:

$$\begin{cases} \partial_t n + \partial_\theta n = -F(t,x,\theta,\tau)n + \nu_n \Delta n + \nu_p \nabla\cdot(n\nabla p) & x\in\mathbb{R}^d,\ \theta>0,\ t>0 \\ n(t,x,0,\tau) = 2\int_0^\infty R(t,x,\theta,\tau)n(t,x,\theta,\tau)\, d\theta & x\in\mathbb{R}^d,\ \ t>0 \end{cases} \tag{3.12}$$

Here, $F(t,x,\theta,\tau)$ characterizes the outflux from the density n at a certain age and resistance level at each time and location, due to proliferation and death. It is defined as

$$F(t,x,\theta,\tau) = R(t,x,\theta,\tau) + D(t,x,\theta,\tau) + C(t,x,\theta,\tau) \tag{3.13}$$

where $R, D,$ and C represents the proliferation rate, natural death rate, and the drug induced death rate respectively. The intrinsic net growth rate $g(t,x,\theta,\tau),$ in the absence of inhibition in growth from high pressure, is defined as

$$g(t,x,\theta,\tau) = r(t,x,\theta,\tau) - D(t,x,\theta,\tau) - C(t,x,\theta,\tau). \tag{3.14}$$

where r is the baseline proliferation rate in the absence of inhibition in proliferation due to high pressure.

The homeostatic pressure p_M is taken as $p_M = \frac{m}{m-1}$ to normalize the maximum packing density $\rho_M(\geq \rho)$ as 1. The inhibition in proliferation and thus in net growth is assumed to happen when pressure p is higher than this homeostatic pressure p_M.

3.2.1 Functional Specifications

The proliferation rate R is taken as

$$R(t,x,\theta,\tau) = h(p,g)r(t,x,\theta,\tau) + (1 - h(p,g))\{D(t,x,\theta,\tau) + C(t,x,\theta,\tau)\}. \tag{3.15}$$

The function $r(t, x, \theta, \tau)$ in (3.15) is given by

$$r(t, x, \theta, \tau) = \beta(\theta) \frac{\phi(\tau)}{1 + \mu_2 c_2(t, x)} s(t, x) \tag{3.16}$$

where $\beta(\theta)$ explains the age dependence of r.

The function $h(p, g)$ in (3.15) is the indicator function

$$h(p, g) = 1 - H(p - p_M)H(g) \tag{3.17}$$

defined with a Heavyside function $H(\cdot)$, and depends on the pressure p and the baseline growth rate $g(t, x, \theta, \tau)$. The indicator function $h(p, g)$ is 0 when the baseline growth rate is non-negative ($g(t, x, \theta, \tau) \geq 0$) and the pressure at this location is too high ($p \geq p_M$) at the same time. The role of h is in restricting the net growth of tumor in the overcrowded regions. The proliferation rate is bounded above by $D + C$ whenever $p > p_M$ (when $h = 0$). Thus the net growth $R - D - C$ is non-positive in regions under high pressure. Meanwhile, in the regions where $p < p_M$, the proliferation rate R equals the baseline rate r.

Thus the proliferation rate R depends on the nutrients $s(t, x)$ and chemostatic drug $c_2(t, x)$, cell age θ, and the consumption of nutrients $\phi(\tau)$ and chemostatic drug μ_2 by tumor cells, under the condition $p < p_M$.

Following the assumption in [**?**], $\phi'(\theta) < 0$ is assumed. This assumption is based on the biological evidences that resistant cells tend to pay a fitness cost, by using more resources in developing resistance programs than the sensitive cells. [**?**, **?**] This is observed as a slower growth of the resistant colony in drug-free environment, due to the shift of resource allocation away from biomass production. The proliferation rate function in (3.15) is a modification from [**?**] by letting proliferation rate depend on the cell age.

For the numerical simulation, the nutrient uptake function is taken as

$$\phi(\tau) = \frac{a_1}{1 + a_2 \tau^5} \tag{3.18}$$

The uptake functions of both nutrient and cytotoxic drug are taken from the functions modeled in [**?**]. The age dependence $\beta(\theta)$ is normalized to take values in $[0, 1)$ so that a_1 represents the maximum proliferation rate per unit nutrient in a unit time. The parameter a_2 is a positive constant and controls the effect of resistance level on the proliferation rate.

The death rate D is taken as

$$D(t, x, \theta, \tau) = d(\theta)\rho(t, x). \tag{3.19}$$

Thus D is proportional to the total density at the location in a fixed age, and is allowed to depend on the cell age by $d(\theta)$. No explicit dependence of the death rate on the drug resistance level of cells is assumed.

The death induced by cytotoxic drug is taken as

$$C(t,x,\theta,\tau) = \mu_1(\tau, c_1)c_1(t,x) \tag{3.20}$$

where μ_1 is the drug uptake function and c_1 is the amount of cytotoxic drug. The drug uptake function is modeled as

$$\mu_1(\tau, c_1) = \frac{b_1}{1 + b_2(c_1)\tau^{b_3(c_1)}}. \tag{3.21}$$

Here, b_1 is the maximum death rate per unit cytotoxic drug in a unit time. The parameters b_2, b_3 are positive functions of the amount of cytotoxic drug c_1. The uptake function μ_2 of the cytostatic drug c_2 is taken as a constant.

The computational domain for the numerical simulation is taken as a two dimensional space $[0,1]^2$. The region where tumor cells are present is identified as $\Omega_0(t) = \{x \in [0,1]^2 \mid \rho(t,x) \leq \bar{\rho}_0\}$ where $\bar{\rho}_0$ is a critical cell density level. The nutrients and drugs are modeled as the steady state solutions of the following equations:

$$\begin{aligned}
\nu_s \Delta s(t,x) &= \left[\gamma_s + \int_0^1 \int_0^\infty \phi(\tau) n(t,x,\theta,\tau)\, d\theta d\tau\right] s(t,x) \\
\nu_{c_1} \Delta c_1(t,x) &= \left[\gamma_{c_1} + \int_0^1 \int_0^\infty \mu_1(\tau) n(t,x,\theta,\tau)\, d\theta d\tau\right] c_1(t,x) \\
\nu_{c_2} \Delta c_2(t,x) &= \left[\gamma_{c_2} + \int_0^1 \int_0^\infty \mu_2(\tau) n(t,x,\theta,\tau)\, d\theta d\tau\right] c_2(t,x)
\end{aligned} \tag{3.22}$$

Here, ν_s, ν_{c_1} and ν_{c_2} are diffusion constants and γ_r, γ_{c_1}, and γ_{c_2} are decay constants of nutrients, cytotoxic drug, and cytostatic drug respectively.

Two different boundary conditions for the nutrients and drugs are considered. A constant infusion at $\partial\Omega_0$ case is given below:

$$s(t,x)|_{\partial\Omega_0} = S_0^b, \quad C_1(t,x)|_{\partial\Omega_0} = C_1^b, \quad C_2(t,x)|_{\partial\Omega_0} = C_2^b \tag{3.23}$$

The other case is when the nutrients and drugs diffuse from the one of the boundaries of the computational domain. Without loss of generality, we impose diffusion from the top of the domain($x_1 \in [0,1]$ and $x_2 = 1$) with Dirichlet boundary conditions.

$$s(t,x_1,1) = S_0^b, \quad c_1(t,x_1,1) = C_1^b, \quad c_2(t,x_1,1) = C_2^b \qquad x_1 \in [0,1] \tag{3.24}$$

In both cases, S_0^b, C_1^b, and C_2^b are constants. Faster diffusion is assumed outside of Ω_0 than in Ω_0 where tumor is present with cell density above $\bar{\rho}_0$. At a location where tumor cells are not present, $n(t,x,\theta,\tau) = 0$, and the uptake of nutrients and drugs do not occur. Allowing for different diffusion constants ($\nu_s^b, \nu_{c_1}^b, \nu_{c_2}^b$) than in Ω_0, drugs and nutrients outside of Ω_0 solve the following equations:

$$\begin{aligned} \nu_s^b \Delta s(t,x) &= \gamma_s s(t,x) \\ \nu_{c_1}^b \Delta c_1(t,x) &= \gamma_{c_1} c_1(t,x) \\ \nu_{c_2}^b \Delta c_2(t,x) &= \gamma_{c_2} c_2(t,x) \end{aligned} \tag{3.25}$$

which match with (3.22).

3.3 Numerical Results

3.3.1 Numerical Settings

For the purpose of simulation, the age variable θ was restricted in $[0, 10]$, with $d\theta = 0.25$. The cells at the maximum age 10 stays without aging further, but were kept in the simulation at the maximum age until their deaths. We note that the dependence of the birth and death rates on age in our simulation becomes negligible at the numerical maximum age. Also, the age distribution converges to show that cells are concentrated at the younger ages, which will be shown in the subsection 3.3.4.

Initial tumor in the simulation is given by the following equation:

$$n(0,x,\theta,\tau) = \begin{cases} \rho_0 \left(1 - \left(1 + \exp\left(-\frac{\|x-x_c\|^2 - r_0^2}{\epsilon_x}\right)\right)^{-1}\right) Q_0(\tau,x) & \text{if } \theta = 7.5 \\ 0 & \text{otherwise} \end{cases} \tag{3.26}$$

where x_c is the center of the domain, r_0 determines the radius of the initial tumor, ρ_0 is the maximum magnitude at the center, and ϵ_x determines the sharpness at the tumor boundary.

The tumor population consists of two subpopulations having distinct resistance levels, $\tau =$

0.2 and 0.8. We take the initial tumor to be divided in four regions, by letting

$$Q_0(0.2, x) = \begin{cases} 1 & x \in [25, 50]^2 \cup [0, 25]^2 \\ 0 & \text{otherwise} \end{cases} \tag{3.27}$$

and

$$Q_0(0.8, x) = \begin{cases} 0 & x \in [25, 50]^2 \cup [0, 25]^2 \\ 1 & \text{otherwise.} \end{cases} \tag{3.28}$$

An example of the graphical representation of the indicator function Q_0 is illustrated in Figure 3.1, along with the initial tumor density and average age prescribed initially. A circular shape of initial tumor is assumed. We also experiment with an initial tumor that is spatially uniform in the resistance distribution. This corresponds to the case when $Q_0(\tau, x) = Q_0 = \frac{1}{2}$ in $[0.2, 0.8] \times \mathbb{R}^2$. The population density distribution and age distribution in this case have no difference from what is illustrated in Figure 3.1.

The critical cell density level $\bar{\rho}_0$ that decides Ω_0 and the boundary of the tumor is taken to be 10^{-3}. For the other parameters in (3.26), $\rho_0 = 10$, $\epsilon_x = 0.01$ and $r_0 = 0.05$ were taken in the subsection 3.3.2. In 3.3.3, we take $\epsilon_x = 0.03$ to observe the dynamics when tumor diffuses toward the upper boundary of the domain where nutrients and drug are highly concentrated under the Dirichlet boundary assumption in shorter time. The choice of the age of cells at time $t = 0$ ($n(0, x, \theta, \tau)$) is concentrated at the age 7.5 and 7.75 to see proliferation faster, but no impact on the simulation results in the long term were observed even if the initial age distribution was uniform or concentrated at different ages.

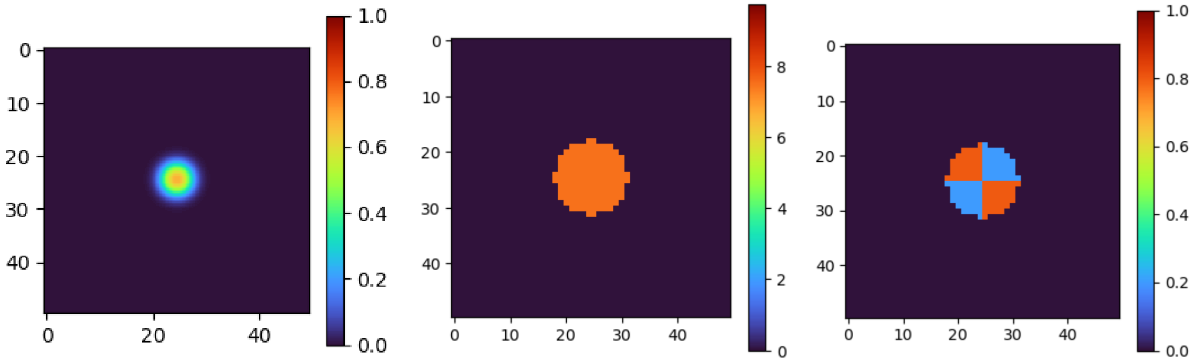

Figure 3.1: The figure on the left shows $\rho(t,x)$ at $t=0$. The middle figure shows the average age across the tumor $\frac{\int_0^\infty \theta \int_0^1 n(t,x,\theta,\tau)\, d\tau d\theta}{\rho(t,x)}$, in the region where $\rho(t,x) > 10^{-3}$. The average resistance level across the tumor $\frac{\int_0^1 \tau \int_0^\infty n(t,x,\theta,\tau)\, d\theta d\tau}{\rho(t,x)}$ is given in the figure on the right, in the region where $\rho(t,x) > 10^{-3}$.

The parameters in (3.22) that concern the nutrients and drug levels in Ω_0 are taken as $\gamma_s = \gamma_{c_1} = \gamma_{c_2} = 1$ and $\nu_s = \nu_{c_1} = \nu_{c_2} = 0.08$. Allowing for faster diffusion of nutrients and drugs outside Ω_0, we take $\nu_s^b = \nu_{c_1}^b = \nu_{c_2}^b = 1.25$ in (3.25). In both the constant (3.23) and Dirichlet boundary condition (3.24), we take $S_0^b = 12$ for nutrients, $C_2^b = 0$ for cytostatic drug and varied the cytotoxic drug levels at $C_1^b = 0, 2, 4, 6, 8$.

The parameters in $\phi(\tau)$ and $\mu_1(\tau, c_1)$ are chosen as $a_1 = 0.22, a_2 = 1.2$, and $b_1 = 1.5$, along with the functions $b_2(c_1) = 5 + 1.25c_1$ and $b_3(c_1) = 2 + 0.5c_1$, which follow the same setting from [**?**]. They are carefully chosen to follow the Norton-Simon hypothesis [**?**] that the rate of regression of tumor under chemotherapy is proportional to the natural growth rate of tumor. This is represented by the following relation in our model:

$$\frac{a_1}{1+a_2} \propto \frac{b_1}{1+b_2}.$$

Further, the assumption in [?] that the average proliferation rate and the average mortality rate due to the effect of cytotoxic drug satisfies

$$\int_0^1 \phi(\tau)\, d\tau \approx 0.2, \quad \int_0^1 \mu_1(\tau, c_1)\, d\tau \approx 0.8.$$

is met. The uptake of cytostatic drug happens at a constant rate 0.8.

The function $\beta(\theta)$ in (3.16) is modeled as

$$\beta(\theta) = \begin{cases} 0 & \text{if } \theta \geq \theta_0 \\ \frac{(\theta-\theta_0)^2}{\sigma(2\sigma^2+2\sigma(\theta-\theta_0)+(\theta-\theta_0)^2)} & \text{if } \theta \geq \theta_0 \end{cases} \tag{3.29}$$

where $\theta_0 = 2$ and $\sigma = 1$. The parameter d in $d(\theta)$ is taken as

$$d(\theta) = \begin{cases} 1.25 & \text{if } \theta < 2.5 \\ 0 & \text{if } \theta \geq 2.5 \end{cases} \tag{3.30}$$

3.3.2 Tumor Growth

The growth of tumor without any intervention of drugs is discussed in this subsection. We first observe the dynamics when the initial data n and ρ are described in (3.26) and Figure 3.1 respectively. It is assumed that $Q_0 = \frac{1}{2}$, implying that no spatial heterogeneity was given at time 0. Dirichlet boundary condition around Ω_0, given in (3.24), is assumed for the nutrient distribution.

Figure 3.2 describes how cell population density $\rho(t, x)$ changes over time. Tumor grows in high density at the upper parts of the tumor, where nutrients are concentrated. The increase in the size of tumor pushes the bottom of the tumor to the region where nutrients are scarce, which inhibits the growth of cell density at the bottom. As the diameter of tumor increases, slower diffusion through the dense tumor region limits nutrient availability at the core, causing lower cell density at the core of the tumor after enough time. The distribution of nutrients is also depicted in the Figure 3.3 to support this growth pattern.

The proliferation events as the density $\int_0^1 n(t, x, 0, \tau)\, d\tau$, shown in Figure 3.4, shows more than the growth pattern observed by the change in $\rho(t, x)$. Regions near the top of the tumor are where mitosis events are highly concentrated until this region reaches high enough pressure due to overcrowding of cells. Then the proliferation becomes inhibited at these regions, causing discontinuities in the number of mitosis events at these points, shown as the discontinuities near the top of the tumor at time $t = 80$ and 100 in Figure 3.4. This is a result of the discontinuity in the proliferation rate function R at the point where pressure reaches the homeostatic pressure level ($p = p_M$). However, mitosis events are less prevalent at the bottom of the tumor overall, which agrees with the observation made in the growth of total cell population density.

Furthermore, the regions where mitosis events are most prevalent are not located at the most outer rim of the tumor. The discontinuity in the proliferation rate at the pressure level $p = p_M$ limits mitosis events at where mitosis would have been most prevalent, if not for the discontinuities. These regions are located near but not at the top outer edge of the tumor, and is precisely where the tumor is most densely populated, as shown in the Figure 3.2.

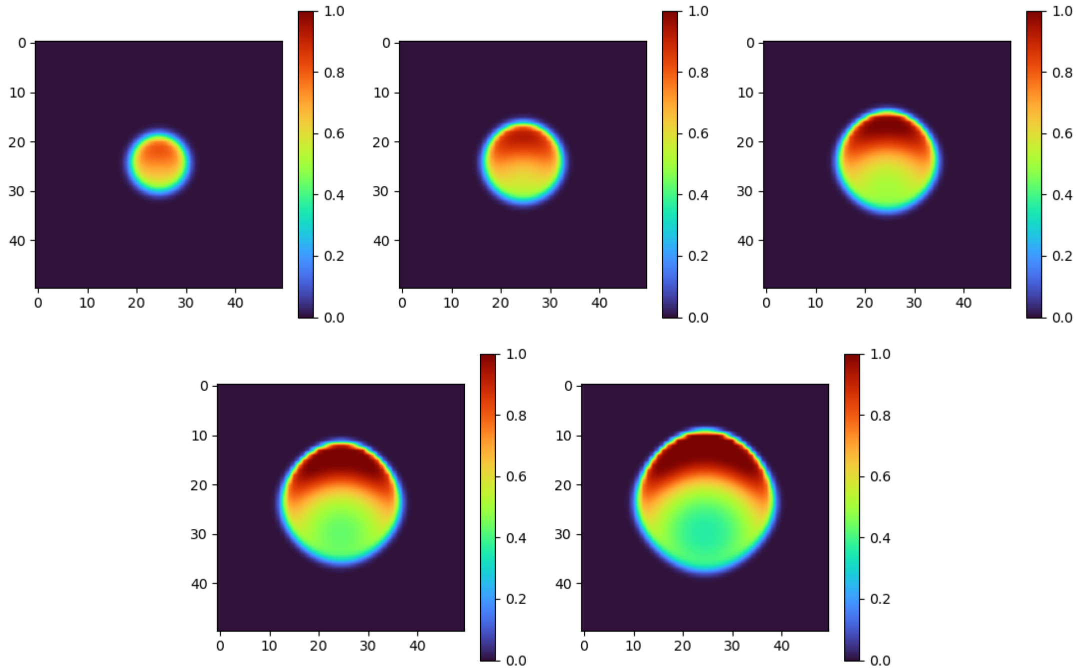

Figure 3.2: Growth of tumor cell population density $\rho(t, x)$ when nutrient diffuses from the top of the domain at time $t = 20, 40, 60, 80, 100$. The initial resistance distribution is uniform in space.

Now we take Q_0 as in (3.27) and (3.28) for the two resistant subpopulations. Nutrient environment is assumed to be constant around the tumor region Ω_0. The nutrient level ($S_0 = 7$) in this case is adjusted to be lower than the influx level ($S_0 = 12$) that was assigned for the Dirichlet condition. This adjustment is coming from the observation that at the constant nutrient

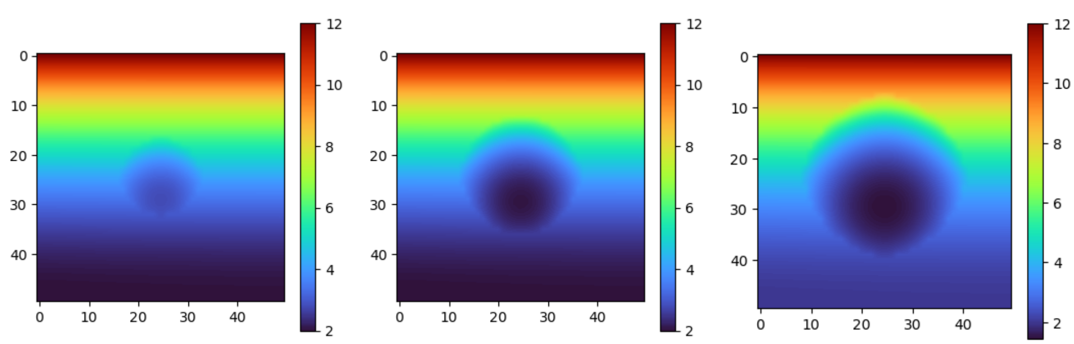

Figure 3.3: Nutrient distribution $S(t, x)$ at time $t = 20, 60, 100$ when the tumor grows as in Figure 3.2. The initial resistance distribution is uniform in space.

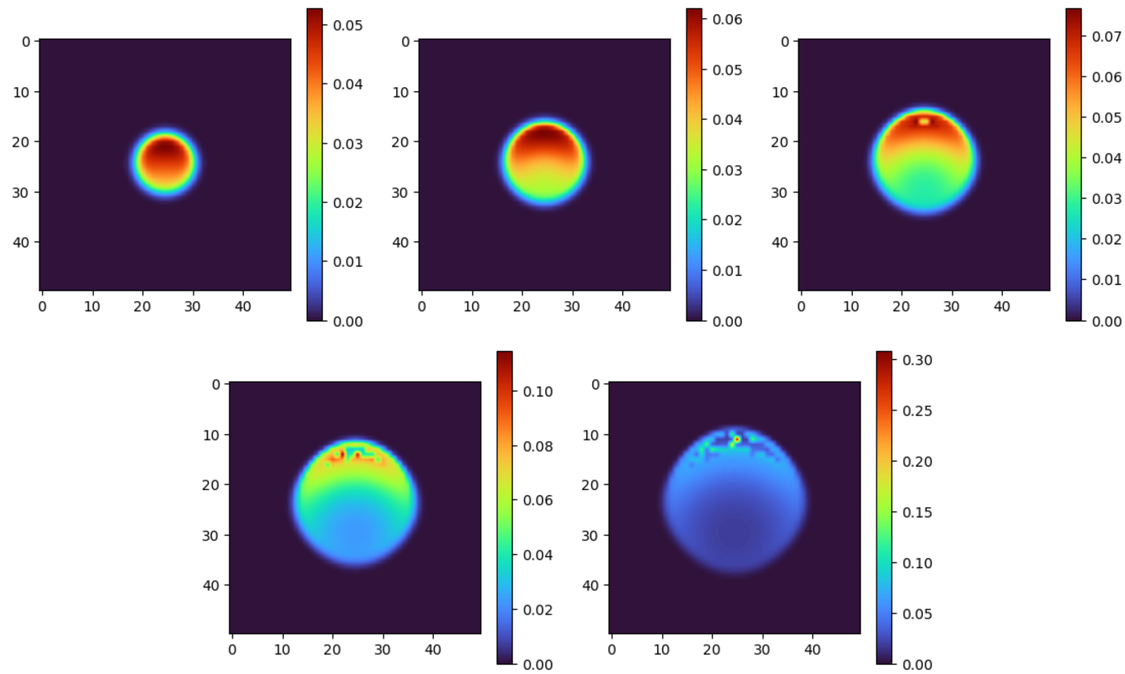

Figure 3.4: Proliferation of the tumor, given by the density $\int_0^1 n(t, x, 0, \tau)\ d\tau$ at time $t = 20, 40, 60, 80, 100$ when nutrient diffuses from the top of the domain.

level $S_0 = 12$ around Ω_0, nutrient supply is abundant enough for both the resistant and sensitive subpopulations. This eliminates any visible difference in the growth between the resistance levels (which becomes available through different speeds of nutrients uptake), and limits interesting growth dynamics.

The growth in cell population density ρ is illustrated in Figure 3.5. After time $t = 60$, nutrients reach tumor core in low density due to the grown diameter of the tumor, as shown in Figure 3.6 which describes how nutrient distribution $S(t, x)$ change over time. The nutrient uptake of more resistant cells are assumed to be slower, and thus the decrease in cell density at the core of the resistant cells is observed more evidently, and also faster in time. Regions with the highest cell density, regardless of the resistance level, appears slightly inside from the tumor edge.

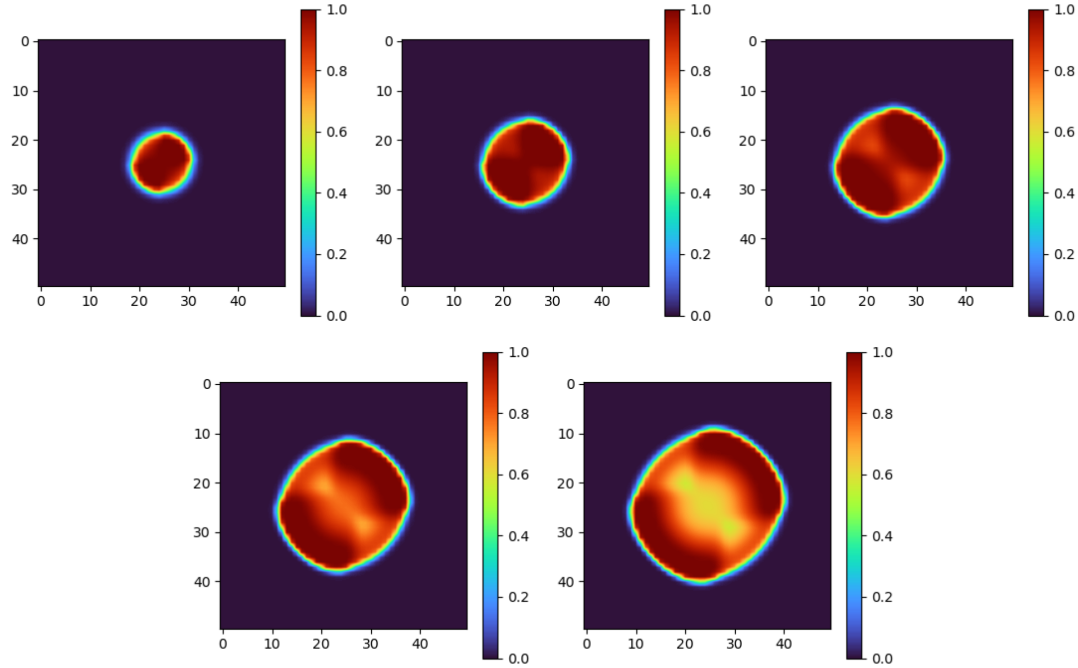

Figure 3.5: Growth of cell population density $\rho(t, x)$ for time $t = 20, 40, 60, 80, 100$ when nutrient and drug are assumed to be constants outside of Ω_0. The initial settings in density, age distribution, and resistant subpopulations are illustrated in Figure 3.1.

Mitosis events across the tumor are tracked by the density $\int_0^1 n(t, x, 0, \tau)\, d\tau$, and are shown in Figure 3.7. In the initial phases, the sensitive cells proliferate much more actively compared to the more resistant cells. The saturation in the sensitive cell population density as shown in the Figure 3.5 from time $t = 20$ limits proliferation in this region. However, locations in the tumor where proliferation is most intense, marked as red in the mitosis event figures, are where the sensitive population is. Agreeing with the observations made about the growth in tumor core from Figure 3.5, nutrient deficiency restricts mitosis events at the tumor core.

Starting from the initial distribution of spatially heterogeneous drug resistance of cells, how resistance affects population growth under the absence of drugs is also studied. Figure 3.8 shows the mean resistance level over time. The sensitive population, represented by the blue regions

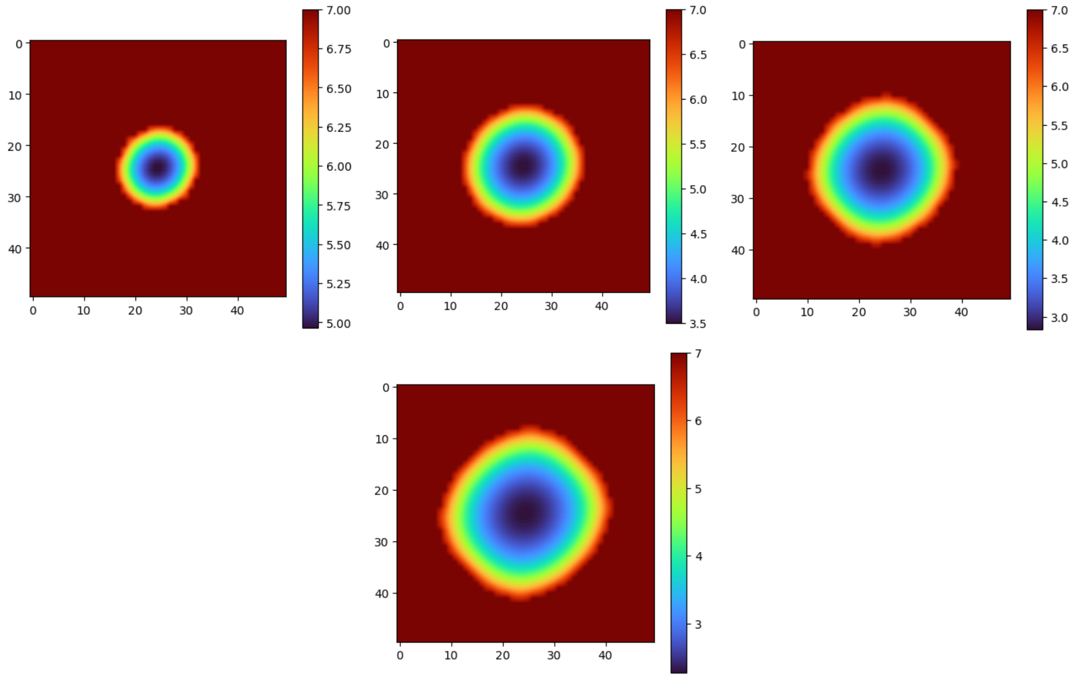

Figure 3.6: Nutrient environment $S(t, x)$ at time $t = 20, 60, 80, 100$ when the tumor cell population density is given in the Figure 3.5. Constant nutrient environment is assumed outside of Ω_0.

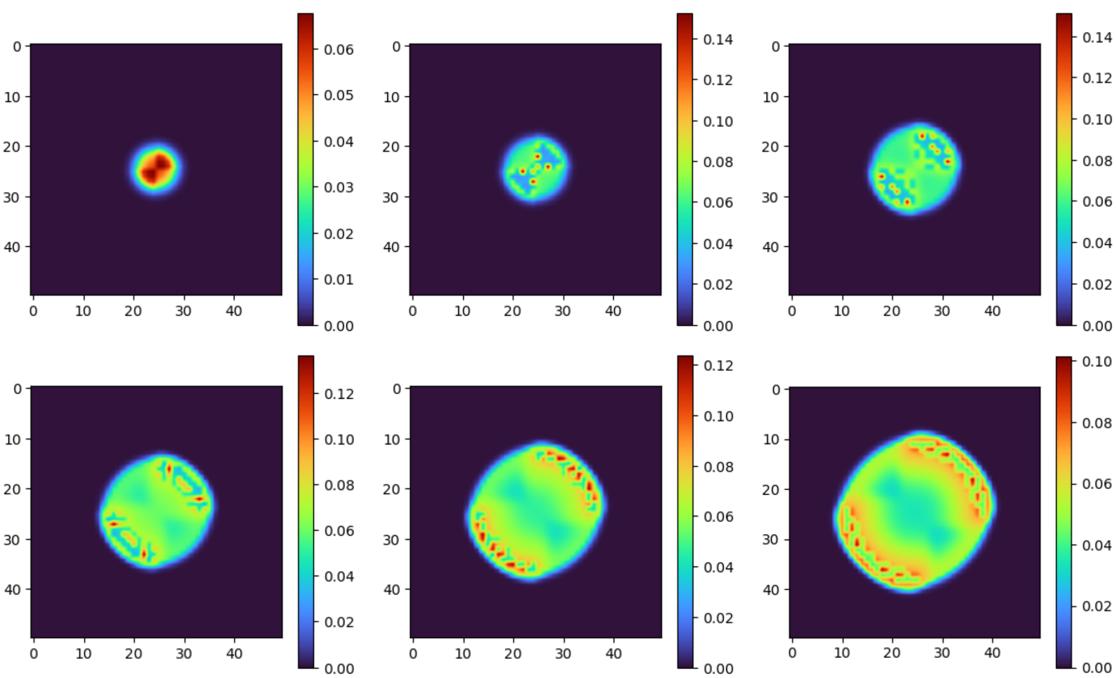

Figure 3.7: Proliferation of the tumor, represented by the density $\int_0^1 n(t, x, 0, \tau)\ d\tau$ at time $t = 5, 20, 40, 60, 80, 100$ when $S(t, x)$ is constant outside of Ω_0.

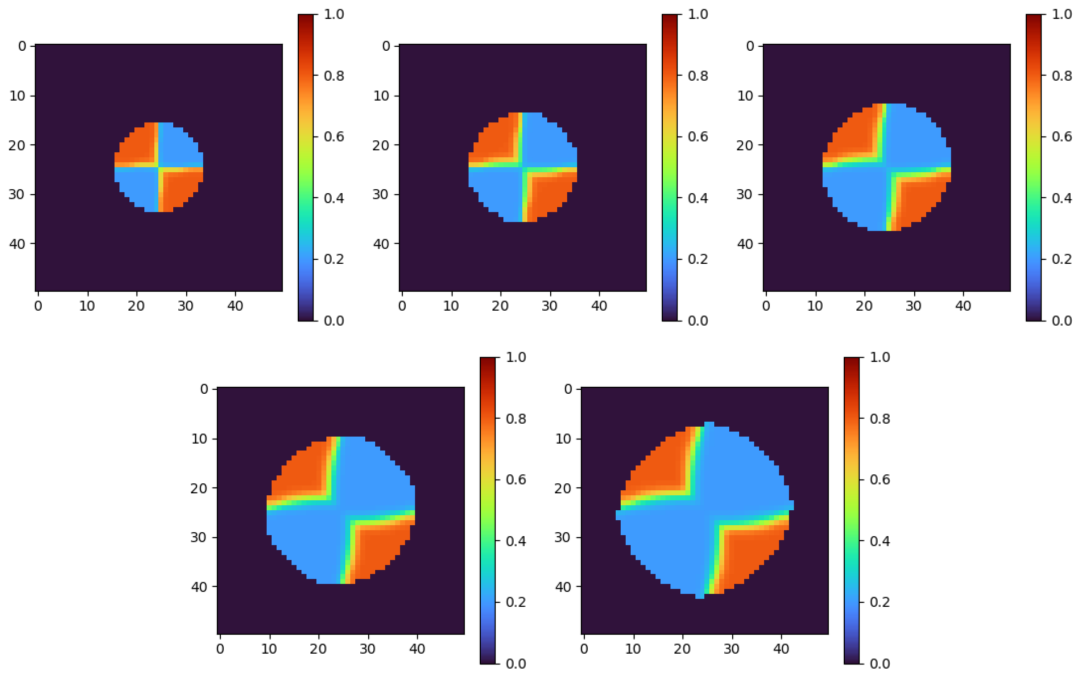

Figure 3.8: Mean resistance level across the tumor at time $t = 20, 40, 60, 80, 100$ when $S(t, x)$ is constant outside of Ω_0.

in the figure, grows to take over and push away the resistant population. This is explained by the dependence of nutrient uptake rate function ϕ on the drug resistance level of a cell, defined in (3.18). Less nutrients are invested in proliferation of more resistant cells, leading to slower growth of these cells. This matches with the biological observation that the resistant cells are likely to pay fitness costs.

Figure 3.9 and Figure 3.10 show how average ages of the sensitive and resistant population across the tumor change over time. Average age is calculated by $\frac{\int_0^\infty \theta\, n(t,x,\theta,\tau_0)\, d\theta}{\int_0^\infty n(t,x,\theta,\tau_0)\, d\theta}$ where $\int_0^\infty n(t, x, \theta, \tau_0)\, d\theta > 10^{-3}$ and is defined as 0 otherwise, for any fixed resistance level τ_0. The average age of resistant cells increased gradually in the tumor core. On the other hand, sensitive cells were older in average at the locations where mitosis was inhibited by high local pressure. This matches with the observations made from the Figure 3.7 about where and in which subpop-

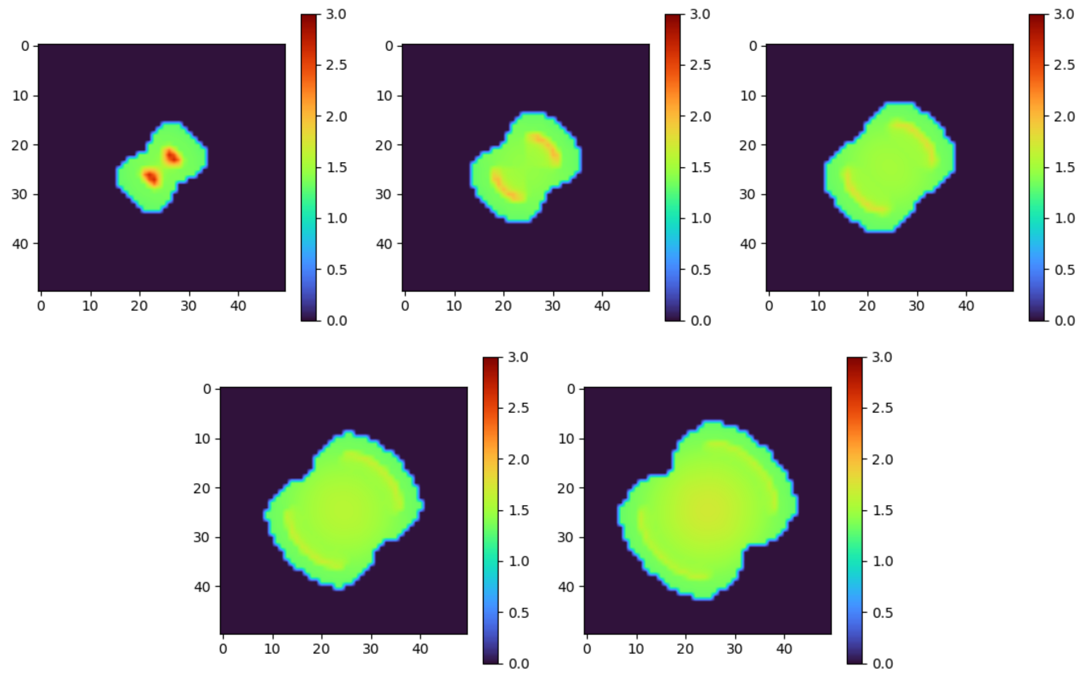

Figure 3.9: Average age of the sensitive (drug resistance level given as $\tau = 0.2$) cells at time $t = 20, 40, 60, 80, 100$ when $S(t, x)$ is constant outside of Ω_0.

ulation the mitosis events were mainly happening.

Finally, we let Q_0 as in (3.27) and (3.28) for the two resistance levels among the tumor cells, and assume Dirichlet boundary condition for the nutrients. The growth in cell population density ρ is shown in Figure 3.11. The nutrient concentration changing from plenty to scarce as we traverse the space vertically downward, the growth of both resistant and sensitive cells .

The nutrient distribution figures are not displayed here, since how nutrients diffuse under the same boundary condition given a similar mass of tumor was already shown in Figure 3.3.

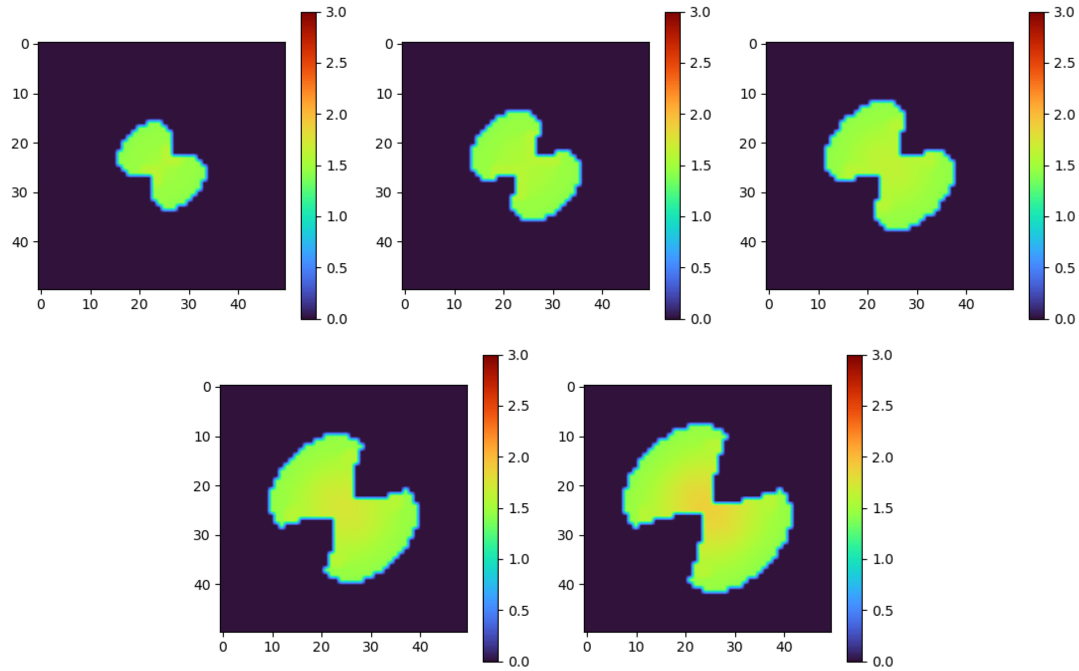

Figure 3.10: Average age of the resistant (drug resistance level given as $\tau = 0.8$) cells at time $t = 20, 40, 60, 80, 100$ when $S(t, x)$ is constant outside of Ω_0.

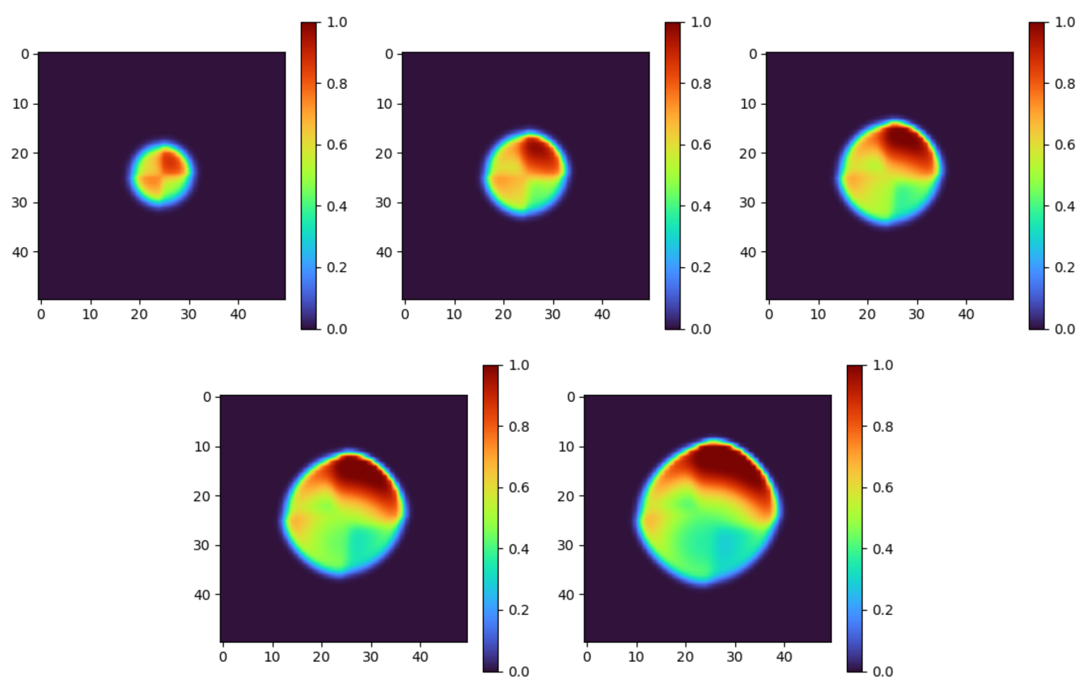

Figure 3.11: Growth of cell population density $\rho(t, x)$ for $t = 20, 40, 60, 80, 100$.

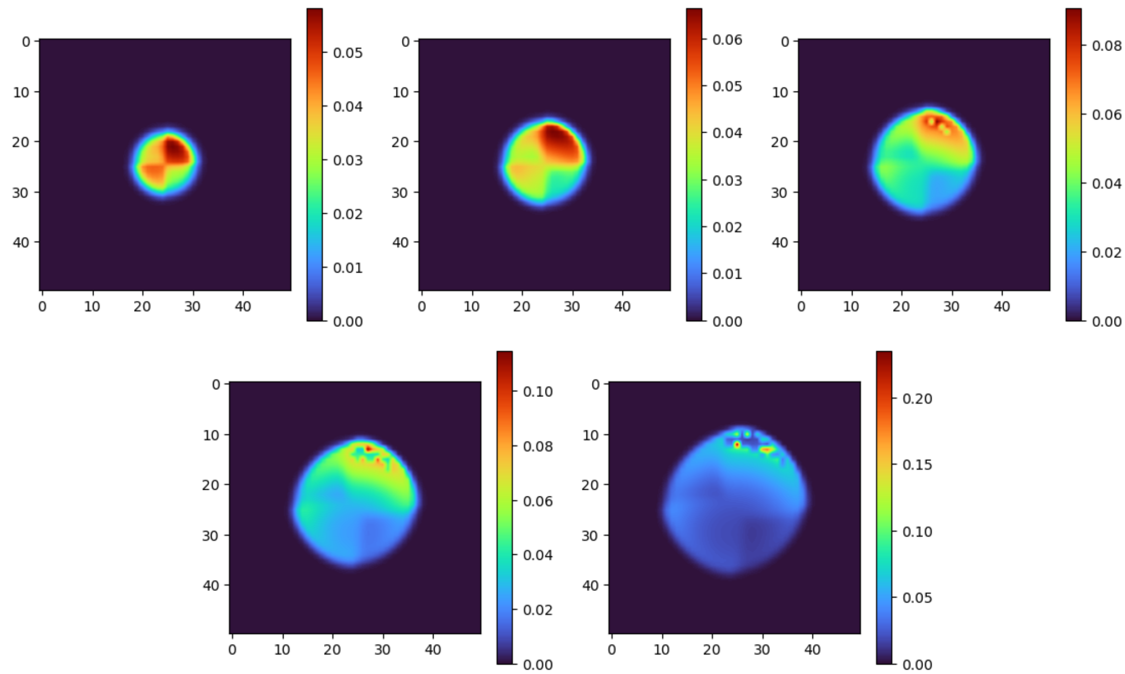

Figure 3.12: Proliferation of the tumor in Figure 3.11, given by the density $\int_0^1 n(t,x,0,\tau)\,d\tau$ at time $t = 20, 40, 60, 80, 100$.

3.3.3 Responses to Chemotherapy and Progression of Drug Resistance in Tumor

In this subsection, $n(0,x,\theta,\tau)$ varies according to the spatial distribution of resistance within the tumor, while the total cell population density $\rho(0,x)$ is kept identical across all simulations. To simulate treatment dynamics, the initial tumor illustrated in Figure 3.13 is first allowed to grow until $t = 30$ without drugs. The resulting state is then taken as the starting point ($t = 0$) for therapies. The subsequent tumor evolution under drug administration is then observed at times $t \in [0, 50]$. The initial tumor was taken slightly different than in section 3.3.2, to observe the dynamics when tumor is affected by high concentration of both drugs and nutrients in shorter time.

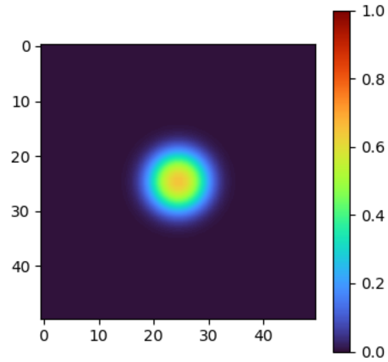

Figure 3.13: Initial tumor cell population density $\rho(0, x)$.

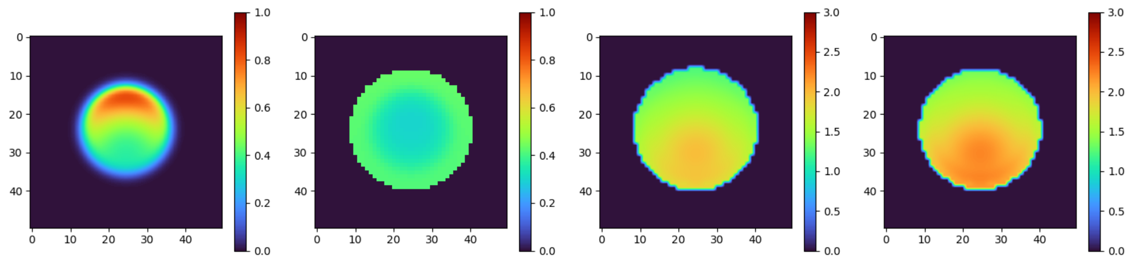

Figure 3.14: From the left to right: Cell population density ρ, mean resistance level $\frac{\int_0^1 \tau \int_0^\infty n(0,x,\theta,\tau)\, d\theta d\tau}{\rho(0,x)}$ across the tumor, average age of the sensitive ($\tau = 0.2$) subpopulation, and average age of the resistant ($\tau = 0.8$) subpopulation of the tumor before the injection of cytotoxic drug.

3.3.3.1 Spatially Homogeneous Initial Resistance with Nutrient and Drug Diffusion from the Top Boundary

We first observe the case when the resistance level begins homogeneous in space and both the cytotoxic drug and nutrients diffuse from the top of the domain. The tumor grown for time length 30 from the cell density shown in Figure 3.13, is illustrated in the Figure 3.14.

The tumor was next observed under different cytotoxic treatments, where the intensity of the drug varied with boundary concentration $C_1^b = 2, 4, 6$, and 8. The resulting cell population density $\rho(t, x)$ under each treatment is illustrated in Figure 3.15. Although the drug diffuses from the top boundary, the higher nutrient concentration in the upper region of the domain continues to promote stronger tumor growth near the top. The effect of cytotoxic drug becomes more apparent as we increase the drug dosage.

The spatial distribution of mitosis events shown in Figure 3.16 explains the concentration of tumor density in the upper region, as proliferation occurs predominantly where nutrients are abundant. For the lower dose ($C_1^b = 2$), the maximum number of proliferation events even increased by $t = 50$. Under $C_1^b = 4, 6$, decrease in this number was observed initially, followed by a rebound, reflecting continued expansion into the regions where nutrient and cytotoxic drug are rich. A substantial decrease in proliferation was observed only at the highest dose $C_1^b = 8$.

Figure 3.17 illustrates the mean resistance level across the tumor, showing how the resistance distribution evolves under different drug concentrations. At the lowest drug dose ($C_1^b = 2$) and at time $t = 50$, the mean resistance is close to the drug resistance level (0.8) of the resistant cells throughout an outer annulus that surrounds the inner region of lower resistance. Under our assumption that nutrients and drugs diffuse more slowly though densely populated tumor regions, this pattern suggest that, when nutrients are sufficient and drug diffusion is limited, resistant cells may form a protective barrier around sensitive cells, reducing drug penetration into the inner core. At higher dosages, resistant cells rapidly dominated the entire tumor, with the takeover occurring faster at increased drug dosages.

The average ages of resistant and sensitive subpopulations are shown in Figures 3.18 and 3.19 respectively, for each cytotoxic drug dosage. Both sensitive and resistant cells at all drug dosages, consistently show an older average age in a region positioned slightly inward from the

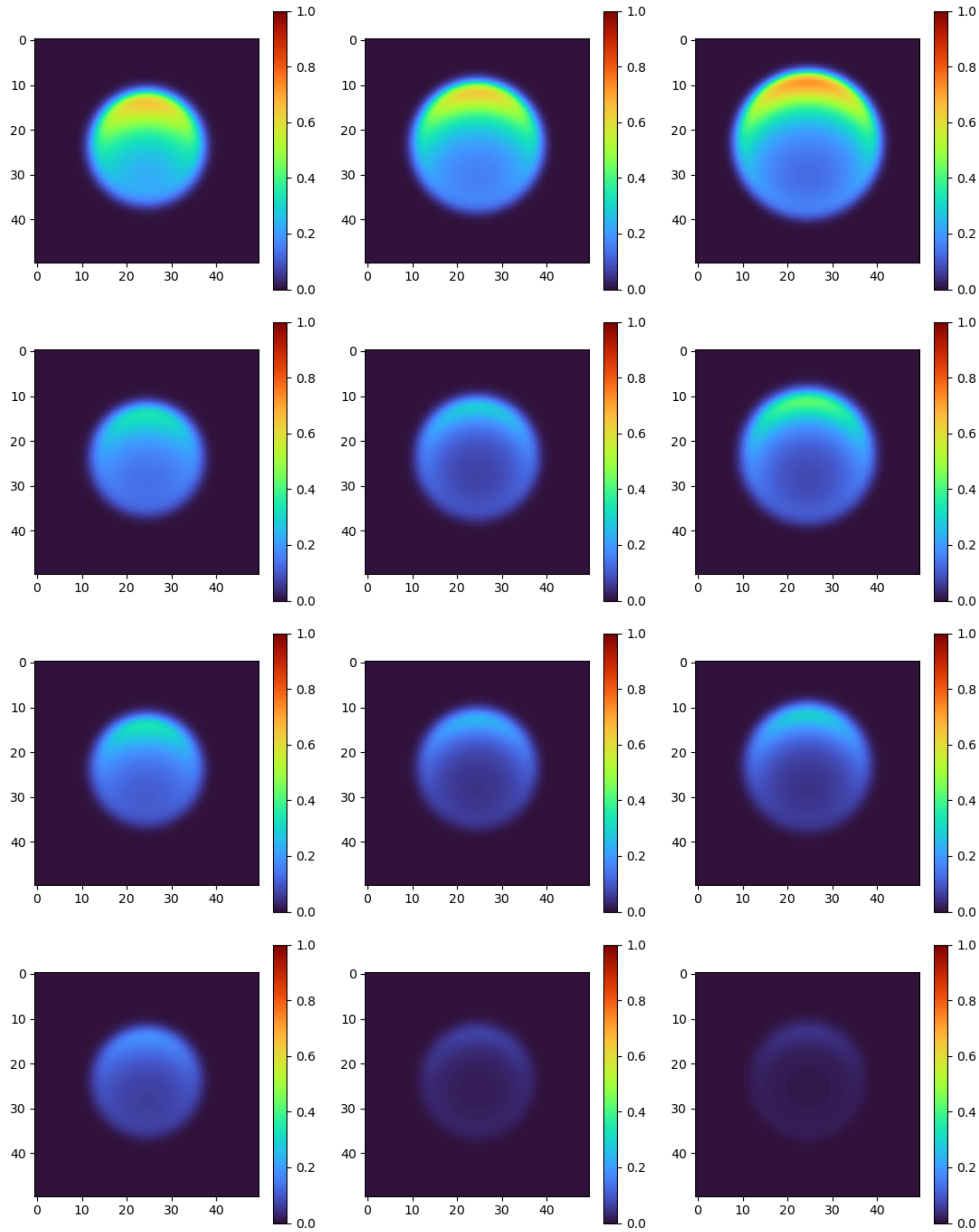

Figure 3.15: Cell population density $\rho(t, x)$ at times (left to right) $t = 10, 30, 50$ when the nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

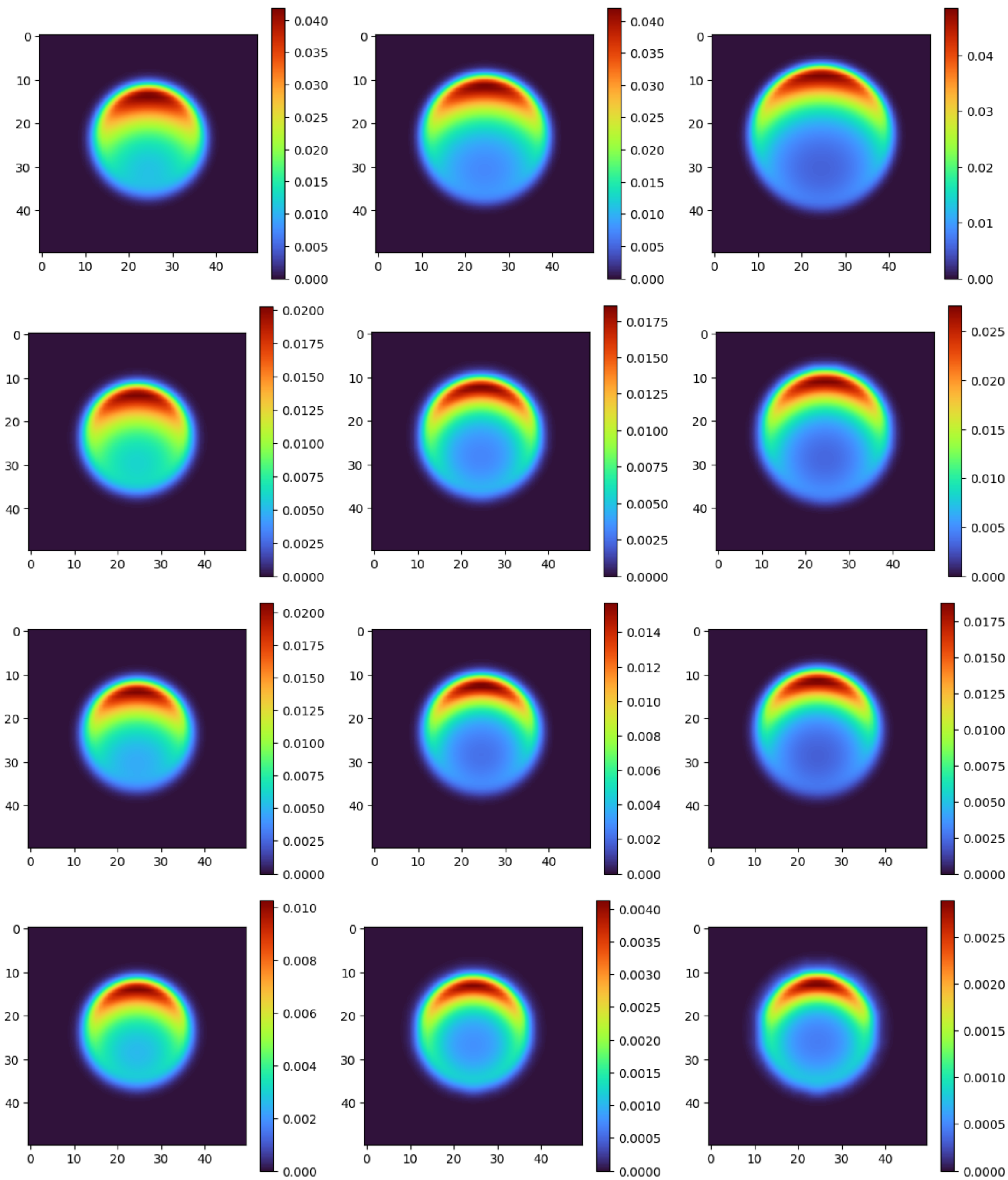

Figure 3.16: Proliferation of the tumor, given by the density $\int_0^1 n(t, x, 0, \tau)\ d\tau$ at times $t = 10, 30, 50$ when the nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

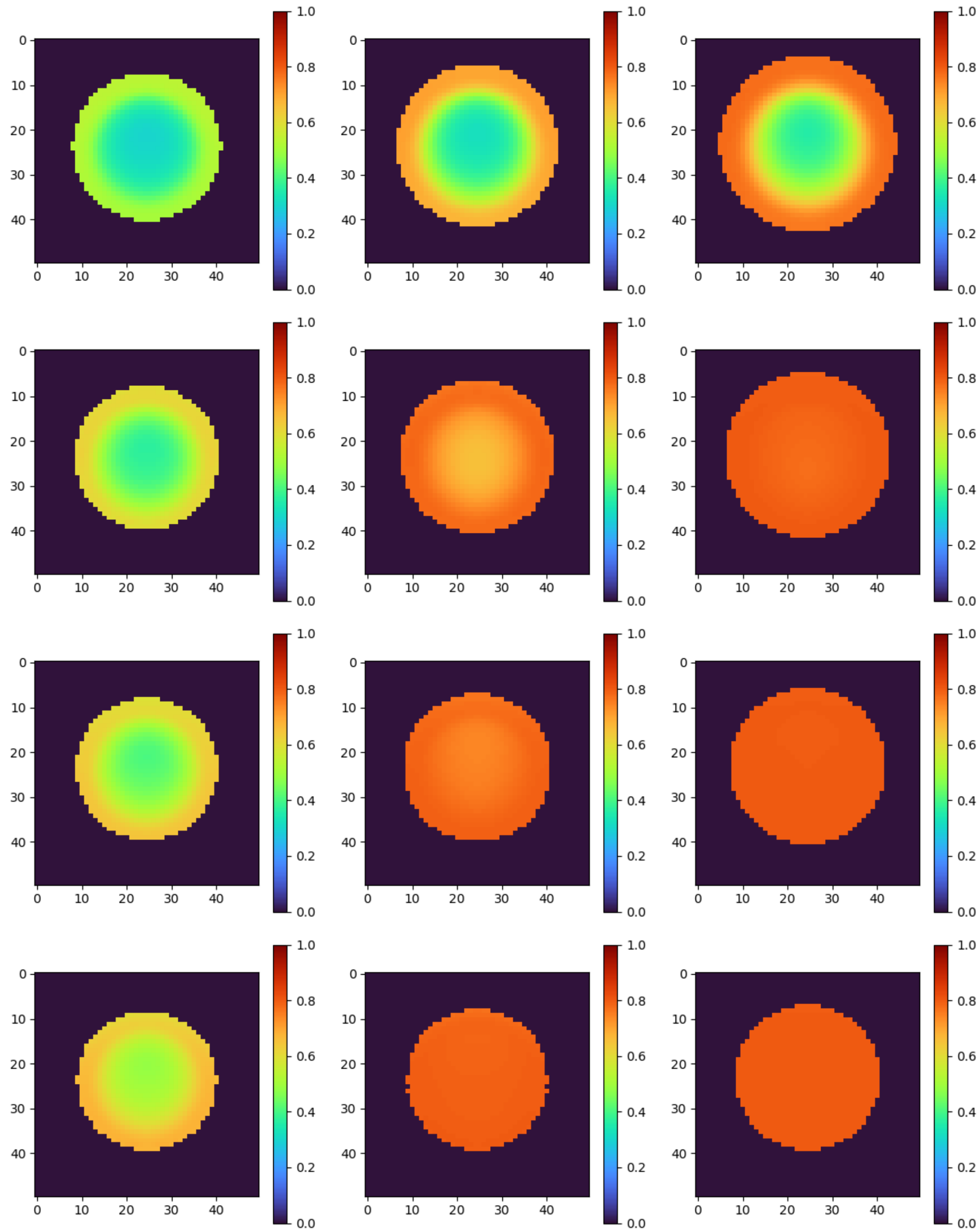

Figure 3.17: Mean resistance level across the tumor at times $t = 10, 30, 50$ when the nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

lower edge of the tumor, where nutrient and drug concentrations are lowest. The higher average age in this region is likely a consequence of both reduced drug induced mortality and fewer proliferation events. As drug dosage increases, the existing sensitive cells die out faster so that the average age is maintained younger throughout the tumor before their clearance. In addition, at higher drug concentrations, the population of sensitive cells decreases in a retracting manner, rather than uniform clearance. Sensitive cells localize to areas where both nutrient availability and drug exposure are low.

A marked difference in behavior is observed in the resistant cells. Although the overall spatial pattern of the average age does not change noticeably across resistance levels, increasing the cytotoxic drug dosage has little effect on the average age distribution of resistant cells. The decrease in maximum average age of the resistant cells is very minimal as drug level increases. Additionally, the average age of resistant cells are consistently older than that of sensitive cells across all cytotoxic drug doses. This can be attributed to the survival advantage of resistant cells under cytotoxic treatment, combined with their proliferation disadvantage.

The total cell population under each drug dosage is shown in Figure 3.20. Tumor rebound is observed when $C_1^b = 2, 4$ and 6 while rebound in the population occurs faster under lower drug dosage. The initial decrease in the tumor volume becomes faster when the cytotoxic drug levels are higher. Under $C_1^b = 8$, the total population only decreases until $t = 50$.

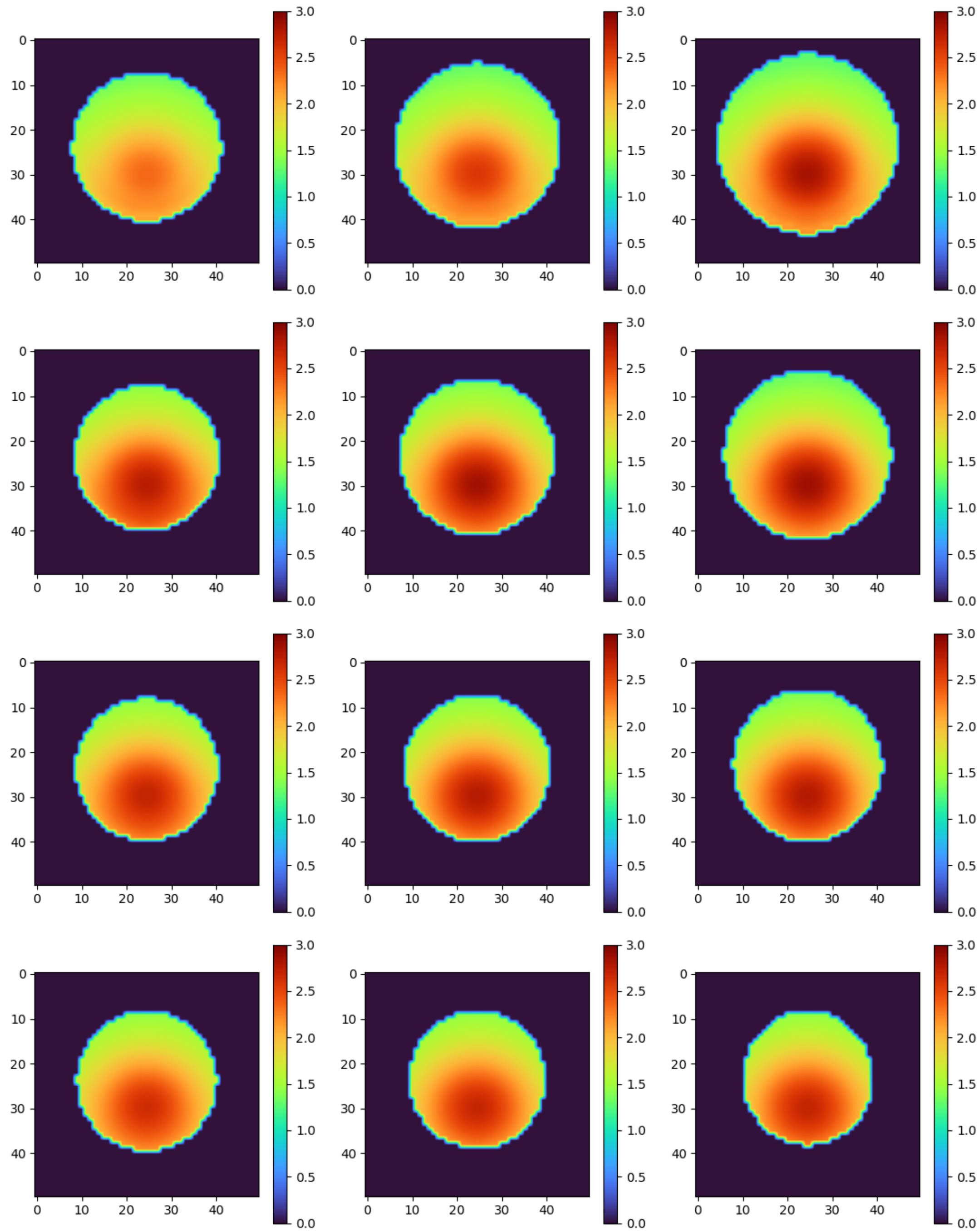

Figure 3.18: Average age of the resistant cells ($\tau = 0.8$) at times $t = 10, 30, 50$ when nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

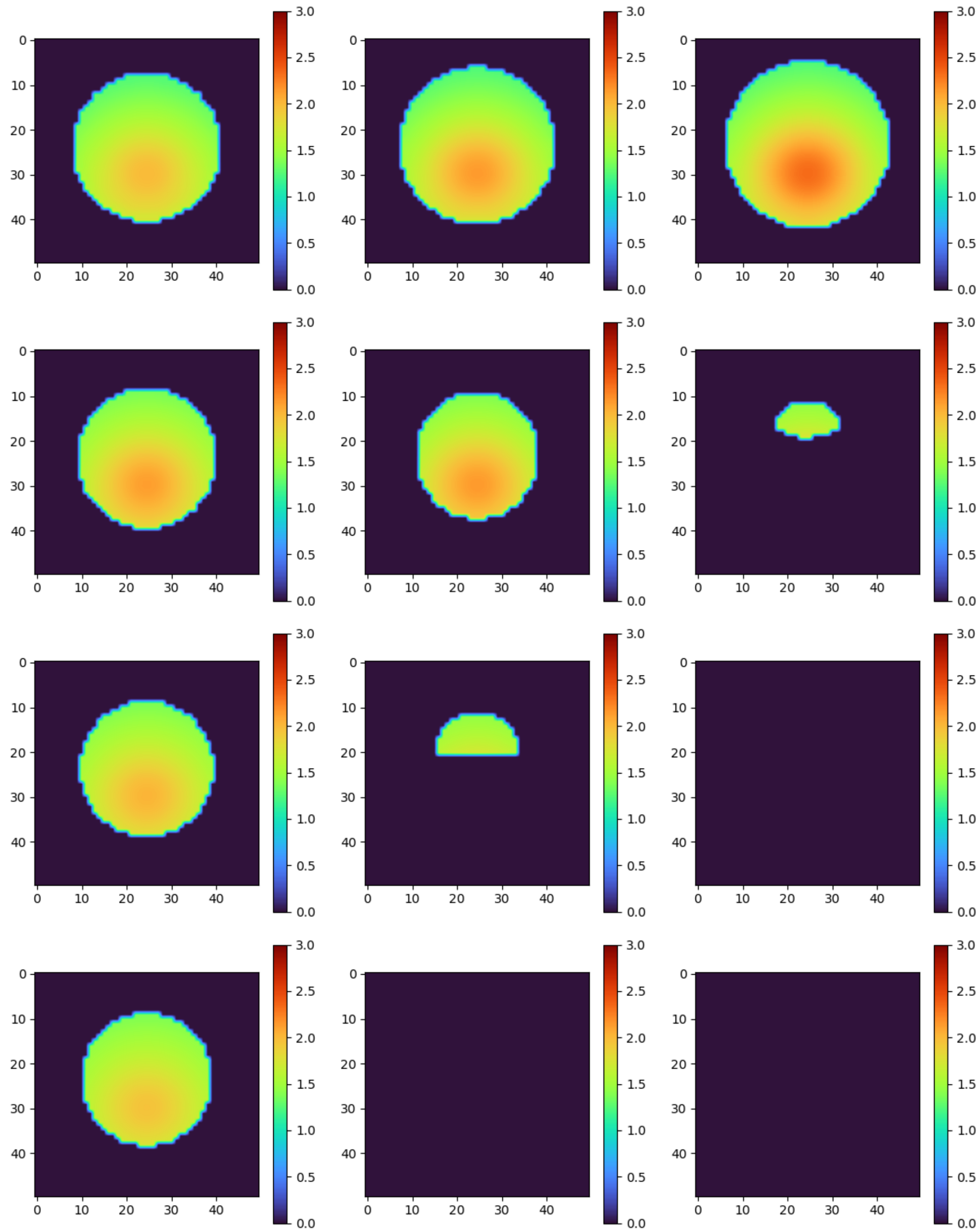

Figure 3.19: Average age of the sensitive cells ($\tau = 0.2$) at times $t = 10, 30, 50$ when nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

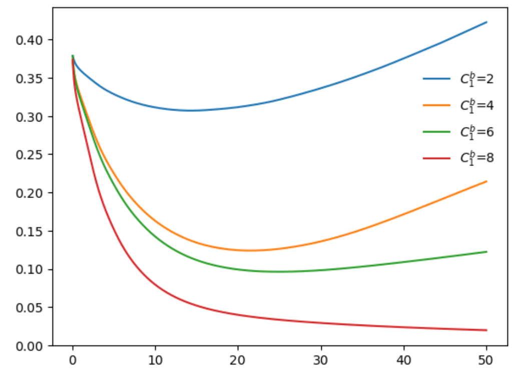

Figure 3.20: Total cell population defined by $\int_{\mathbb{R}^2} \rho(t,x)\, dx$ when nutrient and drug diffuse from the top of the domain, under varying cytotoxic drug levels $C_1^b = 2, 4, 6, 8$.

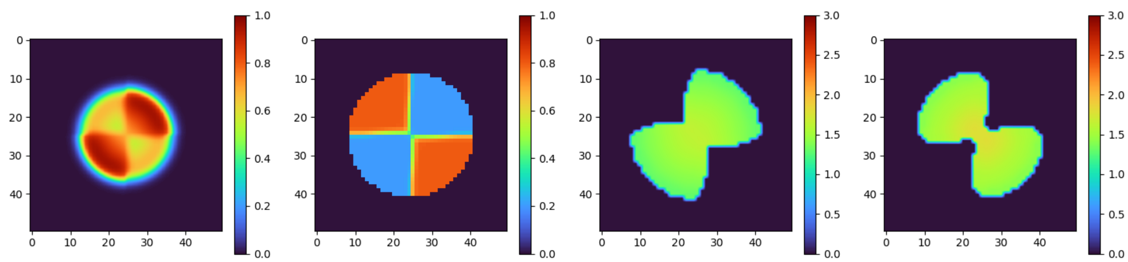

Figure 3.21: Tumor before treatment when constant nutrient distribution is assumed around the tumor. From the left to right: Cell population density ρ, mean resistance level $\frac{\int_0^1 \tau \int_0^\infty n(0,x,\theta,\tau)\, d\theta d\tau}{\rho(0,x)}$ across the tumor, average age of the sensitive ($\tau = 0.2$) subpopulation, and average age of the resistant ($\tau = 0.8$) subpopulation of the tumor before the injection of cytotoxic drug.

3.3.3.2 Spatially Heterogeneous Initial Resistance with Constant Nutrient and Drug around Ω_0

In the next case, the tumor is initialized with a spatially heterogeneous resistance distribution prescribed by the equations (3.27) and (3.28). Both the nutrient and cytotoxic drug concentrations are assumed to be constant outside of the tumor region Ω_0. Starting from the tumor cell density shown in Figure 3.13, tumor grows for 30 time units without drugs initially. The resulting cell population density, mean resistance level, and average ages of resistant and sensitive cells of the tumor before the cytotoxic drug treatment are illustrated in Figure 3.21.

The tumor under varying intensities of the constant concentrations of cytotoxic drug $C_1^b = 2, 4, 6,$ and 8 around Ω_0, is observed. The cell population density $\rho(t, x)$ under each drug dosage is illustrated in Figure 3.22. Extinction of sensitive population occurs much faster than the previous case when the cytotoxic drug diffused from the top boundary, since the level of cytotoxic drug touching tumor is much higher in this case, for each treatment scenario. Sensitive cells die by the time $t = 50$ under $C_1^b = 4, 6, 8$, which is also supported by Figure 3.23 which describes the proliferation events across the tumor. Almost no proliferation is visible among the sensitive population at $t = 50$ except for when $C_1^b = 2$.

Proliferation among the resistant cells appears similar across all drug dosages until $t = 10$. At the lowest dose, resistant cells proliferate more actively near the periphery than in the core, reflecting the preference for regions where nutrients and drug concentrations are both relatively high when the cytotoxic drug level remains insufficient to kill resistant cells effectively. This trend becomes less pronounced as the dosage increases.

The mean resistance level is illustrated in Figure 3.24. Under the lowest drug dosage $C_1^b = 2$, the resistant cells expand progressively, compressing the sensitive cells. In contrast, at higher dosages ($C_1^b = 4, 6, 8$), the sensitive subpopulation is eradicated much earlier, leaving only the cells of resistance level 0.8. Increased level of cytotoxic drug accelerates this pattern, but high dose ($C_1^b = 8$) of cytotoxic drug eliminates the resistant cells over time.

Average age distributions of the resistant and sensitive subpopulations are shown in Fig-

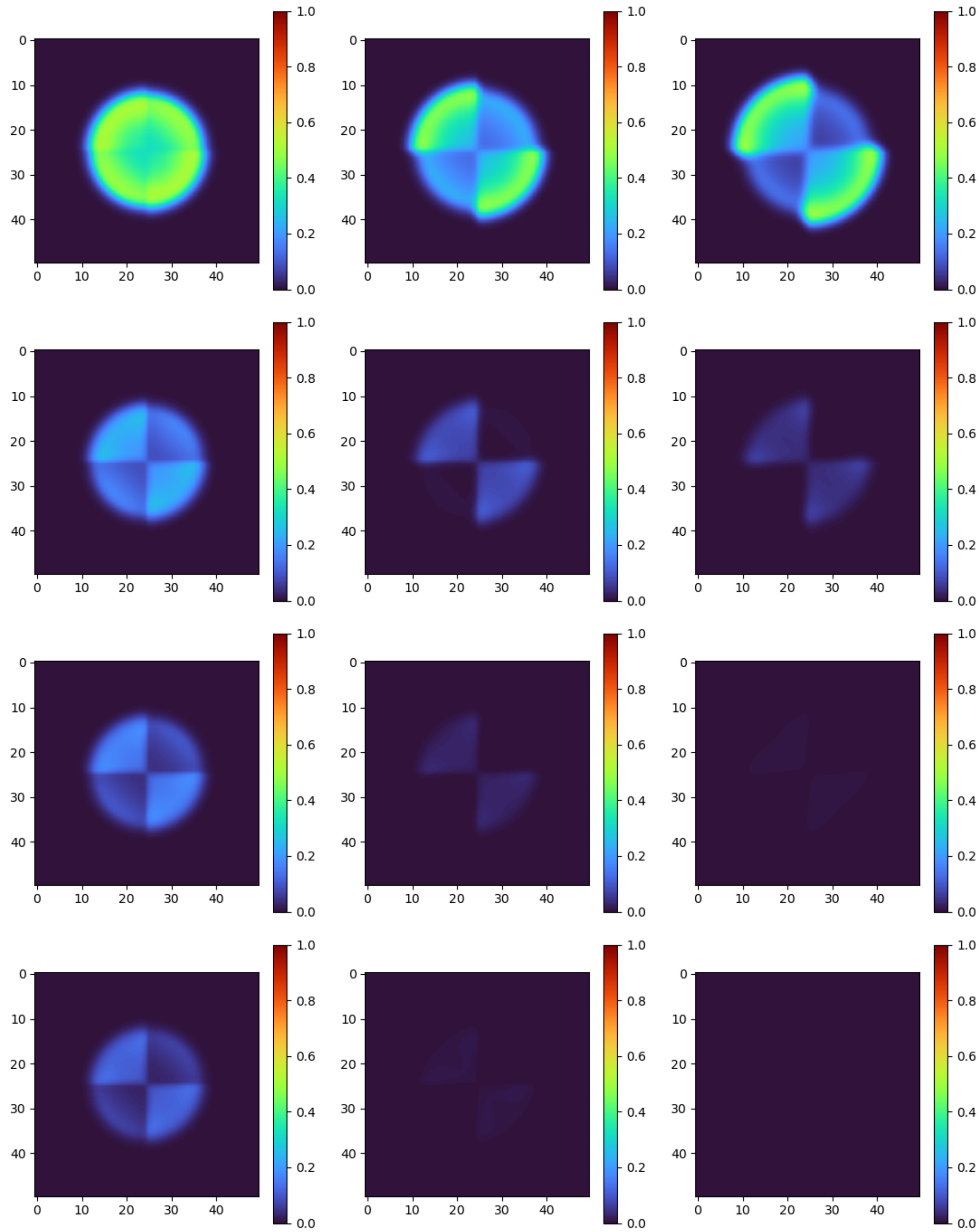

Figure 3.22: Cell population density $\rho(t, x)$ at times (left to right) $t = 10, 30, 50$ when nutrient and drug are constant outside of Ω_0. From top to bottom: $C_1^b = 2, 4, 6, 8$.

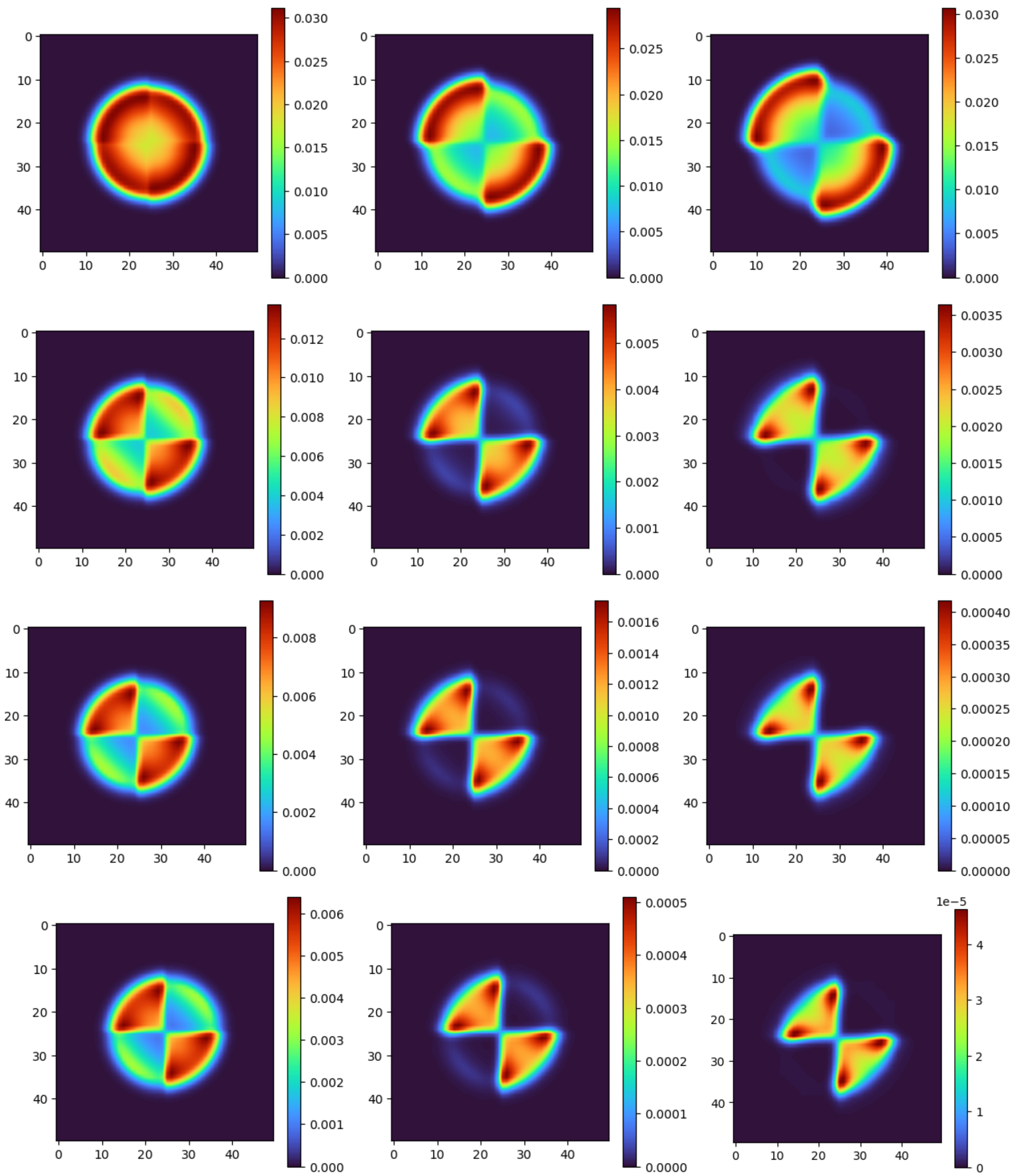

Figure 3.23: Proliferation of the tumor, given by the density $\int_0^1 n(t,x,0,\tau)\ d\tau$ at times $t = 10, 30, 50$ when nutrient and drug are assumed to be constants outside of Ω_0. From top to bottom: $C_1^b = 2, 4, 6, 8$.

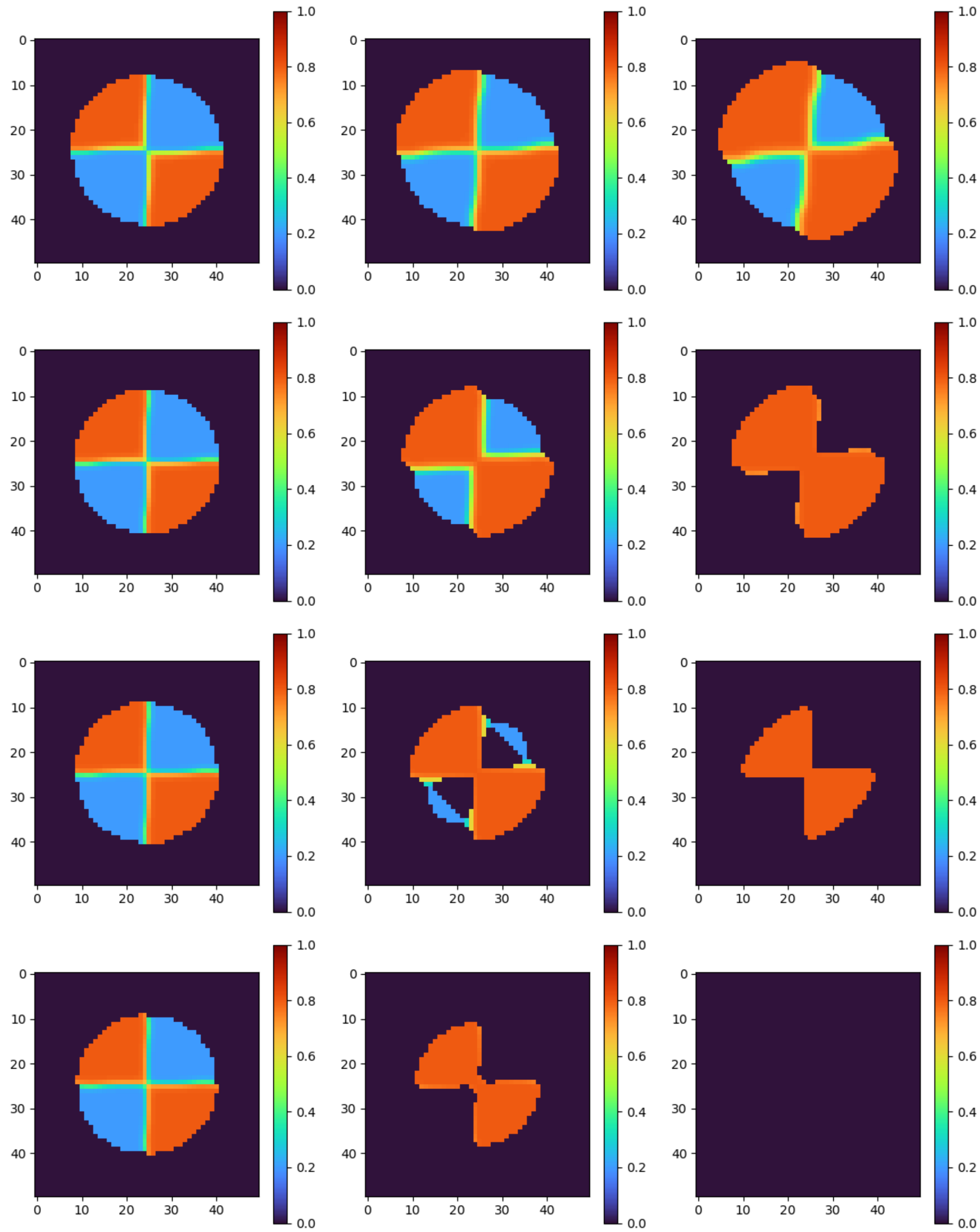

Figure 3.24: Mean resistance level across the tumor at times $t = 10, 30, 50$ when nutrient and drug are assumed to be constants outside of Ω_0. From top to bottom: $C_1^b = 2, 4, 6, 8$.

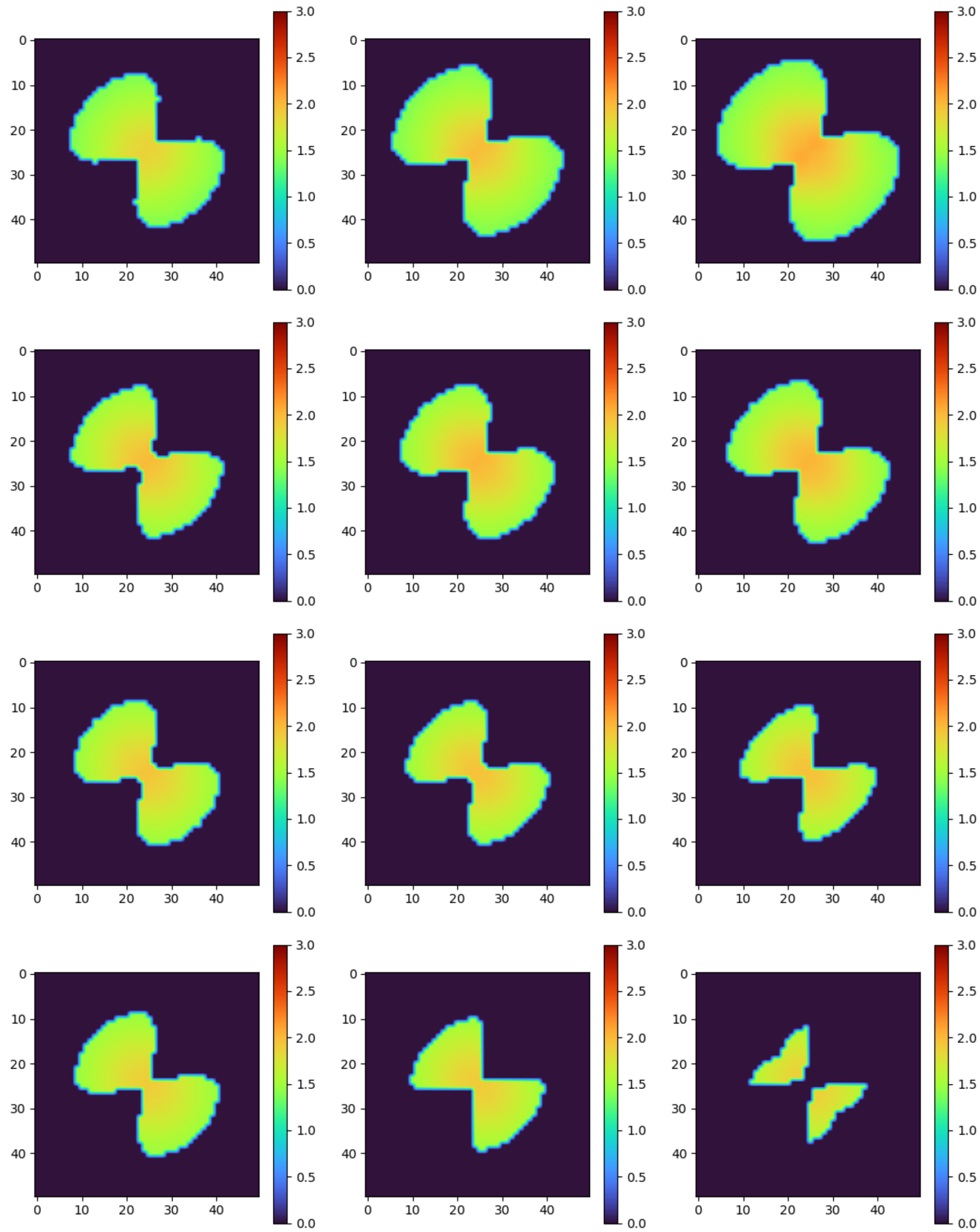

Figure 3.25: Average age of the resistant cells ($\tau = 0.8$) at times $t = 10, 30, 50$ when nutrient and drug are assumed to be constants outside of Ω_0. From top to bottom: $C_1^b = 2, 4, 6, 8$.

ures 3.25 and 3.26, respectively. Across all resistance levels, the average age is higher in the tumor core. This is consistent with the earlier observation that the region of higher average age coincides with the region with low nutrient and drug concentrations. As in the previous case, sensitive cells are younger on average than resistant cells. Both resistant and sensitive cells undergo extinction beginning in the tumor core.

The total cell population under each drug dosage is shown in Figure 3.27. Tumor rebound is observed when $C_1^b = 2$. Under $C_1^b = 4, 6$ and 8, the total population only decreases until $t = 50$.

3.3.3.3 Spatially Heterogeneous Initial Resistance with Nutrient and Drug Diffusion from the Top Boundary

Finally, we study the case when the tumor initialized with a spatially heterogeneous resistance distribution given by equation (3.27) and (3.28), while the nutrient and cytotoxic drug diffuses from the top boundary of the domain. As in all other cases, tumor is grown for 30 time units from the cell density illustrated in Figure 3.13. The resulting cell population density, mean resistance level, and the average ages of resistant and sensitive subpopulations are presented in Figure 3.28.

The tumor cell population densities $\rho(t, x)$ under varying intensities of cytotoxic drug are illustrated in Figure 3.29. The behavior of $\rho(t, x)$ is consistent with what was observed in the previous two cases. Under lower levels of drug ($C_1^b = 2, 4$), cells are concentrated at the upper part

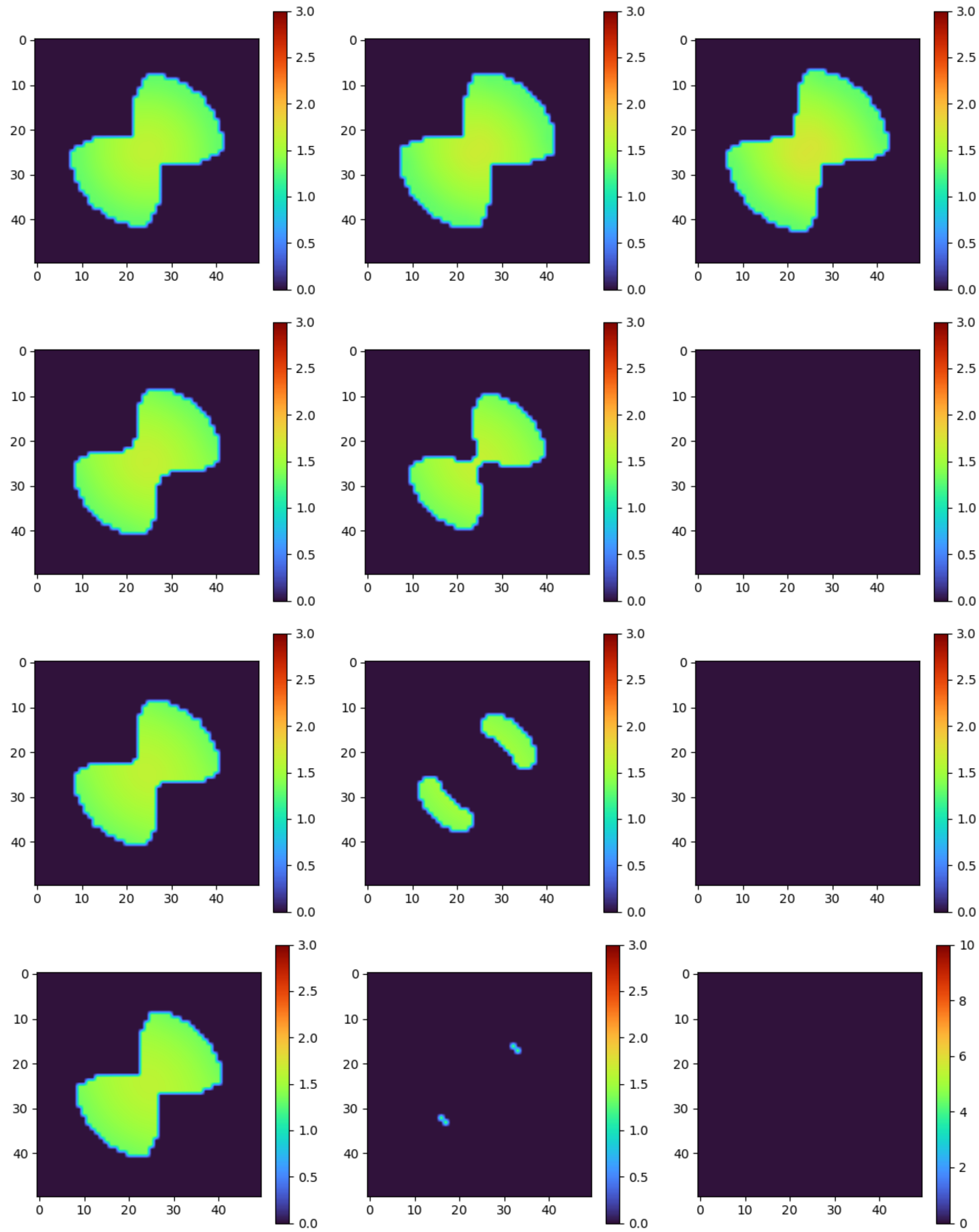

Figure 3.26: Average age of the sensitive cells ($\tau = 0.2$) at times $t = 10, 30, 50$ when nutrient and drug are assumed to be constants outside of Ω_0. From top to bottom: $C_1^b = 2, 4, 6, 8$.

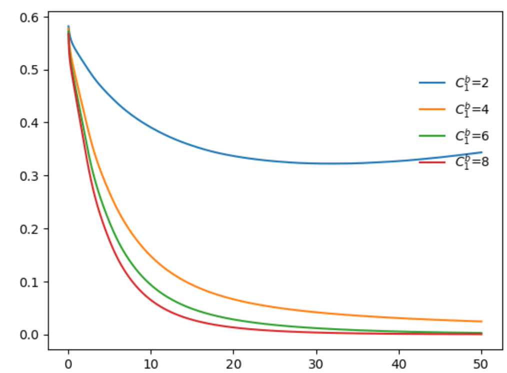

Figure 3.27: Total cell population, defined by $\int_{\mathbb{R}^2} \rho(t,x)\,dx$, when nutrient and drug are assumed to be constants outside of Ω_0, under varying cytotoxic drug levels $C_1^b = 2, 4, 6, 8$.

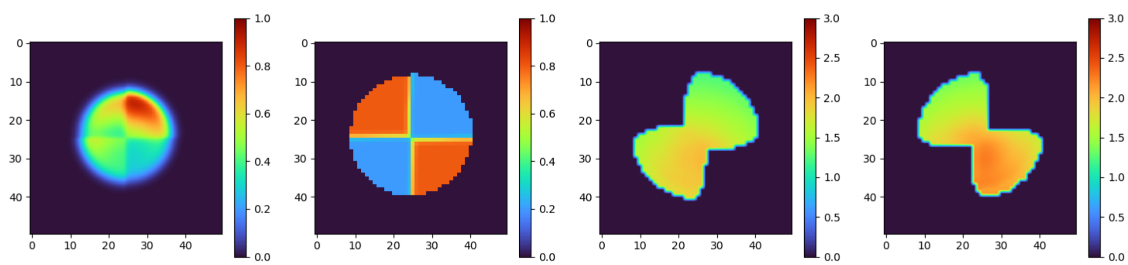

Figure 3.28: Tumor before treatment when nutrient and cytotoxic drug diffuses from the top boundary. From the left to right: Cell population density ρ, mean resistance level $\frac{\int_0^1 \tau \int_0^\infty n(0,x,\theta,\tau)\,d\theta d\tau}{\rho(0,x)}$ across the tumor, average age of the sensitive ($\tau = 0.2$) subpopulation, and average age of the resistant ($\tau = 0.8$) subpopulation of the tumor before the injection of cytotoxic drug.

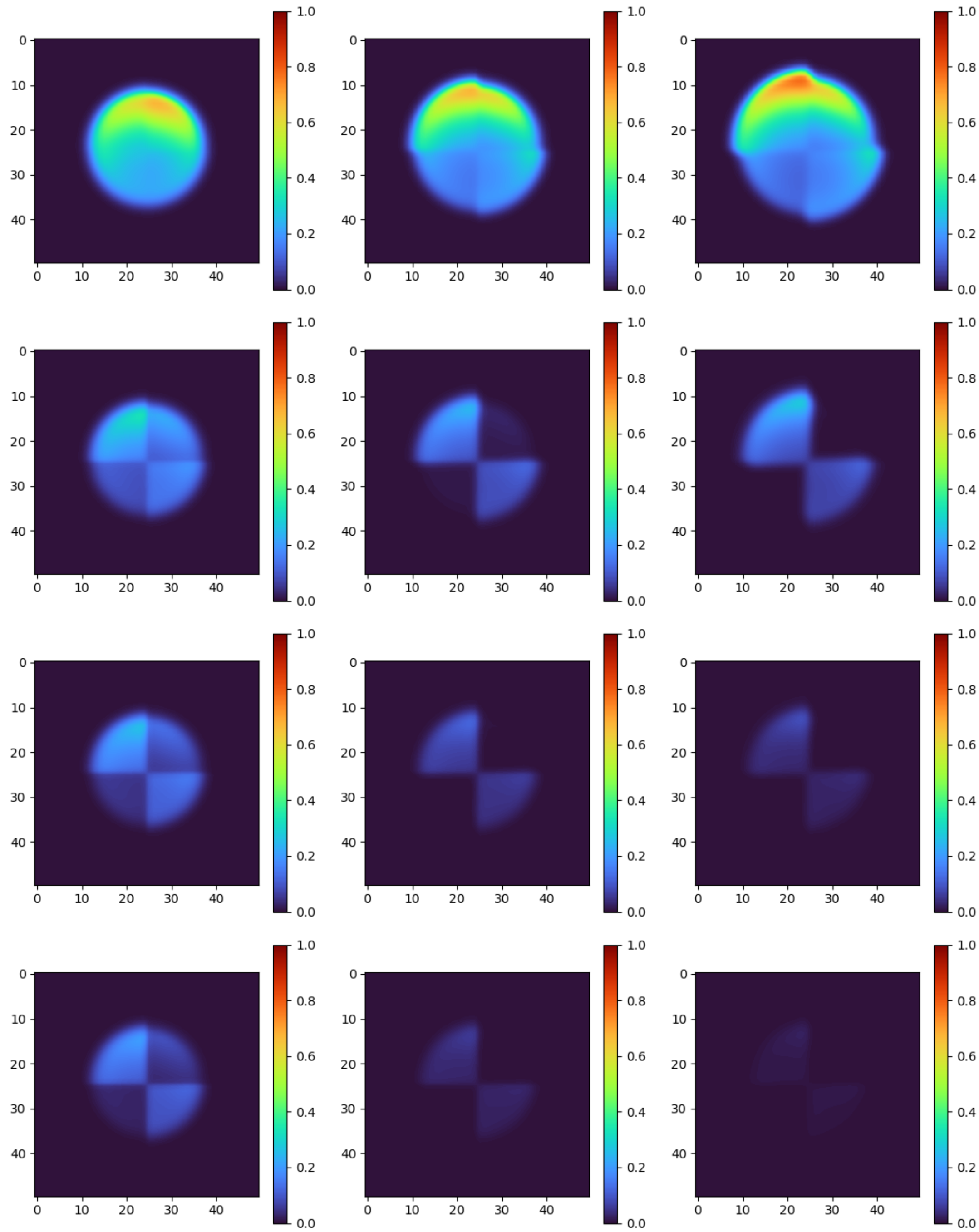

Figure 3.29: Cell population density $\rho(t,x)$ at times (left to right) $t = 10, 30, 50$ when the nutrients and drug diffuses from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

of the tumor where nutrients are most abundant. This pattern becomes negligible when $C_1^b = 6$ and 8. Sensitive cells almost disappear at $t = 50$ under the drug levels $C_1^b = 4, 6,$ and 8. Increased drug dosage results in faster elimination of sensitive cells, and more effective killing of resistant cells.

Proliferation events of the tumor cells are illustrated in Figure 3.30. The concentration of mitosis events are distinct from any of the previous two cases. Proliferation mainly occurs at the top of the tumor where nutrients are highly concentrated, for both the resistant and sensitive cells.

Mean resistance level under each drug dosage is illustrated in Figure 3.31. Expansion of resistant cells that compress the sensitive cells is observed aggressively under the lower drug dosages ($C_1^b = 2, 4$) when the sensitive subpopulation is not extinct. Overall, clearance of sensitive cells occurred faster at the lower part of the tumor where both nutrient and cytotoxic drug concentration are higher, than at the upper part.

Average age of the resistant and sensitive cells are shown in Figures 3.32 and 3.33 respectively. The regions where resistant and sensitive cells are younger coincide with what we observed in the case of spatially homogeneous initial resistance. Cells on average are youngest where proliferation mainly happens (see Figure 3.30). These observations indicate that nutrient distribution is the main factor that shapes the age distribution of tumor cells. The average age distribution of the sensitive cells reveals a more rapid extinction of the population in the lower regions of the tumor. In addition, the resistant cells appear to displace the sensitive cells outward from the center of the tumor.

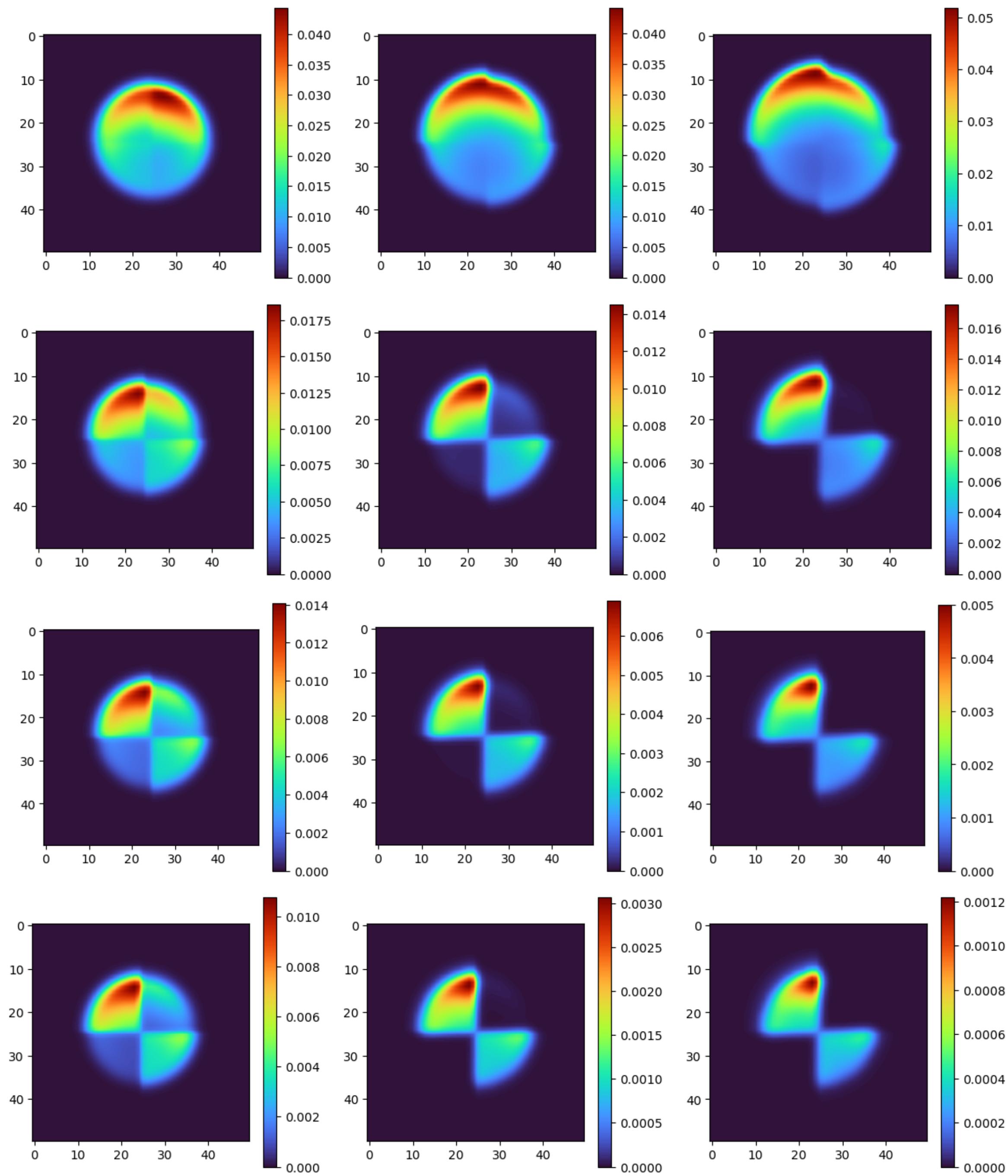

Figure 3.30: Proliferation of the tumor, given by the density $\int_0^1 n(t,x,0,\tau)\ d\tau$ at times $t = 10, 30, 50$ when nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

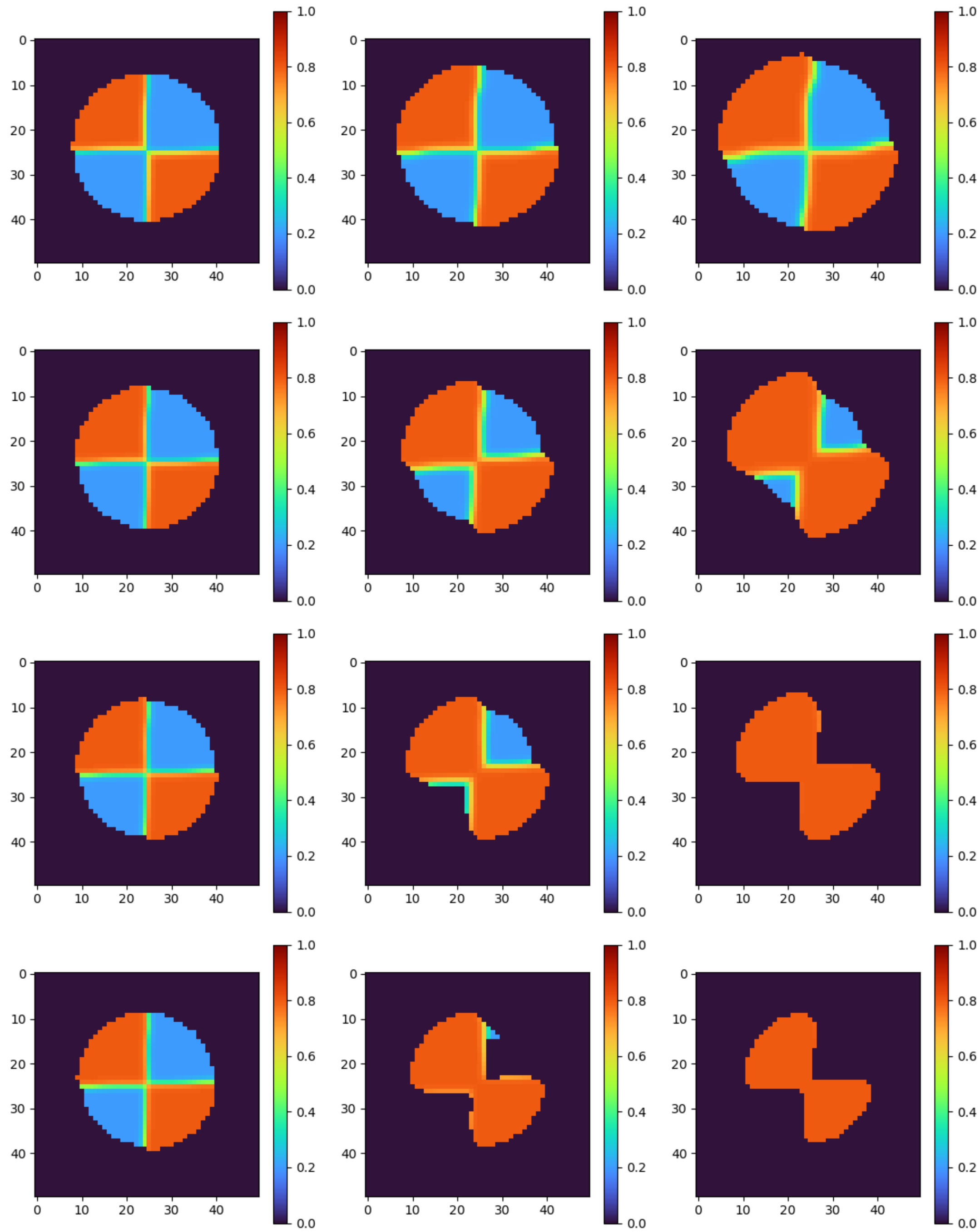

Figure 3.31: Mean resistance level across the tumor at times $t = 10, 30, 50$ when nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

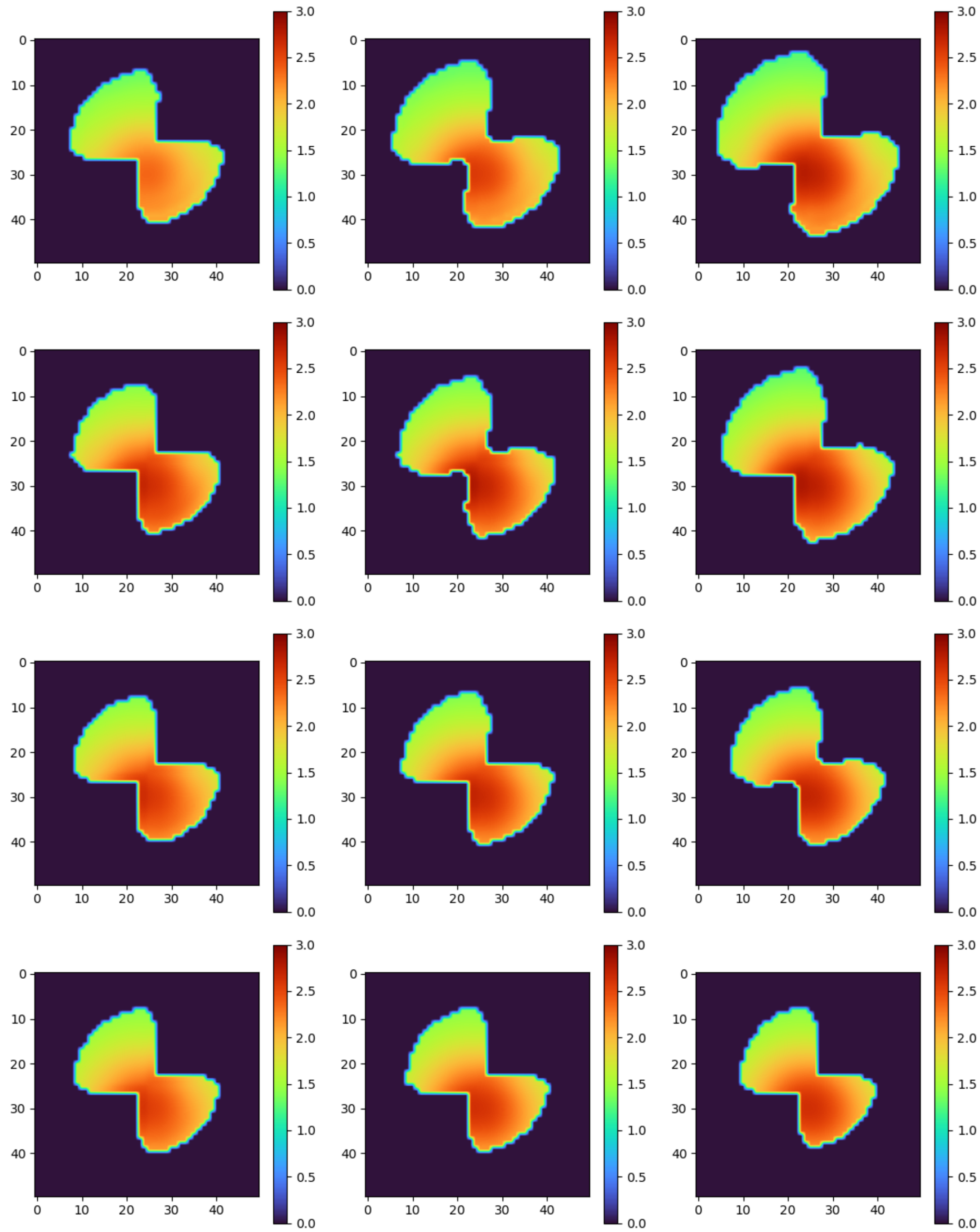

Figure 3.32: Average age of the resistant cells ($\tau = 0.8$) at times $t = 10, 30, 50$ when nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

The total cell population under each drug dosage is shown in Figure 3.34. Tumor rebound is observed when $C_1^b = 2$ and 4, while rebound in the population occurs faster under lower drug dosage. Under $C_1^b = 6$ and 8, the total population only decreases until $t = 50$.

3.4 Discussion

In this chapter, we developed an age and resistance-structured spatio-temporal model that incorporates diffusion of nutrient and drugs. Numerical simulation of the model under no clinical intervention and varying drug doses were compared, assuming constant infusion of drugs and nutrients. We find that sensitive cells have a growth advantage when clinical intervention is absent, but cytotoxic drug selected resistant cells. As seen from many other studies, lower level of cytotoxic drug resulted in faster relapse of tumor even when the drug is being constantly infused, and higher dose had stronger selection force over the resistance dynamics in tumor. An interesting case we observed was when sensitive cells are not extinct under cytotoxic drug, resistant cells formed an outer rim that surrounds the sensitive cells. This may indicate that under low dose of cytotoxic drug from which some sensitive cells may escape, resistant cells can form a protective barrier over the sensitive cells that deters delivery of drug to the sensitive cells, conferring resistance to these cells even without mutations. A competition between resistant and sensitive cells, combined with the proliferation advantage given to the sensitive cells, resulted in different age distribution in resistant cells, under low enough dose of drug to keep certain amount of sensitive cells alive. Some possible future directions are discussed below.

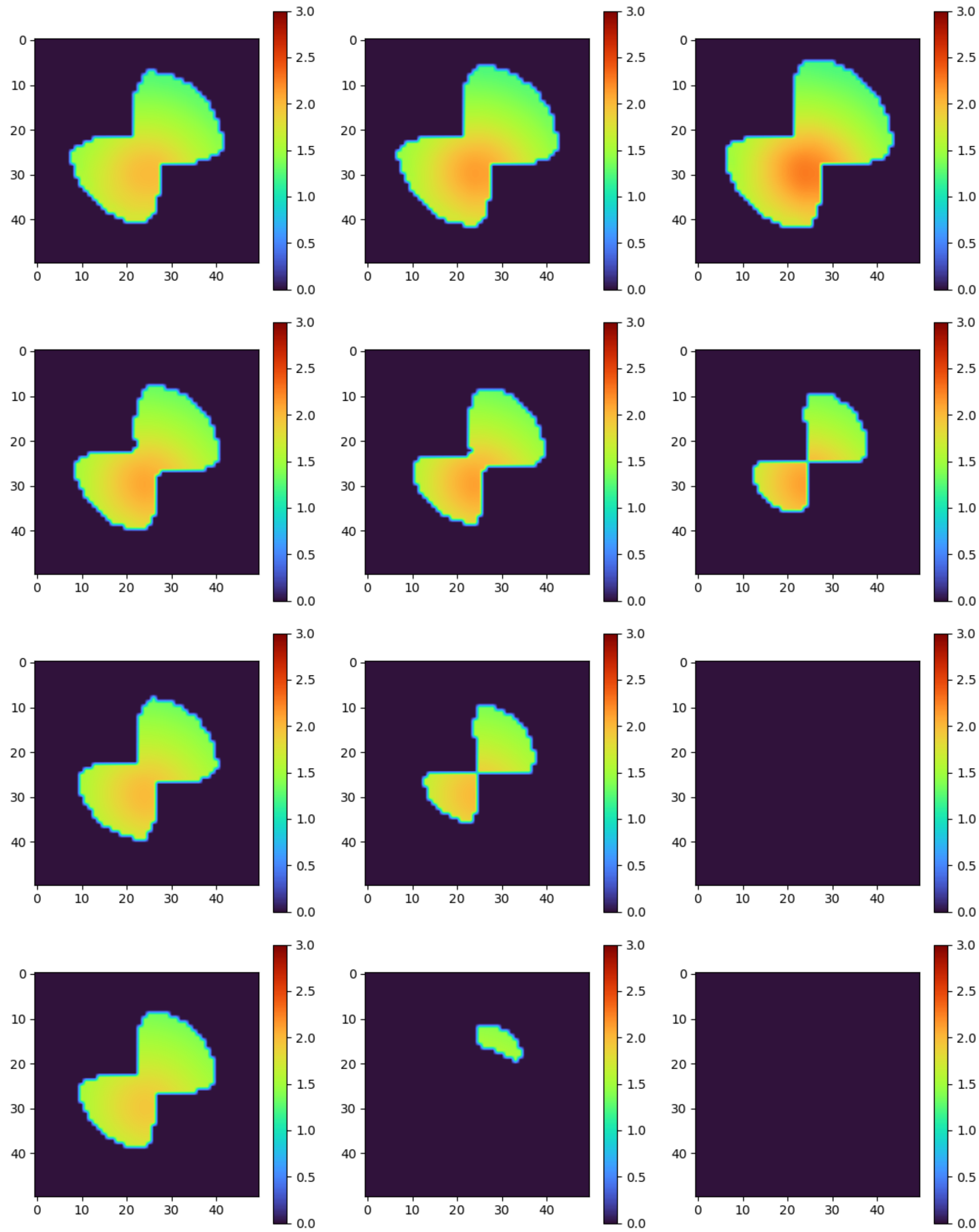

Figure 3.33: Average age of the sensitive cells ($\tau = 0.2$) at times $t = 10, 30, 50$ when nutrient and drug diffuse from the top of the domain. From top to bottom: $C_1^b = 2, 4, 6, 8$.

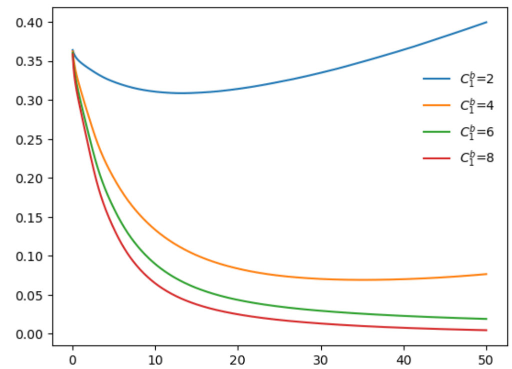

Figure 3.34: Total cell population defined by $\int_{\mathbb{R}^2} \rho(t,x)\, dx$, when nutrient and drug diffuse from the top of the domain, under varying cytotoxic drug levels $C_1^b = 2, 4, 6, 8$.

While the model equation accommodates the use of cytostatic drug, numerical simulation including the use of cytostatic drug was omitted. Combining the use of both cytostatic and cytotoxic drug may be numerically investigated in the future to examine any possible cross effect. In addition, more flexible drug administrations can be considered to compare their efficacy and resistance selection. The nutrient distribution was also largely fixed in the simulation, but there is biological evidence suggesting that nutrient and oxygen availability such as hypoxia can select resistant traits [**?**, **?**]. This can be explored with the same model by varying nutrient concentrations or diffusion coefficient inside the tumor.

It is well established that cytotoxic drug efficacy differs by cell phases. The sensitivity of tumor cells to cytotoxic drugs change over their cell cycles since many cytotoxic drugs target specific cell cycle phase. For instance, antimetabolites such as MTX is most effective for cells at S phase, and microtubule targeting agents such as taxanes, vinca alkaloids are most effective on cells in M phase and most other cytotoxic drugs are effective during S and G2 phases, when DNA structure is more vulnerable and repair mechanisms are active. A natural follow up study would be to allow the drug induced mortality or uptake rate to vary depending on the cell ages, for more accurate dynamics, especially on the age distributions.

Chapter 4: Conclusion

In this work, cell age-structured spatio-temporal models are developed to study tumor growth and the evolution of chemotherapy induced resistance.

The tumor growth model of Chapter 2 combines the age-structure and spatial heterogeneity in 2D. The results about existence of a weak solution and finite speed of propagation of the tumor front in this model are stated here and are proved in [?]. The asymptotic behavior of the model is analyzed both analytically and numerically. Numerical simulations show that model solution asymptotically approaches traveling wave when a death process is not included. The long time behavior of this model including a death term is also analyzed resulting with a condition for unique positive asymptotic profile to exist. It is shown that including death term changes the long-term behavior of the model in a significant way. The tumor volume grows exponentially in the initial phase and slows down to show linear growth of the tumor radius. Numerical simulations about the spatial distribution change of cell density, mitosis events, oldest cells, and average age all coherently show surfing front, proliferating rim near but not the edge of the tumor, and non-proliferating tumor core.

Although therapeutic intervention is not directly considered in this model, the age-structured spatio-temporal model developed in this chapter allows to observe the age distributions and regions where cells are actively proliferating or seemingly quiescent. In this regard, the model can provide clinically meaningful insights, contributing to treatment timing and injection locations, considering that the efficacy of therapy often depends on the cell cycle or dormancy.

The model presented in the next chapter, Chapter 3, builds on the model of Chapter 2 [**?**] and the work of Cho and Levy (2018) [**?**]. This model combines a resistance phenotype variable to the age-structure and spatial heterogeneity. The model also incorporates the diffusion of drugs and nutrient to study their effects on the dynamics inside a tumor. The spatial distribution of cell density, mitosis events, mean resistance level, and the average age of cells in each resistance level is simulated numerically under varying initial resistance distributions and diffusion of nutrient and drugs, to exhibit how resistance cells pay fitness cost and nutrient depletion limits proliferation and cell population growth at the core of tumor. Under chemotherapy, drug-induced selection of resistant traits was observed, and smaller dose of drugs resulted in a faster relapse of the tumor. The average age across the tumor is mostly shaped by the nutrient and drug concentrations, and when cells of different resistance levels occupy the same location, the average age of more resistant cells under lower drug dose showed a different spatial pattern than what was observed in the other cases. We hypothesize that this difference is the result of a competition between cells of different resistance levels, and may be attributed to the setting of a higher uptake rate of the drug in the more sensitive cells.

The model in Chapter 3 directly observes how different doses of cytotoxic drugs can affect treatment outcomes (including the time of relapse), and shape the selection of resistance traits.

In particular, the model demonstrates how the competition between cells of different resistance levels influence average age of resistant cells at different locations of the tumor, which may contribute to understand the regions where resistant cells are proliferating most. Since chemotherapies depend heavily on cell cycles and targets actively cycling cells, this implies that knowledge about the locations inside tumor where resistant cells are better targeted can be pursued with models of this type.

One of the major differences between the properties of the model in Chapter 2 and 3 is that convergence of overall age distribution could not be noticed in the simulation in Chapter 3 until time $t = 100$ unlike what is observed in the Chapter 2. Especially, given the distinct way of how pressure dictates cell proliferation in the two works, and considering the introduction of cytotoxic drug and nutrient in this work, the result of Chapter 2 about age distributions may have little direct implications to what was studied in Chapter 3. However, there remains possibly interesting dynamics that were left uninvestigated in the model of Chapter 3. It is generally accepted that resistant tumor cells may proliferate slower than the sensitive cells, and this is incorporated in the model of Chapter 3. In this setting, different age distribution dynamics are expected across cell subpopulations with distinct drug resistance levels. There has been a recent study that reported the difference in age distribution across drug resistance levels as well. In He *et al.* (2024), they observed that majority of drug-resistant cells were in the G0/G1 phase [**?**]. Further investigation in this direction may open an useful discussion for engineering anti-tumor therapies that can target specific cell cycle phases.

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